Chapter 11

Conic Sections-I

Solutions

SECTION - A

Objective Type Questions (One Option is correct)

[Equation of Circles in Different Form]

1. The equation of the circle passing through the points (0, 0), (2, 0), (0, 3) is

(1) $x^2 + y^2 - 2x - 3y = 0$ (2) $x^2 + y^2 + 2x + 3y = 0$

(3) $x^2 + y^2 + 4x - 6y = 0$ (4) $x^2 + y^2 - 3x - xy = 0$

Sol. Answer (1)

Let the circle is $x^2 + y^2 + 2gx + 2fy + c = 0$

Putting (x,y) = (0, 0), (2, 0), (0, 3) respectively we get $c = 0$, $g = -1$, $f = \frac{-3}{2}$

Hence equation becomes $x^2 + y^2 - 2x - 3y = 0$

2. The radius of the circle passing through the points (1, 2), (5, 2) and (5, –2) is

(1) $2\sqrt{2}$ (2) $3\sqrt{2}$ (3) $2\sqrt{5}$ (4) $5\sqrt{2}$

Sol. Answer (1)

$A(1, 2)$, $B(5, 2)$, $C(5, -2)$

Abscissae of B and C are equal

$\therefore$ BC is parallel to x-axis.

y coordinates of A and B are equal.

$\therefore$ AB is parallel to y-axis.

$\therefore$ ΔABC is right angled triangle at B.

$\therefore$ Hypotenuse AC = diameter of the circle

$$2r = \sqrt{16+16} = 4\sqrt{2} \quad \Rightarrow \quad r = 2\sqrt{2}$$

3. The lines $2x - 3y = 5$ and $3x - 4y = 7$ are diameters of a circle of area 154 cm^2. Then the equation of the circle is

(1) $x^2 + y^2 + 2x - 2y = 62$ (2) $x^2 + y^2 + 2x - 2y = 47$

(3) $x^2 + y^2 - 2x + 2y = 62$ (4) $x^2 + y^2 - 2x + 2y = 47$

Sol. Answer (4)

Solve equation $2x - 3y = 5$ and $3x - 4y = 7$

$(1, -1)$

Centre of circle = $(1, -1)$

$\pi r^2 = 154$

$r = 7$

Equation of circle $(x-1)^2 + (y+1)^2 = 49$

$x^2 + y^2 - 2x + 2y - 47 = 0$

4. The equation of diameter of a circle $x^2 + y^2 + 2x - 4y = 4$, that is parallel to $3x + 5y = 4$ is

(1) $3x + 5y = 7$ (2) $3x - 5y = 7$

(3) $3x + 5y = -7$ (4) $3x - 5y = -7$

Sol. Answer (1)

Equation of diameter $3x + 5y = \lambda$

Centre $(-1, 2)$ lie on the diameter

$-3 + 10 = \lambda \Rightarrow \lambda = 7$

Diameter $3x + 5y = 7$

5. The intercept on the line $y = x$ by the circle $x^2 + y^2 - 2x = 0$ is AB. Equation of the circle with AB as the diameter is

(1) $x^2 + y^2 + x - y = 0$ (2) $x^2 + y^2 + x + y = 0$

(3) $x^2 + y^2 - x - y = 0$ (4) $x^2 + y^2 - x + y = 0$

Sol. Answer (3)

Equation of circle $S + \lambda L = 0$

$x^2 + y^2 - 2x + \lambda(x - y) = 0$

$x^2 + y^2 + (\lambda - 2)x - \lambda y = 0 \quad \ldots(1)$

Centre $\left(\frac{2-\lambda}{2}, \frac{\lambda}{2}\right)$ lie on the line $y = x$

$\frac{2-\lambda}{2} = \frac{\lambda}{2} \Rightarrow \lambda = 1$

$\therefore$ From (1) $x^2 + y^2 - x - y = 0$

6. If x_1, x_2 are the roots of the equation $x^2 + bx + c = 0$ and y_1 and y_2 are the roots of $y^2 + qy + r = 0$ then the equation of the circle having (x_1, y_1) and (x_2, y_2) as ends of diameter is

(1) $x^2 + y^2 + bx + qy + c - 2r = 0$ (2) $x^2 + y^2 + bx + qy + 2c + r = 0$

(3) $x^2 + y^2 + bx + qy + c + r = 0$ (4) $x^2 + y^2 - bx - qy - c - r = 0$

Sol. Answer (3)

$x_1 + x_2 = -b,\ x_1x_2 = c$

$y_1 + y_2 = -q,\ y_1y_2 = r$

Circle is

$x^2 - x(x_1 + x_2) + y^2 - y(y_1 + y_2) + x_1x_2 + y_1y_2 = 0$

$x^2 + y^2 + bx + qy + c + r = 0$

7. The shortest distance form the point $P(-7, 2)$ to the circle $x^2 + y^2 - 10x - 14y - 151 = 0$ is in units

(1) 4 (2) 3 (3) 2 (4) 1

Sol. Answer (3)

$S_1 \equiv (-7)^2 + (2)^2 - 10(-7) - 14(2) - 151 < 0$

Point P lie inside the circle

Centre of circle = C (5, 7), P (–7, 2)

Radius of circle = 15

$CP = \sqrt{144 + 25} = 13$

Shortest distance of circle from $P = r - CP = 15 - 13 = 2$

8. The length of intercept of x-axis by the circle $x^2 + y^2 + 6x + 2y + 1 = 0$ is

(1) $\sqrt{2}$ (2) 2 (3) $8\sqrt{2}$ (4) $4\sqrt{2}$

Sol. Answer (4)

$2\sqrt{g^2 - c} = 2\sqrt{(3)^2 - 1}$

$= 4\sqrt{2}$

9. The number of normals from any point to a circle cannot be

(1) 0 (2) 1 (3) 2 (4) 3

Sol. Answer (1)

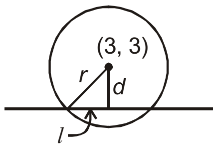
Number of normals cannot be zero.

10. The length of intercept on the straight line $3x + 4y - 1 = 0$ by the circle $x^2 + y^2 - 6x - 6y - 7 = 0$ is

(1) $2\sqrt{2}$ (2) 6 (3) $4\sqrt{2}$ (4) $\sqrt{2}$

Sol. Answer (2)

$d = \dfrac{3\times 3 + 4\times 3 - 1}{\sqrt{9 + 16}} = 4$



$2\sqrt{5^2 - 4^2} = 6$

11. Circles are drawn through the point (2, 0) to cut intercept of length 5 units on x-axis. If their centre lies in the first quadrant, then their equation is

(1) $x^2 + y^2 - 9x + 2ky + 14 = 0,\ k \in R^+$ (2) $3x^2 + 3y^2 + 27x - 2ky + 42 = 0,\ k \in R^+$

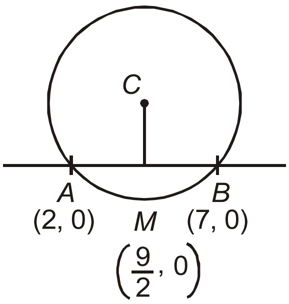
(3) $x^2 + y^2 - 9x - 2ky + 14 = 0,\ k \in R^+$ (4) $x^2 + y^2 - 2kx - 9y + 14 = 0,\ k \in R^+$

Sol. Answer (3)

$AB = 5$

$A(2, 0)\ \therefore B(7, 0)$

Middle point $M\left(\frac{9}{2},\ 0\right)$



$\therefore$ Centre is $\left(\frac{9}{2},\ k\right)$ and radius = $AC = \sqrt{\left(\frac{5}{2}\right)^2 + k^2}$

Eq. of circle $\left(x - \frac{9}{2}\right)^2 + \left(y - k\right)^2 = \left(\frac{5}{2}\right)^2 + k^2$

$x^2 + y^2 - 9x - 2yk + 14 = 0$

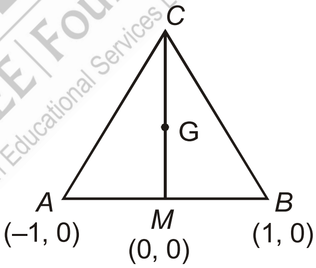
12. Two vertices of an equilateral triangle are (–1, 0) and (1, 0) and its third vertex lies above the x-axis. The equation of circumcircle is

(1) $x^2 + y^2 - \frac{2y}{\sqrt{3}} - 1 = 0$ (2) $x^2 + y^2 - \frac{y}{\sqrt{3}} - 1 = 0$ (3) $x^2 + y^2 - \frac{2y}{3} - 1 = 0$ (4) $x^2 + y^2 + x + y = 0$

Sol. Answer (1)

$CM = \frac{\sqrt{3}}{2}(AB) = \sqrt{3}$

$\therefore\ C\,(0,\ \sqrt{3})$



Centroid $G\left(0,\ \frac{\sqrt{3}}{3}\right)$ = circumcentre $\left(0,\ \frac{1}{\sqrt{3}}\right)$

Radius $r = AG = \sqrt{1 + \frac{1}{3}} = \frac{2}{\sqrt{3}}$

Equation of circumcircle, $x^2 + \left(y - \frac{1}{\sqrt{3}}\right)^2 = \frac{4}{3}$

$x^2 + y^2 - \frac{2y}{\sqrt{3}} - 1 = 0$

13. The centre of a circle passing through the origin and cutting of intercepts 3 and 4 on the x and y-axes is

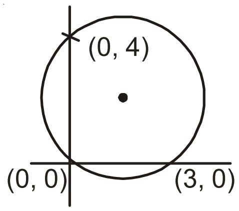
(1) $\left(2,\ \frac{3}{2}\right)$ (2) $\left(\frac{3}{2},\ \frac{3}{2}\right)$ (3) $\left(\frac{3}{2},\ 2\right)$ (4) (2, 2)

Sol. Answer (3)

Equation of a circle is

$x^2 + y^2 - 3x - 4y = 0$

Centre is $\left(\frac{3}{2},\ 2\right)$



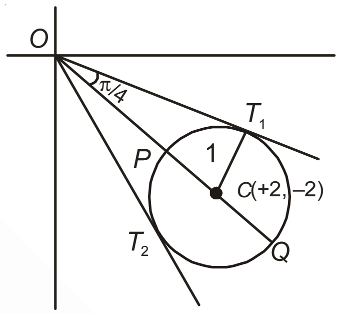
14. The co-ordinates of a point P, which lies on the circle $x^2 + y^2 - 4x + 4y + 7 = 0$ in such a way that OP is minimum, are

(1) $\left(2+\frac{1}{\sqrt{2}}, -2+\frac{1}{\sqrt{2}}\right)$ (2) $\left(2-\frac{1}{\sqrt{2}}, -2+\frac{1}{\sqrt{2}}\right)$ (3) $\left(-2, -2+\frac{1}{\sqrt{2}}\right)$ (4) $\left(\frac{1}{\sqrt{2}}, -2+\frac{1}{\sqrt{2}}\right)$

Sol. Answer (2)

Let $S : x^2 + y^2 - 4x + 4y + 7 = 0$

$\Rightarrow (x-2)^2 + (y+2)^2 = 1$



$\therefore \quad \angle COT_1 = -\frac{\pi}{4}$

$\Rightarrow \quad \angle POT_1 = -\frac{\pi}{4}$

$\therefore \quad OP = OC - CP$

$= \sqrt{4+4} - 1 = 2\sqrt{2} - 1$

As we know that,

$z = r(\cos\theta + i\sin\theta)$

$= |z|(\cos\theta + i\sin\theta)$

$= |OP|\left(\cos\left(-\frac{\pi}{4}\right) + i\sin\left(-\frac{\pi}{4}\right)\right)$

$= |OP|\left(\frac{1}{\sqrt{2}} - i\frac{1}{\sqrt{2}}\right)$

$= \frac{|OP|}{\sqrt{2}} - i.\frac{OP}{\sqrt{2}}$

$= \left(\frac{2\sqrt{2}-1}{\sqrt{2}}\right) - i.\left(\frac{2\sqrt{2}-1}{\sqrt{2}}\right)$

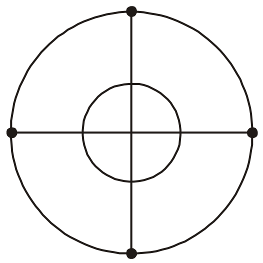
$= \left(\frac{2\sqrt{2}-1}{\sqrt{2}}, -\frac{2\sqrt{2}-1}{\sqrt{2}}\right)$

$= \left(2-\frac{1}{\sqrt{2}}, -2+\frac{1}{\sqrt{2}}\right)$

15. The number of points $(a + 1, a)$ where $a \in I$, lying inside the region bounded by the circles $x^2 + y^2 - 2x - 1 = 0$ and $x^2 + y^2 - 2x - 15 = 0$ is

(1) 2 (2) 3 (3) 4 (4) 6

Sol. Answer (3)



Let $S_1 : x^2 + y^2 - 2x - 1 = 0$

$\Rightarrow (x-1)^2 + y^2 = 1$

and $S_2 : x^2 + y^2 - 2x - 15 = 0$

$\Rightarrow (x-1)^2 + y^2 = 4^2$

Number of required points are 4, there are (2, 1), (3, 2) (–2, –1), (–2, –3)

16. Four distinct points $\left(a, \frac{1}{a}\right)$, $\left(b, \frac{1}{b}\right)$, $\left(c, \frac{1}{c}\right)$ and $\left(d, \frac{1}{d}\right)$ are lie on a circle, where a, b, c, $d \neq 0$ then the value of $abcd$ is

(1) 2 (2) 1 (3) 3 (4) 4

Sol. Answer (2)

Let the equation of a circle is $x^2 + y^2 + 2gx + 2fy + c = 0$

Since $\left(a, \frac{1}{a}\right)$ lies on the circle, we get,

$$a^2 + \frac{1}{a^2} + 2g.a + 2f.\frac{1}{a} + c = 0$$

$\Rightarrow$ $a^4 + 2ga^3 + 2fa + ca^2 + 1 = 0$

$\Rightarrow$ $a^4 + 2ga^3 + ca^2 + 2fa + 1 = 0$

$\therefore$ $\Sigma abcd = 1$

[Tangent to Circle]

17. The length of the tangents from any point of the circle $15x^2 + 15y^2 - 48x + 64y = 0$ to the two circles $5x^2 + 5y^2 - 24x + 32y + 75 = 0$, $5x^2 + 5y^2 - 48x + 64y + 300 = 0$ are in the ratio

(1) 1 : 2 (2) 1 : 4 (3) 2 : 3 (4) 3 : 4

Sol. Answer (1)

O (0 0) lie on the first circle.

Length of tangent from 0 to the 2nd circle $OT_1 = \sqrt{\frac{75}{5}}$

Length of tangent from 0 to the 3rd circle $OT_2 = \sqrt{\frac{300}{5}}$

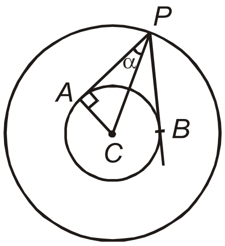
$\therefore$ $OT_1 : OT_2 = 1 : 2$

18. If from any point P on the circle $x^2 + y^2 + 2gx + 2fy + c = 0$, tangents are drawn to the circle $x^2 + y^2 + 2gx + 2fy + c\sin^2\theta + (g^2 + f^2)\cos^2\theta = 0$, then the angle between the tangents is

(1) $\frac{\theta}{4}$ (2) $\frac{\theta}{2}$ (3) θ (4) 2θ

Sol. Answer (4)

Circles are concentric and centre is $C(-g, -f)$



$$R_1 = \sqrt{g^2 + f^2 - c}$$

$$R_2 = \sqrt{g^2 + f^2 - c\sin^2\theta - (g^2 + f^2)\cos^2\theta}$$

$$R_2 = \sin\theta\sqrt{g^2 + f^2 - c} = \sin\theta \cdot R_1$$

$$\sin\alpha = \frac{AC}{PC} = \frac{R_2}{R_1} = \sin\theta$$

$\therefore$ $\alpha = \theta$

$2\alpha = 2\theta$

$\therefore$ $\angle APB = 2\theta$

19. The length of the tangent drawn from any point on the circle $x^2 + y^2 + 2gx + 2fy + a = 0$ to the circle $x^2 + y^2 + 2gx + 2fy + b = 0$ is

(1) $\sqrt{b-a}$ (2) $\sqrt{a-b}$ (3) $\sqrt{a+b}$ (4) $\sqrt{ab}$

Sol. Answer (1)

Length of tangent from any point on the circle S_1 to the circle S_2 = $\sqrt{S_2 - S_1} = \sqrt{b-a}$

20. If the line $y = 3x + c$ is a tangent to $x^2 + y^2 = 4$ then the value of c is

(1) ±4 (2) $\pm 2\sqrt{10}$ (3) $\pm 10\sqrt{2}$ (4) $\pm\sqrt{10}$

Sol. Answer (2)

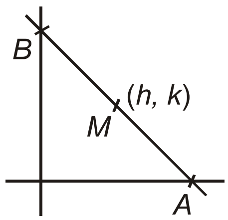
$c = \pm a\sqrt{1+m^2} = \pm 2\sqrt{1+9} = \pm 2\sqrt{10}$

21. Locus of middle point of intercept of any tangent with respect to the circle $x^2 + y^2 = 4$ between the axis is

(1) $x^2 + y^2 - x^2y^2 = 0$ (2) $x^2 + y^2 + x^2y^2 = 0$ (3) $x^2 + y^2 - 2x^2y^2 = 0$ (4) $x^2 + y^2 - 3x^2y^2 = 0$

Sol. Answer (1)

Let the tangent is $x\cos\theta + y\sin\theta = 2$



Clearly $A = \left(\dfrac{2}{\cos\theta}, 0\right), B = \left(0, \dfrac{2}{\sin\theta}\right)$

$\Rightarrow \quad h = \dfrac{1}{\cos\theta}, k = \dfrac{1}{\sin\theta}$

$\Rightarrow \quad x^2 + y^2 = x^2y^2$

22. Two perpendicular tangents to the circle $x^2 + y^2 = a^2$ meet at P. Then the locus of P has the equation :

(1) $x^2 + y^2 = 2a^2$ (2) $x^2 + y^2 = 3a^2$ (3) $x^2 + y^2 = 4a^2$ (4) $x^2 + y^2 = 5a^2$

Sol. Answer (1)

Let $P(h, k)$

Equation of line through $P(h, k)$

$y - k = m(x - h)$

$mx - y + k - mh = 0 \quad \ldots(i)$

Distance of line from centre = radius

$$\frac{k - mh}{\sqrt{m^2+1}} = a$$

$(h^2 - a^2)m^2 - 2mkh + k^2 - a^2 = 0$

Tangents are perpendicular

$\therefore \quad m_1m_2 = -1$

$$\frac{k^2 - a^2}{h^2 - a^2} = -1$$

$\therefore \quad h^2 + k^2 = 2a^2$

Locus of $P(h, k)$

$x^2 + y^2 = 2a^2$

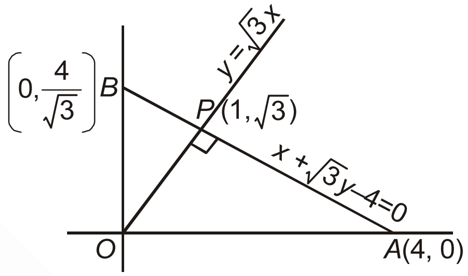
23. The area of the triangle formed by the +ve x-axis and the normal and the tangent to the circle $x^2 + y^2 = 4$ at $(1, \sqrt{3})$ is

(1) $2\sqrt{3}$ (2) $\sqrt{3}$ (3) $\frac{1}{\sqrt{3}}$ (4) 1

Sol. Answer (1)

Equation of tangent at $P(1, \sqrt{3})$ is

$x + \sqrt{3}y - 4 = 0$ ...(i)



Equation of normal is

$\sqrt{3}x - y + \lambda = 0$

It passes through $(1, \sqrt{3})$ so $\lambda = 0$

Equation of normal is,

$\sqrt{3}x - y = 0$ ...(ii)

Area of the triangle formed by x-axis, tangent at $(1, \sqrt{3})$ and normal at $(1, \sqrt{3})$ is

$= \text{Area of } \Delta OPA$

$= \frac{1}{2} \times OP \times AP = \frac{1}{2} \times 2 \times 2\sqrt{3} = 2\sqrt{3}$ sq. units.

24. If $3x + y = 0$ is a tangent to the circle which has its centre at the point $(2, -1)$, then equation of the other tangent to the circle from the origin, is

(1) $x + 3y = 0$ (2) $3x - y = 0$ (3) $x - 3y = 0$ (4) $x + 2y = 0$

Sol. Answer (3)

Centre $C(2, -1)$

Radius of circle = distance of line $3x + y = 0$ from centre

$$r = \frac{6-1}{\sqrt{9+1}} = \frac{5}{\sqrt{10}} = \sqrt{\frac{5}{2}}$$

Eq. of line from origin $y = mx$

Distance of line from centre = r

$$\frac{2m+1}{\sqrt{m^2+1}} = \sqrt{\frac{5}{2}}$$

$$3m^2 + 8m - 3 = 0 \Rightarrow m = -3 \text{ and } \frac{1}{3}$$

$$y = \frac{1}{3}x \Rightarrow x - 3y = 0$$

25. If equation of one tangent drawn from (0, 0) to the circle with centre (2, 4) is $4x + 3y = 0$, then equation of the other tangent from (0, 0) is

(1) $4x - 3y = 0$ (2) $x = 0$ (3) $y = 0$ (4) $x + 4y = 0$

Sol. Answer (3)

Radius of circle = $\dfrac{8+12}{\sqrt{16+9}} = \dfrac{20}{5} = 4$

Equation of line through origin $y = mx$

Distance from centre = radius

$$\frac{2m-4}{\sqrt{m^2+1}} = 4 \Rightarrow m(3m+4) = 0$$

$m = -\dfrac{4}{3}$ already considered

$\therefore$ $m = 0$

$y = 0$

26. The area of the triangle formed by the tangents from the point (4, 3) to the circle $x^2 + y^2 = 9$ and the line joining their points of contact is

(1) 12 (2) 6 (3) 4 (4) $\dfrac{192}{25}$

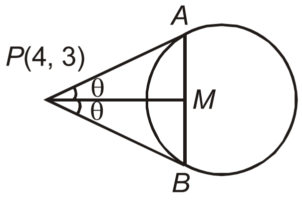
Sol. Answer (4)

Equation of chord of contact AB

$$x \cdot 4 + y \cdot 3 - 9 = 0$$

$$PM = \left|\frac{16+9-9}{\sqrt{25}}\right| = \frac{16}{5}$$

(Length of perpendicular from $P(4, 3)$ to $AB \equiv 4x + 3y - 9 = 0$)



($\because$ Length of tangent is $\sqrt{S_1}$)

$PA = \sqrt{16+9-9} = 4$

$\cos\theta = \dfrac{PM}{PA} = \dfrac{4}{5}$

Area of $\Delta APB = \dfrac{1}{2}(PA)^2 \cdot \sin 2\theta$ [$\therefore$ Area of $\Delta = \dfrac{1}{2}\times(PA)\times(PB)\sin\angle APB$]

$$\Delta = \frac{1}{2} bc \sin A$$

$$= \frac{1}{2} \cdot 16 \cdot 2 \sin\theta \cdot \cos\theta$$

$$= 16 \cdot \frac{4}{5} \cdot \frac{3}{5} = \frac{192}{25}$$

27. If two tangents are drawn from a point to the circle $x^2 + y^2 = 32$ to the circle $x^2 + y^2 = 16$, then the angle between the tangents is

(1) $\frac{\pi}{4}$ (2) $\frac{\pi}{3}$ (3) $\frac{\pi}{2}$ (4) $\frac{\pi}{6}$

Sol. Answer (3)

$S_1 : x^2 + y^2 = 32$

$S_2 : x^2 + y^2 = 16$

$\Rightarrow$ $S_1 = 0$ is the director circle of $S_2 = 0$

$\Rightarrow$ Director circle is the locus of two perpendicular tangents

Angle is $\frac{\pi}{2}$

28. The point of intersection of the lines $x - y + 1 = 0$ and $x + y + 5 = 0$ is P. A circle with centre at (1, 0) passes through P. The tangent to the circle at P meets the x-axis at (k, 0). The value of k is

(1) 2 (2) –3 (3) –2 (4) –4

Sol. Answer (4)

$L_1 : x - y + 1 = 0$

$L_2 : x + y + 5 = 0$

$\Rightarrow$ $x = -3, y = -2$

$\Rightarrow$ $P \equiv (-3, -2)$

Equation of circle is $(x - 1)^2 + y^2 = 20$

$\Rightarrow$ $x^2 + y^2 - 2x - 19 = 0$

Equation of tangent at P is

$2x + y + 8 = 0$

Put $y = 0$

$\Rightarrow$ $x = -4$

Point is (–4, 0)

$\Rightarrow$ $k = -4$

29. The equation of one of the circles which touch the pair of lines $x^2 - y^2 + 2y - 1 = 0$ is

(1) $x^2 + y^2 + 2x + 1 = 0$ (2) $x^2 + y^2 - 2x + 1 = 0$

(3) $x^2 + y^2 + 2y - 1 = 0$ (4) $x^2 + y^2 - 2y - 1 = 0$

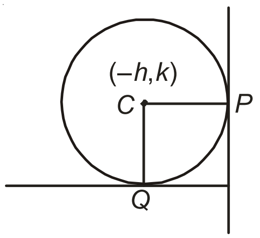
Sol. Answer (3)

Now $x^2 - y^2 + 2y - 1 = 0$

$\Rightarrow$ $x^2 - (y - 1)^2 = 0$

$\Rightarrow$ $(x - y + 1)(x + y - 1) = 0$

$\Rightarrow$ $x - y + 1 = 0,\ x + y - 1 = 0$



Now, $CP = CQ$

$$\Rightarrow \left|\frac{h+k-1}{\sqrt{2}}\right| = \left|\frac{h-k+1}{\sqrt{2}}\right|$$

$\Rightarrow$ $h + k - 1 = \pm (h - k + 1)$

$\Rightarrow$ $h = 0, k = -1$

$\therefore$ $C \equiv (0, -1)$

Now, radius = $CQ = \left|\frac{0-1-1}{\sqrt{2}}\right| = \sqrt{2}$

Equation of circle is

$x^2 + (y+1)^2 = (\sqrt{2})^2$

$x^2 + y^2 + 2y - 1 = 0$

30. The equations of common tangents to the circles $x^2 + y^2 = 1$ and $(x-1)^2 + (y-3)^2 = 4$ are

(1) $3x + 4y - 5 = 0,\ 4x - 3y + 5 = 0$
(2) $3x + 4y - 5 = 0,\ 4x - 3y - 5 = 0$
(3) $3x - 4y + 5 = 0,\ 4x + 3y - 5 = 0$
(4) $3x + 4y + 5 = 0,\ 4x + 3y + 5 = 0$

Sol. Answer (2)

External point = (–1, –3)

Equation of tangent

$y + 3 = m(x+1)$

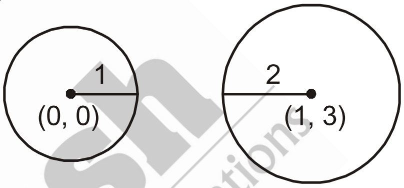
$mx - y + m - 3 = 0$

Distance from (0, 0) is 1

Hence m = not defined, $m = \frac{4}{3}$

Equation of tangent

$\boxed{x = -1}$ $\boxed{4x - 3y - 5 = 0}$



Internal division = $\left(\frac{1}{3}, 1\right)$

Equation of tangent

$y - 1 = m\left(x - \frac{1}{3}\right)$

Distance from (0, 0) is 1

Hence, $m = 0$, $m = -\frac{3}{4}$

Equation of tangent

$y = 1$ and $3x + 4y - 5 = 0$

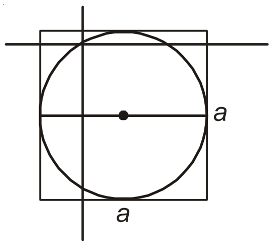
31. The area of a square circumscribing the circle $3(x^2 + y^2) - 6x + 8y = 0$ is

(1) $\frac{50}{9}$ sq. units
(2) $\frac{75}{9}$ sq. units
(3) $\frac{25}{9}$ sq. units
(4) $\frac{100}{9}$ sq. units

Sol. Answer (4)

$S : x^2 + y^2 - 2x + \frac{8}{3}y = 0$

$\Rightarrow\ (x-1)^2 + \left(y + \frac{4}{3}\right)^2 = \frac{16}{9} + 1 = \left(\frac{5}{3}\right)^2$



Let side of a square = a

Now diameter = a

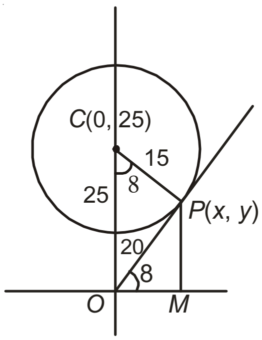
$\therefore\ a = 2.\frac{5}{3} = \frac{10}{3}$

$\Rightarrow\ a^2 = \frac{100}{9}$

32. Let a point P lie on the circle $x^2 + y^2 - 50y + 400 = 0$ such that $\angle POX$ is minimum. Then the co-ordinates of P are

(1) (12, 10) (2) (12, 16) (3) (12, 18) (4) (18, 12)

Sol. Answer (2)



Now, $S : x^2 + y^2 - 30y - 400 = 0$

$\Rightarrow \; x^2 + (y - 15)^2 = 625 = (25)^2$

Now, $OP = \sqrt{25^2 - 15^2}$

$= \sqrt{625 - 225} = \sqrt{400} = 25$

Let $\angle POX = \theta$

$\therefore \quad \cos\theta = \dfrac{x}{20} \quad \Rightarrow \quad \dfrac{15}{25} = \dfrac{x}{20}$

$\Rightarrow \quad x = 12$

$\sin\theta = \dfrac{y}{20} \quad \Rightarrow \quad \dfrac{20}{25} = \dfrac{y}{20}$

$\Rightarrow \quad y = 16$

$\therefore \quad P \equiv (x, y) \equiv (12, 16)$

33. If the circle $x^2 + y^2 + 4x + 2y + c = 0$ bisects the circumference of the circle $x^2 + y^2 - 2x - 8y - d = 0$ then $c + d =$

(1) 60 (2) –46 (3) 40 (4) 56

Sol. Answer (2)

Eq. of common chord $S_1 - S_2 = 0$

$6x + 10y + c + d = 0 \quad \ldots\text{(i)}$

Line (i) is a diameter of circle (ii)

$\therefore$ centre (1, 4) lie on the line (i)

$6 + 40 + c + d = 0$

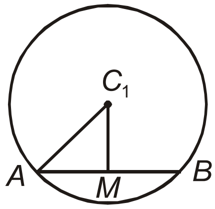
$c + d = -46$

[Analysis of Two Circles and Locus]

34. The length of the common chord of the circles $x^2 + y^2 + 2x + 3y + 1 = 0$ and $x^2 + y^2 + 4x + 3y + 2 = 0$ is

(1) $\dfrac{9}{2}$ (2) $\dfrac{3}{2}$ (3) $3\sqrt{2}$ (4) $2\sqrt{2}$

Sol. Answer (4)



Eq. of common chord $S_2 - S_1 = 0$

$2x + 1 = 0$

$C_1\left(-1, -\dfrac{3}{2}\right)$

$r_1 = \sqrt{1 + \dfrac{9}{4} - 1} = \dfrac{3}{2}$

$C_1M = \left(\dfrac{-2+1}{\sqrt{4}}\right) = \dfrac{1}{2}$

$$AM^2 = r^2 - C_1M^2 = \frac{9}{4} - \frac{1}{4} = \frac{8}{4} = 2$$

$AM = \sqrt{2}$

$AB = 2\sqrt{2}$

35. The distance between the chords of contact of the tangents to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ from the origin and from the point (g, f) is

(1) $g^2 + f^2 - c$ (2) $\sqrt{g^2 + f^2 - c}$ (3) $\frac{g^2 + f^2 - c}{\sqrt{g^2 + f^2}}$ (4) $\frac{1}{2}\frac{|g^2 + f^2 - c|}{\sqrt{g^2 + f^2}}$

Sol. Answer (4)

Equation of chord of contact from $P(g, f)$

$T \equiv 0$

$$x \cdot g + y \cdot f + g(x + g) + f(y + f) + c = 0$$

$$2gx + 2fy + g^2 + f^2 + c = 0 \quad \text{...(i)}$$

Similarly eq. of chord of contact from origin $O(0, 0)$ is

$$gx + fy + c = 0$$

$$\therefore \quad 2gx + 2fy + 2c = 0 \quad \text{...(ii)}$$

Distance between line (i) and (ii)

$$= \left|\frac{g^2 + f^2 - c}{\sqrt{4g^2 + 4f^2}}\right| = \frac{|g^2 + f^2 - c|}{2\sqrt{g^2 + f^2}}$$

36. Through a fixed point (h, k) secants are drawn to the circle $x^2 + y^2 = a^2$. The locus of the mid points of the secants intercepted by the given circle is

(1) $2(x^2 + y^2) = hx + ky$ (2) $x^2 + y^2 = hx + ky$

(3) $x^2 + y^2 + hx + ky = 0$ (4) $x^2 + y^2 - hx + ky + 13 = 0$

Sol. Answer (2)

Let mid-point of the chord be $P(x_1, y_1)$.

$\therefore$ Equation of chord $S_1 = T$

$$x_1^2 + y_1^2 - a^2 = xx_1 + yy_1 - a^2$$

This chord passes through (h, k).

$$x_1^2 + y_1^2 = hx_1 + ky_1$$

$\therefore$ Locus of $P(x_1, y_1)$ be $x_1^2 + y_1^2 = hx_1 + ky_1$

37. The equation of circle passing through the point (1, 1) and point of intersection of $x^2 + y^2 = 6$ and $x^2 + y^2 - 6x + 8 = 0$, is

(1) $x^2 + y^2 - 6x + 4 = 0$ (2) $x^2 + y^2 - 3x + 1 = 0$

(3) $x^2 + y^2 - 4y + 2 = 0$ (4) $x^2 + y^2 - 2x - 2y + 2 = 0$

Sol. Answer (2)

Required circle be $S_1 + \lambda S_2 = 0$

$$x^2 + y^2 - 6x + 8 + \lambda(x^2 + y^2 - 6) = 0$$

$$(1+\lambda)x^2 + (1+\lambda)y^2 - 6x + (8 - 6\lambda) = 0 \quad \ldots(i)$$

Circle (i) passes through (1, 1) $\therefore$ $\lambda = 1$

Circle be $x^2 + y^2 - 3x + 1 = 0$

38. If the circle $O : x^2 + y^2 = 16$ intersects another circle C of radius 5 units in such a way that the common chord is of maximum length and has a slope equal to $\frac{3}{4}$, then the coordinates of the centre of C are

(1) $\left(\frac{9}{5}, \frac{-12}{5}\right)$ (2) $\left(\frac{-9}{5}, \frac{-12}{5}\right)$ (3) $\left(\frac{9}{5}, \frac{12}{5}\right)$ (4) $\left(\frac{-9}{15}, \frac{12}{15}\right)$

Sol. Answer (1)

Length of common chord is maximum and therefore common chord is diameter of small circle.

$\therefore$ Eq. of common chord $y = \frac{3}{4}x$ ...(i)

Let centre of circle S_2 is $C(a, b)$.

Eq. of perpendicular line to (i) is $y = \frac{-4}{3}x$ centre (a, b) lie as this line.

$\therefore$ $b = -\frac{4}{3}a$ ...(ii)

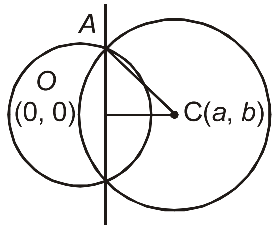
$AO = 4$

$AC = 5$

$\therefore$ $OC = 3$

$\sqrt{a^2 + b^2} = 3$

$a^2 + b^2 = 9$ ...(iii)



Solve eq. (ii) and (iii)

$a = \frac{9}{5}$

$b = -\frac{12}{5}$

Centre $\left(\frac{9}{5}, \frac{-12}{5}\right)$

39. A variable chord is drawn through origin to the circle $x^2 + y^2 - 2ax = 0$. Locus of the centre of the circle described on the chord as diameter is

(1) $x^2 + y^2 - ax = 0$ (2) $x^2 + y^2 + ax = 0$ (3) $x^2 + y^2 - ay = 0$ (4) $x^2 + y^2 \;\; ax - ay = 0$

Sol. Answer (1)

Let variable chord be $y = mx$

Equation of a circle passes through the intersection of a circle and a line

$$x^2 + y^2 - 2ax + \lambda(mx - y) = 0$$

$$x^2 + y^2 - (2a - \lambda m)x - \lambda y = 0$$

Centre of this circle $\left(\dfrac{2a - \lambda m}{2}, \dfrac{\lambda}{2}\right)$

Lie on the line $y = mx$

$$\therefore \quad \frac{\lambda}{2} = m\left(\frac{2a - \lambda m}{2}\right) \quad \ldots\text{(i)}$$

Let centre be (h, k)

$$\therefore \quad h = \frac{2a - \lambda m}{2} \quad \ldots\text{(ii)}$$

$$k = \frac{\lambda}{2} \quad \ldots\text{(iii)}$$

Eliminate λ and m from eq. (i), (ii) and (iii)

$$h^2 + k^2 - ah = 0$$

$\therefore$ Locus of (h, k)

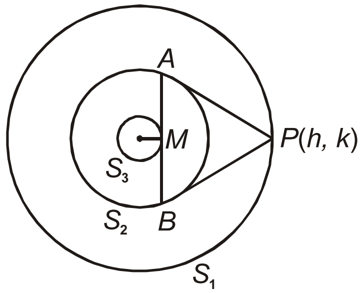
$$x^2 + y^2 - ax = 0$$

40. If the chord of contact of the tangents from a point on the circle $x^2 + y^2 = a^2$ to the circle $x^2 + y^2 = b^2$ touches the circle $x^2 + y^2 = c^2$, then a, b, c are in

(1) A.P (2) G.P (3) H.P (4) AGP

Sol. Answer (2)

Let $P(h, k)$ lie on S_1 $\therefore$ $h^2 + k^2 = a^2$ …(i)



Equation of chord of contact from P to S_2 is $x \cdot h + y \cdot k = b^2$ …(ii)

This line is a tangent to the circle S_3.

$\therefore$ Distance from centre = radius

$$\frac{(-b^2)}{\sqrt{h^2 + k^2}} = c$$

$$\frac{|\, b^2 \,|}{\sqrt{a^2}} = c \quad \Rightarrow \quad b^2 = ac$$

$\therefore$ a, b, c … in G.P.

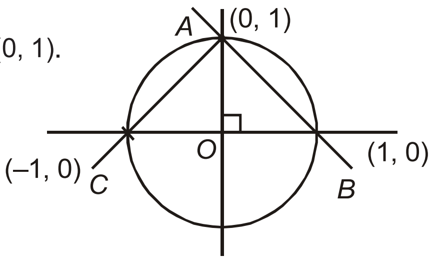
41. If the chord $y = mx + 1$ subtends an angle of measure of 45° at the major segment of the circle $x^2 + y^2 = 1$, then the value of m is

(1) $1 \pm \sqrt{2}$ (2) $-2 \pm \sqrt{2}$ (3) $-1 \pm \sqrt{2}$ (4) ± 1

Sol. Answer (4)

Chord subtends 90° at the centre. Chord must pass through (0, 1).

Second point may be (±1, 0).



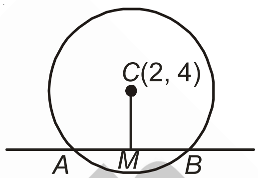
Chord cuts x-axis at $\left(-\dfrac{1}{m}, 0\right)$.

$\therefore \quad m = \pm 1$

42. The line $3x - 4y = k$ will cut the circle $x^2 + y^2 - 4x - 8y - 5 = 0$ at distinct points if

(1) $-35 < k < 35$ (2) $-35 < k < 15$ (3) $-15 < k < 15$ (4) $15 < k < 35$

Sol. Answer (2)



Given $x^2 + y^2 - 4x - 8y - 5 = 0$

$\Rightarrow \quad (x-2)^2 + (y-4)^2 = 4 + 16 + 5 = (5)^2$

$\therefore \quad$ Centre = $C(2, 4)$,

Radius = 5

The line AB will cut the circle in two distinct points if

Now, $CM < 5$

$$\Rightarrow \quad \left|\frac{6-16-k}{\sqrt{25}}\right| < 5$$

$$\Rightarrow \quad -5 < \frac{-10-k}{5} < 5$$

$\Rightarrow \quad -25 < -10 - k < 25$

$\Rightarrow \quad -15 < -k < 35$

$\Rightarrow \quad -35 < k < 15$

43. The equation of the circle, orthogonal to both the circles $x^2 + y^2 + 3x - 5y + 6 = 0$ and $4x^2 + 4y^2 - 28x + 29 = 0$ and whose centre lies on the line $3x + 4y + 1 = 0$ is

(1) $4x^2 + 4y^2 + 2y - 29 = 0$ (2) $4x^2 + 4y^2 + 6y + 5 = 0$

(3) $2x^2 + 2y^2 + 3x + 7y = 0$ (4) $x^2 + y^2 + 3x - 7y + 3 = 0$

Sol. Answer (1)

Let equation of circle be $S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$

Condition of orthogonality $2g_1g_2 + 2f_1f_2 = c_1 + c_2$

Circles S and S_1 are orthogonal

$$\therefore \quad 2g\left(\frac{3}{2}\right) + 2f\left(-\frac{5}{2}\right) = c + 6$$

$$3g - 5f - c = 6 \qquad \ldots(i)$$

Circles S and S_2 are orthogonals.

$\therefore \quad 2g\left(-\frac{7}{2}\right)+2f\,(0)=c+\frac{29}{4}$

$-7g-c=\frac{29}{4}$ ...(ii)

(i)-(ii) $10g-5f=-\frac{5}{4}$ ...(iii)

Centre of S $(-g, -f)$ lie on the line $3x + 4y + 1 = 0$

$\therefore \quad -3g-4f+1=0$ ...(iv)

Solve eq. (iii) and (iv)

$g = 0,\ f=\frac{1}{4},\ c=-\frac{29}{4}$

$\therefore \quad S \equiv x^2+y^2+\frac{1}{2}y-\frac{29}{4}=0$

$4x^2+4y^2+2y-29=0$

44. The equation of a circle which touches the line $x + y = 5$ at the point $(-2, 7)$ and cuts the circle $x^2 + y^2 + 4x - 6y + 9 = 0$ orthogonally is

(1) $x^2 + y^2 + 7x - 11y + 38 = 0$
(2) $x^2 + y^2 + 7x + 11y + 38 = 0$
(3) $x^2 + y^2 + 7x - 11y - 38 = 0$
(4) $x^2 + y^2 - 7x - 11y + 39 = 0$

Sol. Answer (1)

Let equation of circle $S \equiv x^2+y^2+2gx+2fy+c=0$

$(-2, 7)$ lie on the circle

$\therefore \quad -4g + 14f + c = -53$ ...(i)

Equation of a line through $(-2, 7)$ and perpendicular to line $x + y = 5$ is

$y-7=1(x+2)$

$x-y+9=0$

Centre $(-g, -f)$ lie on this line $-g+f+9=0$ ...(ii)

Circles S and S_1 are orthogonal

$\therefore \quad 2g(2)+2f(-3)=c+9$

$4g-6f-c=9$ ...(iii)

Solve eq. (i) and (iii), $f=\frac{-11}{2}$

$g=\frac{7}{2},\ c=38$

$\therefore$ Equation of circle $x^2+y^2+7x-11y+38=0$

45. The locus of the centres of the circles which cut the circles $x^2 + y^2 + 4x - 6y + 9 = 0$ and $x^2 + y^2 - 4x + 6y + 4 = 0$ orthogonally is

(1) $8x - 12y + 5 = 0$ (2) $8x + 12y - 5 = 0$ (3) $12x - 8y + 5 = 0$ (4) $3x + 4y + 7 = 0$

Sol. Answer (1)

Let circle $S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$ circle S intersect S_1 and S_2 orthogonally

$\therefore$ $2g(2) + 2f(-3) = c + 9$ ... (i)

$2g(-2) + 2f(3) = c + 4$ ... (ii)

(i) – (ii) $8g - 12f = 5$

Locus of $(-g, -f)$

$-8x + 12y = 5$

$8x - 12y + 5 = 0$

46. If centre of a circle lies on the line $2x - 6y + 9 = 0$ and it cuts the circle $x^2 + y^2 = 2$ orthogonally then the circle passes through two fixed points

(1) $\left(\frac{1}{2}, \frac{3}{2}\right), \left(\frac{-2}{5}, \frac{6}{5}\right)$ (2) $(2, 3), (-2, 6)$ (3) $\left(\frac{-1}{2}, \frac{3}{2}\right), \left(\frac{-2}{5}, \frac{6}{5}\right)$ (4) $(-2, 3)$ $(-2, 6)$

Sol. Answer (3)

Centre $(-g, -f)$ lie on $2x - 6y + 9 = 0$

$\therefore$ $-2g + 6f + 9 = 0$ ...(i)

cuts orthogonally $x^2 + y^2 = 2$

$\therefore$ $0 = c - 2$

$c = 2$

$\therefore$ Circle be $x^2 + y^2 + 2gx + 2fy + 2 = 0$

Eliminate $x^2 + y^2 + (6f + 9)x + 2fy + 2 = 0$

$(x^2 + y^2 + 9x + 2) + f(6x + 2y) = 0$

This circle passes through the point of intersection of

$x^2 + y^2 + 9x + 2 = 0$ ...(i)

and $6x + 2y = 0$ ...(ii)

Solve these equation intersection points are $\left(-\frac{1}{2}, \frac{3}{2}\right)$ and $\left(-\frac{2}{5}, \frac{6}{5}\right)$.

47. The equation of a circle which passes through $(2a, 0)$ and whose radical axis in relation to the circle $x^2 + y^2 = a^2$ is $x = \frac{a}{2}$, is

(1) $x^2 + y^2 - ax = 0$ (2) $x^2 + y^2 + 2ax = 0$

(3) $x^2 + y^2 - 2ax = 0$ (4) $x^2 + y^2 + ax = 0$

Sol. Answer (3)

The required circle is $s + \lambda L = 0$

$$(x^2 + y^2 \;\; a^2) + \lambda\left(x - \frac{a}{2}\right) = 0$$

This passes through $(2a, 0)$ $\Rightarrow$ $\lambda = -2a$

Hence the required circle is

$$(x^2 + y^2 \;\; a^2) - 2a\left(x - \frac{a}{2}\right) = 0$$

$\Rightarrow$ $x^2 + y^2 - 2ax = 0$

48. The radical centre of three circles described on the three sides $4x - 7y + 10 = 0$, $x + y - 5 = 0$ and $7x + + 4y - 15 = 0$ of a triangle as diameters is

(1) (2, 3) (2) (2, 1) (3) (3, 2) (4) (1, 2)

Sol. Answer (4)

Here, radical centre = orthocenter

$S_1 = 4x - 7y + 10 = 0$ ... (1)

$S_2 = x + y - 5 = 0$ ... (2)

$S_3 = 7x + 4y - 15 = 0$ ... (3)

Thus, the sides (1) and (3) are perpendicular to each other. So the point of intersection of these two lines will be orthocenter.

$$\therefore \begin{array}{r} 28x - 49y + 70 = 0 \\ 28x + 16y - 60 = 0 \\ - \quad - \qquad + \quad \\ \hline -65y + 130 = 0 \end{array}$$

$\Rightarrow$ $y = \frac{130}{65} = 2$

When $y = 2$, $4x - 7.2 + 10 = 0$

$\Rightarrow$ $4x = 14 - 10 = 4$

$\Rightarrow$ $x = 1$

$\therefore$ Point is (1, 2)

$\Rightarrow$ Orthocentre = (1, 2)

49. The equation of a circle passing through (1, 0) and (0, 1) having the smallest possible radius is

(1) $x^2 + y^2 + x - y = 0$ (2) $x^2 + y^2 - x - y = 0$ (3) $x^2 + y^2 + x + y = 0$ (4) $x^2 + y^2 - x + 2y = 0$

Sol. Answer (2)

Let $S : x^2 + y^2 + 2gx + 2fy + c = 0$ ... (1)

(1) passes through (1, 0) and (0, 1),

Then $1 + 2g + c = 0$

$\Rightarrow$ $g = -\frac{1+c}{2}$ g

Also, $1 + 2f + c = 0$

$\Rightarrow$ $f = -\frac{1+c}{2}$

Radius $= \sqrt{g^2 + f^2 - c}$

$$= \sqrt{\frac{(1+c)^2}{4} + \frac{(1+c)^2}{4} - c}$$

$$= \sqrt{\frac{c^2 + 2c + 1 + c^2 + 2c + 1 - 4c}{4}}$$

$$= \sqrt{\frac{2c^2 + 2}{4}}$$

$$= \sqrt{\frac{c^2 + 1}{2}}$$

$\Rightarrow$ For minimum radius $c = 0$

$\therefore \quad f = -\frac{1}{2}, g = -\frac{1}{2}$

Equation of a circle is $x^2 + y^2 - x - y = 0$

50. If two circles $(x - 1)^2 + (y - 3)^2 = r^2$ and $x^2 + y^2 - 8x + 2y + 8 = 0$ intersects in the distinct points, then

(1) $1 < r < 4$ (2) $-2 < r < 2$ (3) $2 < r < 8$ (4) None of these

Sol. Answer (3)

$S_1 : (x - 1)^2 + (y - 3)^2 = r^2$

$S_2 : x^2 + y^2 - 8x + 2y + 8 = 0$

$\Rightarrow (x - 4)^2 + (y + 1)^2 = 16 + 1 - 8 = 9 = 3^2$

$C_1 : (1, 3)$

$C_2 : (4, -1)$

$C_1C_2 = \sqrt{(1-4)^2 + (3+1)^2} = 5$

$|r - 3| < C_1C_2 < r + 3$

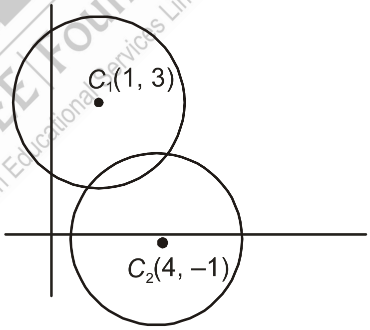
$\Rightarrow |r - 3| < 5 < r + 3$

$\Rightarrow |r - 3| < 5 \; r + 3 > 5$

$\Rightarrow -5 < r - 3 < 5$

$\Rightarrow r > 2$

$\Rightarrow 2 < r < 8$



51. Two rods of lengths a and b slide along the axes in such a way that their ends are concyclic. The locus of the centre of the circle passing through these points is

(1) $4(x^2 + y^2) = a^2 + b^2$ (2) $x^2 - y^2 = a^2 - b^2$ (3) $4(x^2 - y^2) = a^2 - b^2$ (4) $x^2 + y^2 = a^2 + b^2$

Sol. Answer (3)

Let A_1A_2 and B_1B_2 be two rods of lengths a and b which slide along OX and OY respectively.

Let $x^2 + y^2 + 2gx + 2fy + c = 0$ be the circle passing through A_1, A_2, B_1, B_2. Then

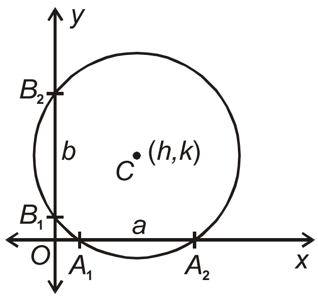
$A_1 A_2$ = Intercept of x-axis $= 2\sqrt{g^2 - c}$

$\Rightarrow a = 2\sqrt{g^2 - c}$ ... (i)

$B_1 B_2$ = Intercept of y-axis

$2\sqrt{f^2 - c}$

$\Rightarrow \quad b = 2\sqrt{f^2 - c}$ $\quad \dots$ (ii)



Squaring (i) and (ii), we get

$a^2 = 4(g^2 \;\; c)$

$b^2 = 4(f^2 \;\; c)$

Subtracting $a^2 - b^2 = 4(g^2 - f^2)$

Hence locus of centre is $a^2 - b^2 = 4(x^2 - y^2)$

52. The locus of the centre of a circle, which touches externally to the circle $x^2 + y^2 - 6x - 6y + 14 = 0$ and also touches the y-axis, is given by

(1) $x^2 - 6x - 10y + 14 = 0$ (2) $x^2 - 10x - 6y + 14 = 0$

(3) $y^2 - 6x - 10y + 14 = 0$ (4) $y^2 - 10x - 6y + 14 = 0$

Sol. Answer (4)

Let centre of circle S_1 be $C_1(a, b)$.

Circle touch y-axis. $\therefore r_1 = a$

Centre of S_2 is C_2 (3, 3) and $r_2 = 2$

S_1 and S_2 touch externally

$C_1C_2 = r_1 + r_2$

$$\sqrt{(a-3)^2 + (b-3)^2} = a + 2$$

$$\Rightarrow \quad b^2 - 10a - 6b + 14 = 0$$

Locus of $C_1(a, b)$ is $y^2 - 10x - 6y + 14 = 0$

53. The locus of centre of a circle of radius 2 which rolls outside of the circle $x^2 + y^2 + 3x - 6y - 9 = 0$ is

(1) $x^2 + y^2 + 3x - 6y + 5 = 0$ (2) $x^2 + y^2 + 3x - 6y + 31 = 0$

(3) $x^2 + y^2 + 3x - 6y + \frac{29}{4} = 0$ (4) $x^2 + y^2 + 3x - 6y - 31 = 0$

Sol. Answer (4)

Let centre $C_1(h, k)$, $r_1 = 2$

Centre of S_2 is $C_2\left(-\frac{3}{2}, 3\right)$, $r_2 = \sqrt{\frac{9}{4} + 9 + 9} = \sqrt{\frac{81}{4}} = \frac{9}{2}$

Distance between centres = sum of radii

$$\sqrt{\left(h + \frac{3}{2}\right)^3 + (k-3)^2} = 2 + \frac{9}{2}$$

Locus of (h, k)

$$x^2 + y^2 + 3x - 6y - 31 = 0$$

54. The locus of the point of intersection of the tangents to the circle $x = a\cos\theta$, $y = a\sin\theta$ at the points, whose parametric angles differ by $\frac{\pi}{3}$ is

(1) Straight line (2) Ellipse (3) Circle of radius $2a$ (4) Circle of radius $\frac{2a}{\sqrt{3}}$

Sol. Answer (4)

Equation of tangent at A ($a\cos\theta$, $a\sin\theta$) is $x\cos\theta + y\sin\theta = a$ ...(i)

Equation of tangent at $B\left[a\cos\left(\theta+\frac{\pi}{3}\right)\cdot a\sin\left(\theta+\frac{\pi}{3}\right)\right]$

$$x\cos\left(\theta+\frac{\pi}{3}\right)+y\sin\left(\theta+\frac{\pi}{3}\right)=a$$

$$x\left[\frac{1}{2}\cos\theta-\frac{\sqrt{3}}{2}\sin\theta\right]+y\left[\frac{1}{2}\sin\theta+\frac{\sqrt{3}}{2}\cos\theta\right]=a$$

$$x\cos\theta+y\sin\theta+\sqrt{3}\,(-x\sin\theta+y\cos\theta)=2a$$

$$a+\sqrt{3}\,(-x\sin\theta+y\cos\theta)=2a$$

$$-x\sin\theta+y\cos\theta=\frac{a}{\sqrt{3}} \quad \text{...(ii)}$$

Squaring equation (i) and (ii) and adding

$$x^2+y^2=a^2+\frac{a^2}{3}=\frac{4a^2}{3}$$

Locus is a circle of radius $\frac{2a}{\sqrt{3}}$.

55. A circle of constant radius $2r$ passes through the origin and meets the axes in 'P' and 'Q'. Locus of the centroid of the ΔPOQ is :

(1) $x^2 + y^2 = r^2$ (2) $9(x^2 + y^2) = 16r^2$ (3) $2(x^2 + y^2) = r^2$ (4) $3(x^2 + y^2) = 8r^2$

Sol. Answer (2)

Circle passes through origin.

$\therefore$ $c = 0$

Radius = $\sqrt{g^2+f^2}$

$g^2+f^2=4r^2$ ...(1)

Let circle intersect x axis at $P(a, 0)$ and y axis at $Q(0, b)$

$\therefore$ $-g=\frac{a}{2}$ and $-f=\frac{b}{2}$

From (1) $\frac{a^2}{4}+\frac{b^2}{4}=4r^2$

$a^2 + b^2 = 16r^2 \quad \ldots(2)$

Let centroid of $\Delta\, POQ$ be $C(h, k)$.

$\therefore \quad h = \frac{a}{3},\ k = \frac{b}{3} \qquad \therefore \quad 9h^2 + 9k^2 = 16r^2$

$\therefore$ Locus of centroid is $9(x^2 + y^2) = 16r^2$

[Miscellaneous]

56. Equation of a circle passing through (2, 8) and touching the lines $4x - 3y - 24 = 0$ and $4x + 3y - 42 = 0$ and having x-coordinates of the centre less than or equal to 8 is

(1) $(x-2)^2 + (y+3)^2 = 25$
(2) $(x-2)^2 + (y-3)^2 = 16$
(3) $(x-2)^2 + (y-3)^2 = 25$
(4) $(x+2)^2 + (y-3)^2 = 25$

Sol. Answer (3)

Let centre of circle be $C(a, b)$.

Circle touch both lines.

$$r = \left|\frac{4a - 3b - 24}{5}\right| = \left|\frac{4a + 3b - 42}{5}\right| = \sqrt{(a-2)^2 + (b-8)^2}$$

Solve these pair wise $b = 3$ and $a = 2$

$r = 5$

Equation of circle $(x-2)^2 + (y-3)^2 = 25$

57. A square is inscribed in the circle $x^2 + y^2 - 2x + 4y + 3 = 0$. Its sides are parallel to the co-ordinate axes, then one vertex of the square is

(1) $(1+\sqrt{2}, -2)$
(2) $(1-\sqrt{2}, -2)$
(3) $(1, -2+\sqrt{2})$
(4) $(2, -1)$

Sol. Answer (4)

Centre $(1, -2)$, $r = \sqrt{2}$

AB and CD are parallel to x-axis.

$\therefore$ Equation of $AB \rightarrow y = -3$

Equation of $CD \rightarrow y = -1$

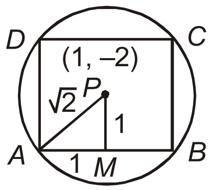
AD and BC are parallel to y-axis.

$\therefore$ Equation of $AD \rightarrow x = 0$

Equation of $BC \rightarrow x = 2$

$\therefore$ $A(0, -3)$, $B(2, -3)$

$C(2, -1)$, $D(0, -1)$



58. The area of the triangle formed by joining the origin to the points of intersection of $\sqrt{5}x + 2y = 3\sqrt{5}$ and $x^2 + y^2 = 10$ is

(1) 6
(2) 5
(3) 4
(4) 3

Sol. Answer (2)

Combined equation of pair of lines joining origin and point of intersection of circle and line $x^2 + y^2 = 10\left(\frac{\sqrt{5}x+2y}{3\sqrt{5}}\right)^2$

$\frac{1}{9}x^2 - \frac{1}{9}y^2 + \frac{8}{9}\sqrt{5}xy = 0$

$a + b = \frac{1}{9} - \frac{1}{9} = 0$

$\therefore$ Δ is right angled triangle at origin (0, 0)

$\therefore$ Area = $\frac{1}{2}r^2 = \frac{1}{2}\left(\sqrt{10}\right)^2 = 5$

59. From the origin O tangents OP and OQ are drawn to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$. Then the circumcentre of the triangle OPQ lies at

(1) $\left(\frac{-g}{2}, \frac{-f}{2}\right)$ (2) (g, f) (3) $(-f, -g)$ (4) (f, g)

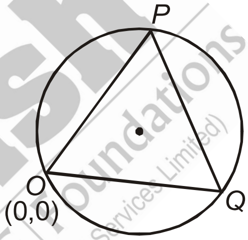
Sol. Answer (1)

Origin (0, 0) lies on the given circle

Circumcircle of ΔOPQ = Given circle

$= x^2 + y^2 + 2gx + 2fy = 0$

Centre $(-g, -f)$



60. The equation of the locus of the mid-points of chords of the circle $4x^2 + 4y^2 - 12x + 4y + 1 = 0$ that subtend an angle of $\frac{2\pi}{3}$ at its centre, is

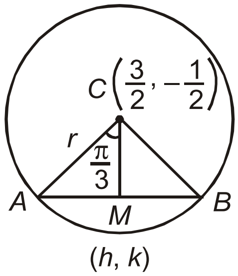
(1) $16x^2 + 16y^2 - 48x + 16y + 31 = 0$
(2) $16x^2 + 16y^2 + 48x + 48y + 31 = 0$
(3) $16x^2 - 16y^2 + 48x + 48y + 31 = 0$
(4) $16x^2 + 16y^2 - 48x - 16y - 31 = 0$

Sol. Answer (1)

$r = \sqrt{\frac{9}{4} + \frac{1}{4} - \frac{1}{4}} = \frac{3}{2}$

$CM = r\cos\frac{\pi}{3} = \frac{r}{2}$

$\sqrt{\left(h - \frac{3}{2}\right)^2 + \left(k + \frac{1}{2}\right)^2} = \frac{3}{4}$



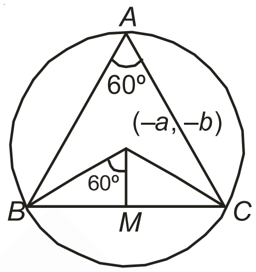
Locus of $M(h, k)$

$16x^2 + 16y^2 - 48x + 16y + 31 = 0$

61. The area of an equilateral triangle inscribed in the circle $x^2 + y^2 + 2ax + 2by + c = 0$ is $\frac{m}{n}(a^2 + b^2 - c)$ then the value of $\frac{1}{\sqrt{3}}m+n$ is

(1) 5 (2) 7 (3) 6 (4) 8

Sol. Answer (2)



$$\therefore \quad \sin 60^\circ = \frac{BM}{OB} = \frac{BM}{\sqrt{a^2+b^2-c}}$$

$$\Rightarrow \quad \frac{\sqrt{3}}{2} = \frac{BM}{\sqrt{a^2+b^2-c}}$$

$$\Rightarrow \quad BM = \frac{\sqrt{3}}{2} \times \sqrt{a^2+b^2-c}$$

$$\Rightarrow \quad BC = 2BM = \sqrt{3} \times \sqrt{a^2+b^2-c}$$

$$\therefore \quad \text{Area of } \Delta ABC = \frac{\sqrt{3}}{4} \times (BC)^2$$

$$= \frac{\sqrt{3}}{4} \times 3(a^2+b^2-c)$$

$$= \frac{3\sqrt{3}}{4} \times (a^2+b^2-c)$$

$$\Rightarrow \quad \frac{m}{n}(a^2+b^2-c) = \frac{3\sqrt{3}}{4} \times (a^2+b^2-c)$$

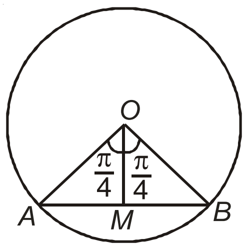
$$\Rightarrow \quad \frac{m}{n} = \frac{3\sqrt{3}}{4}$$

$$\therefore \quad \frac{1}{\sqrt{3}}m+n = \frac{1}{3} \times 3\sqrt{3} + 4 = 3 + 4 = 7$$

62. Area of a circle in which a chord of length $\sqrt{2}$ makes an angle $\frac{\pi}{2}$ at the centre is

(1) $\frac{\pi}{4}$ (2) $\frac{\pi}{2}$ (3) π (4) 2π

Sol. Answer (3)



$$\therefore \quad \sin\frac{\pi}{4} = \frac{AM}{OA} = \frac{\frac{1}{2}\times\sqrt{2}}{OA} = \frac{\sqrt{2}}{2OA}$$

$$\Rightarrow \quad \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2.OA}$$

$\Rightarrow$ OA = 1

$\Rightarrow$ Radius = 1

Area of a circle = $\pi \times (1)^2 = \pi$ sq. units

63. If $3x + b_1y + 5 = 0$ and $4x + b_2y + 10 = 0$ cut the x-axis and y-axis in four concyclic points, then the value of b_1b_2 is

(1) 15 (2) 30 (3) 20 (4) 12

Sol. Answer (4)

As we know that, if the lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ cut x-axis and y-axis in four con-cyclic points, then $a_1a_2 = b_1b_2$.

$\therefore\ b_1b_2 = 3 \times 4 = 12$

64. If $(1 + bx)^n = 1 + 8x + 24x^2 + ...$ and a line $P(b, n)$ cuts the circle $x^2 + y^2 = 9$ in C and D, then $PC.\ PD = (\lambda^2 + 2)$, then λ is

(1) ± 2 (2) ± 4 (3) ± 11 (4) ± 3

Sol. Answer (4)

We have,

$(1 + bx)^n = 1 + 8x + 24x^2 + \ldots\ldots$

$\Rightarrow\ 1+nbx+\dfrac{n(n-1)}{2}\times b^2x^2+\ldots\ldots$

$= 1 + 8x + 24x^2 + \ldots\ldots$

$\Rightarrow\ nb = 8,\ \dfrac{n(n-1)}{2}\times b^2 = 24$

$\dfrac{n(n-1)}{2}\times b^2 = 24$

$\Rightarrow\ n^2b^2 - nb.b = 48$

$\Rightarrow\ (nb)^2 - (nb)b = 48$

$\Rightarrow\ 8^2 - 8b = 48$

$\Rightarrow\ 8b = 64 - 48 = 16$

$\Rightarrow\ b = 2$

$\therefore$ Now, $nb = 8$

$\Rightarrow\ n = 4$

$\therefore\ P \equiv (2, 4)$

Now, $PC.PD = PT^2$

$= \sqrt{4+16-9}$

$= \sqrt{11}$

$\Rightarrow\ \sqrt{\lambda^2+2} = \sqrt{11}$

$\Rightarrow\ \lambda^2 = 9$

$\lambda = \pm 3$

65. If the circle $x^2 + y^2 - 4x - 8y + 16 = 0$ rolls up the tangent to it at $(2+\sqrt{3}, 3)$ by 2 units (assumes x-axis as horizontal), then the centre of the circle in the new position is

(1) (3, 4) (2) $(3\sqrt{3}, 4+\sqrt{3})$ (3) $(3, 4+\sqrt{3})$ (4) $(3+\sqrt{3}, 4+\sqrt{3})$

Sol. Answer (3)

$S : x^2 + y^2 - 4x - 8y + 16 = 0$

$\Rightarrow\ (x - 2)^2 + (y - 4)^2 = 2^2$

$\therefore \quad P \equiv (2+\sqrt{3}, 3)$

Equation of tangent to the circle at P is

$$(2+\sqrt{3})x + 3y - 2\,(x + 2 + \sqrt{3}) - 4(y+3) + 16 = 0$$

$\Rightarrow \quad \sqrt{3}x - y - 2\sqrt{3} = 0$

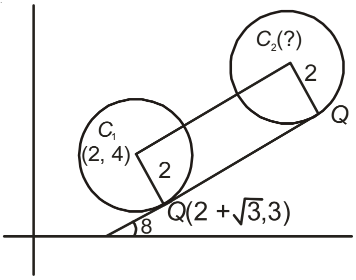
$\therefore \quad m = \sqrt{3}$

$\Rightarrow \quad \theta = 60^{\circ}$

$\therefore \quad B \equiv (2 + 2\cos 60^{\circ},\ 4 + 2\sin 60^{\circ})$

$\quad = (2+1,\ 4+\sqrt{3})$

$\quad = (3,\ 4+\sqrt{3})$



66. The equation of the diameter of the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ which corresponds to the chord $ax + by + d = 0$ is $\lambda x - ay + \mu g + k = 0$ then $\lambda + \mu$ is

(1) $2a$ (2) $2b$ (3) $2c$ (4) $2d$

Sol. Answer (2)

$\therefore \quad AB : ax + by + d = 0$

Diameter = $CD : bx - ay + m = 0$

Which is passes through $(-g, -f)$

$\therefore \quad -bg + af + m = 0$

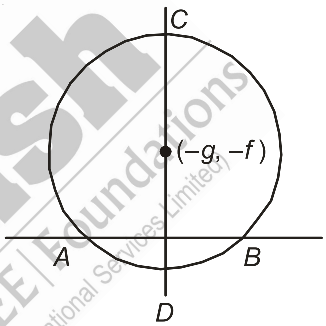
$\Rightarrow \quad m = bg - af$

Equation of diameter is

$bx - ay + bg - af = 0$

$\Rightarrow \quad \lambda x - ay + \mu g + k = 0$

$\therefore \quad \lambda + \mu = b + b = 2b$



67. A line meets the co-ordinate axes at $A(a, 0)$ and $B(0, b)$. A circle is circumscribed about the triangle OAB. If the distance of the points A and B from the tangent at origin to the circle are 3 and 4 respectively, then the value of $a^2 + b^2 + 1$ is

(1) 20 (2) 30 (3) 40 (4) 50

Sol. Answer (4)

$\therefore \quad \angle AOB = \dfrac{\pi}{2}$

$\Rightarrow \quad AB$ is the diameter

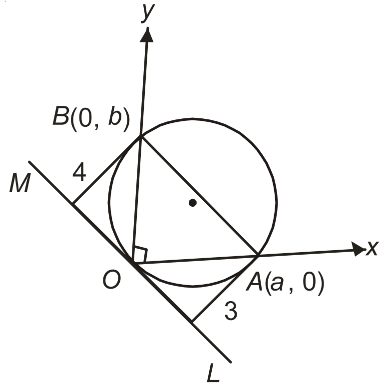
Equation of circle is $(x - a)\,(x - 0) + (y - 0)\,(y - b) = 0$

$\Rightarrow \quad x^2 + y^2 - ax - by = 0$

Tangent at O,

$$x - O + y.O - \frac{a}{2}(x+O) - \frac{b}{2}(y+O) = 0$$

$\Rightarrow \quad ax + by = 0$



$\therefore \quad AL = \dfrac{a.a+0}{\sqrt{a^2+b^2}} = 3$

$\Rightarrow \quad 3\sqrt{a^2+b^2} = a^2$

$\Rightarrow \quad 3 = \dfrac{a^2}{\sqrt{a^2+b^2}}$

Similarly $4 = \dfrac{b^2}{\sqrt{a^2+b^2}}$

$\Rightarrow \quad 3+4 = \dfrac{a^2+b^2}{\sqrt{a^2+b^2}} = \sqrt{a^2+b^2}$

$\Rightarrow \quad a^2 + b^2 = 7^2 = 49$

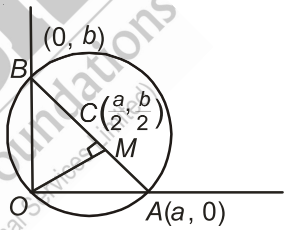
$\Rightarrow \quad a^2 + b^2 + 1 = 50$

68. A circle of constant radius r passes through the origin O, and cuts the axes at A and B. The locus of the foot of the perpendicular from O to AB is $(x^2 + y^2)^k = 4r^2x^2y^2$, Then the value of k is

(1) 2 (2) 1 (3) 3 (4) 4

Sol. Answer (3)

Let $A \equiv (a, 0)$ and $B \equiv (0, b)$ respectively



$\therefore \quad AB = \dfrac{x}{a} + \dfrac{y}{b} = 1$ … (1)

Since AB is a diameter of a circle, center of a circle lie on AB

$\therefore \quad C \equiv \left(\dfrac{a}{2}, \dfrac{b}{2}\right)$

Radius = $\sqrt{\dfrac{a^2}{4} + \dfrac{b^2}{4}} = \dfrac{\sqrt{a^2+b^2}}{2}$

$\Rightarrow \quad a^2 + b^2 = 4r^2$ …. (*)

Now, $OM \perp AB$

$\Rightarrow \quad OM = ax - by = 0$ … (2)

Now solving (1) and (2), we get

$a = \dfrac{x^2+y^2}{x}, \; b = \dfrac{y^2+x^2}{y}$

Put the values of a and b in (*), we get,

$\dfrac{(x^2+y^2)^2}{x^2} + \dfrac{(x^2+y^2)^2}{y^2} = 4r^2$

$\Rightarrow \quad (x^2+y^2)^2\left(\dfrac{1}{x^2} + \dfrac{1}{y^2}\right) = 4r^2$

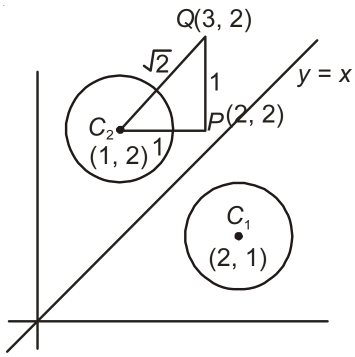
$\Rightarrow \quad (x^2 + y^2)^3 = 4r^2x^2y^2$

$\Rightarrow \quad k = 3$

69. The image of the centre of a circle $(x-2)^2 + (y-1)^2 = 9$ w.r.t. the line $y = x$ moves $\sqrt{2}$ units along (N - E) direction then the co-ordinates of the centre in the new position is

(1) (1, 2) (2) (2, 2) (3) (3, 2) (4) (2, 3)

Sol. Answer (3)



The image of (2, 1) w.r.t. the line $y = x$ is (1, 2),

Now, $P \equiv (2, 2)$

and $Q \equiv (3, 2)$

70. Four circles are inscribed in a square of side 10 cm in such a way that the minimum radius is 2.5 cm. The radius of the smallest circle which touches all those four circles externally is

(1) $\frac{5}{2}(\sqrt{3}-1)$ (2) $\frac{3}{2}(\sqrt{2}-1)$ (3) $\frac{5}{2}(\sqrt{2}-1)$ (4) $\frac{5}{2}(\sqrt{5}-1)$

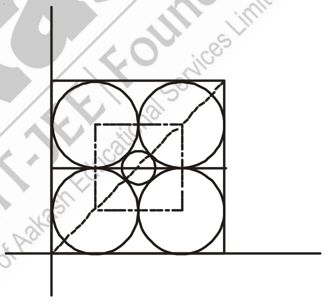
Sol. Answer (3)

Centre of the smallest circle = (5, 5)

Here $QT = \frac{5}{2} = 2.5$

$PT = \frac{5}{2}$

$\therefore \quad PQ = \sqrt{\frac{50}{4}} = \sqrt{\frac{25}{2}} = \frac{5}{\sqrt{2}} = \frac{5}{2} \times \sqrt{2}$



71. Four distinct points $(a, 0)$, $(0, b)$, $(c, 0)$ and $(0, d)$ are lie on a plane in such a way that $ac = bd$, then they will

(1) Form a trapezium (2) Form a triangle

(3) Lie on a circle (4) Form a quadrilateral, whose area is zero

Sol. Answer (3)

As we know that, $a_1x + b_1y + c_1 = 0$

and $a_2x + b_2y + c_2 = 0$ cuts the axes in four cyclic points, then $a_1a_2 = b_1b_2$

Here, $L_1 : \frac{x}{a} + \frac{y}{b} = 1$

$L_2 : \frac{x}{c} + \frac{y}{d} = 1$

$\Rightarrow \quad \frac{1}{a} \times \frac{1}{c} = \frac{1}{b} \times \frac{1}{d}$

$\Rightarrow \quad ac = bd$

Four distinct points are lie on a circle

72. A circle of radius 2 units touches the co-ordinate axes in first quadrant. If the circle moves one complete roll on x-axis along the positive direction of x-axis and then centre is rotated about the origin at $\frac{\pi}{3}$ angle in anti-clockwise direction. Then the co-ordinates of the centre in the new position is

(1) $(2\pi+1-\sqrt{3},\ 1+(1+2\pi)\sqrt{3})$ (2) $(2\pi-1-\sqrt{3},\ (1+2\pi)\sqrt{3})$

(3) $((1+2\pi)\sqrt{3},\ (1+2\pi)\sqrt{3})$ (4) $((1-2\pi)\sqrt{3},\ (1-2\pi)\sqrt{3})$

Sol. Answer (1)

Centre of a circle is (2, 2)

Circumference of the circle = $2\pi r$

$= 2\pi \times 2$

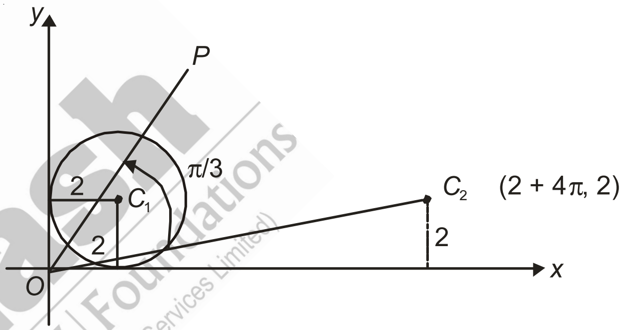
$= 4\pi$

Centre of the circle in the new position is $(2 + 4\pi, 2)$

Let $z_1 = (2 + 4\pi, 2)$

$P \equiv (z)$

Applying rotation, we have



$$\frac{z-0}{z_1-0} = e^{i\frac{\pi}{3}}$$

$$\Rightarrow \quad z = z_1 \times \left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$$

$$= \frac{1}{2}((2+4\pi)+2i)\ (1+i\sqrt{3})$$

$$= ((1+2\pi)+i)\ (1+i\sqrt{3})$$

$$= \left((1+2\pi)+i+i(1+2\pi)\sqrt{3}-\sqrt{3}\right)$$

$$= \left((1-\sqrt{3}+2\pi)+i(1+(1+2\pi)\sqrt{3}\right)$$

$$= \left(2\pi+1-\sqrt{3},\ 1+(1+2\pi)\sqrt{3}\right)$$

73. Tangents drawn from the point $P(1, 8)$ to the circle

$x^2 + y^2 - 6x - 4y - 11 = 0$

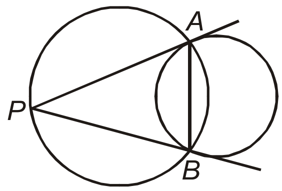
touch the circle at the point A and B. The equation of the circumcircle of the triangle PAB is **[IIT-JEE 2009]**

(1) $x^2 + y^2 + 4x - 6y + 19 = 0$ (2) $x^2 + y^2 - 4x - 10y + 19 = 0$

(3) $x^2 + y^2 - 2x + 6y - 29 = 0$ (4) $x^2 + y^2 - 6x - 4y + 19 = 0$

Sol. Answer (2)

The equation of AB is given by



$$x(1) + y(8) - 6\left(\frac{x+1}{2}\right) - 4\left(\frac{y+8}{2}\right) - 11 = 0$$

$\Rightarrow \quad x + 8y - 3(x + 1) - 2(y + 8) - 11 = 0$

$\Rightarrow$ $-2x + 6y - 3 - 16 - 11 = 0$

$\Rightarrow$ $-2x + 6y - 30 = 0$

$\Rightarrow$ $2x - 6y + 30 = 0$

$\Rightarrow$ $x - 3y + 15 = 0$

The equation of circum circle of ΔPAB may be written as

$(x^2 + y^2 - 6x - 4y - 11) + \lambda(x - 3y + 15) = 0$ ...(i)

The circle passes through $P(1, 8)$, hence

$\Rightarrow$ $(1 + 64 - 6 - 32 - 11) + \lambda(1 - 24 + 15) = 0$

$\Rightarrow$ $(65 - 49) + \lambda(-23 + 15) = 0$

$16 + \lambda(-8) = 0$

$\lambda = 2$

Consequently the equation of the circle is

$x^2 + y^2 - 6x - 4y - 11 + 2(x - 3y + 15) = 0$

$x^2 + y^2 - 4x - 10y + 19 = 0$

74. The circle passing through the point (–1, 0) and touching the y-axis at (0, 2) also passes through the point **[IIT-JEE 2011]**

(1) $\left(-\frac{3}{2}, 0\right)$ (2) $\left(-\frac{5}{2}, 2\right)$ (3) $\left(-\frac{3}{2}, \frac{5}{2}\right)$ (4) $(-4, 0)$

Sol. Answer (4)

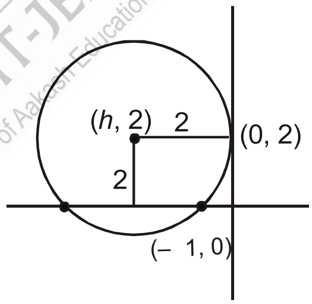
The equation of the circle directing y-axis at (0, 2) can be put in the form

$(x - h)^2 + (y - 2)^2 = h^2$

which will pass through (– 1, 0) if

$(-1 - h)^2 + 4 = h^2$

$\Rightarrow$ $h = -\frac{5}{2}$



Thus the equation of the circle is

$$\left(x + \frac{5}{2}\right)^2 + (y - 2)^2 = \left(\frac{5}{2}\right)^2$$

which passes through (– 4, 0)

75. The locus of the mid-point of the chord of contact of tangents drawn from points lying on the straight line $4x - 5y = 20$ to the circle $x^2 + y^2 = 9$ is **[IIT-JEE 2012]**

(1) $20(x^2 + y^2) - 36x + 45y = 0$ (2) $20(x^2 + y^2) + 36x - 45y = 0$

(3) $36(x^2 + y^2) - 20x + 45y = 0$ (4) $36(x^2 + y^2) + 20x - 45y = 0$

Sol. Answer (1)

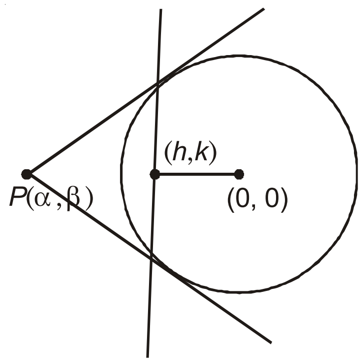
Let, $P(\alpha, \beta)$ be a point on the line $4x - 5y = 20$.

i.e., $4\alpha - 5\beta = 20$ ... (1)

If (h, k) be the mid point of chord of contact, then $T = S_1$, gives

$\Rightarrow \quad xh + yk - 9 = h^2 + k^2 - 9$

$\Rightarrow \quad xh + yk = h^2 + k^2.$ ... (2)



Also, equation of chord of contact w.r.t. $P(\alpha, \beta)$ is

$\alpha x + \beta y = 9$ ... (3)

as equations (2) & (3) are identical,

So, $\dfrac{\alpha}{h} = \dfrac{\beta}{k} = \dfrac{9}{h^2 + k^2}$

$\Rightarrow \quad \alpha = \dfrac{9h}{h^2 + k^2}, \beta = \dfrac{9k}{h^2 + k^2}$

Using equation (1), we get

$$4\left(\frac{9h}{h^2 + k^2}\right) - 5\left(\frac{9k}{h^2 + k^2}\right) = 20$$

i.e., locus of (h, k) is given by

$20(x^2 + y^2) - 36x + 45y = 0$

76. Let complex numbers α and $\dfrac{1}{\alpha}$ lie on circles $(x - x_0)^2 + (y - y_0)^2 = r^2$ and $(x - x_0)^2 + (y - y_0)^2 = 4r^2$, respectively. If $z_0 = x_0 + iy_0$ satisfies the equation $2|z_0|^2 = r^2 + 2$, then $|\alpha| =$ **[JEE(Advanced) 2013]**

(1) $\dfrac{1}{\sqrt{2}}$ (2) $\dfrac{1}{2}$ (3) $\dfrac{1}{\sqrt{7}}$ (4) $\dfrac{1}{3}$

Sol. Answer (3)

α, lies on $(x - x_0)^2 + (y - y_0)^2 = r^2$ ... (1)

$\dfrac{1}{\alpha} = \dfrac{\alpha}{|\alpha|^2}$ lies on $(x - x_0)^2 + (y - y_0)^2 = 4r^2$ ... (2)

Let α be $\alpha_1 + i\beta_1$

α lies on $(x^2 + y^2) + (x_0^2 + y_0^2) - 2(xx_0 + yy_0) = r^2$

$$|\alpha|^2 + |z_0|^2 - 2(\alpha_1 x_0 + \beta_1 y_0) = r^2 \quad \text{... (3)}$$

From eq. (2),

$$\frac{1}{|\alpha|^2} + |z_0|^2 - 2\frac{(\alpha_1 x_0 + \beta_1 y_0)}{|\alpha|^2} = 4r^2$$

$\Rightarrow \quad 1 + |\alpha|^2 |z_0|^2 - 2(\alpha_1 x_0 + \beta_1 y_0) = 4r^2 |\alpha|^2$ ... (4)

From (3) – (4),

$$|\alpha|^2 - 1 + |z_0|^2(1 - |\alpha|^2) = r^2\left(1 - 4|\alpha|^2\right)$$

$$(|\alpha|^2 - 1)(1 - |z_0|^2) = r^2\left(1 - 4|\alpha|^2\right) \qquad \text{...(5)}$$

Now, $r^2 = 2\left(|z_0|^2 - 1\right)$

$$\frac{r^2}{|z_0|^2 - 1} = 2$$

Substituting in (5),

$$|\alpha|^2 - 1 = -2(1 - 4|\alpha|^2)$$

$$\Rightarrow \ |\alpha|^2 - 1 = -2 + 8|\alpha|^2$$

$$\Rightarrow \ 7|\alpha|^2 = 1$$

$$\Rightarrow \ |\alpha| = \frac{1}{\sqrt{7}}$$

SECTION - B

Objective Type Questions (More than one options are correct)

1. The equation $x^2 + y^2 + 4x - 6y + 13 = 0$ represents

(1) A circle (2) A point

(3) A pair of two distinct straight lines (4) A pair of coincident straight lines

Sol. Answer (1, 2)

$r = \sqrt{4 + 9 - 13} = 0$

Equation represents a point circle

$x^2 + y^2 + 4x - 6y + 13 = 0$

$(x+2)^2 + (y-3)^2 = 0$

$\Rightarrow \ x = -2, y = 3$

2. If α is the angle subtended at $P(x_1, y_1)$ by the circle $S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$, then

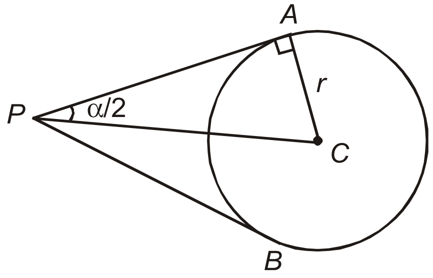
(1) $\cot\alpha = \dfrac{\sqrt{S_1}}{\sqrt{g^2 + f^2 - c}}$ (2) $\cot\dfrac{\alpha}{2} = \dfrac{\sqrt{S_1}}{\sqrt{g^2 + f^2 - c}}$

(3) $\tan\alpha = \dfrac{2\sqrt{g^2 + f^2 - c}}{\sqrt{S_1}}$ (4) $\tan\dfrac{\alpha}{2} = \left(\dfrac{\sqrt{g^2 + f^2 - c}}{\sqrt{S_1}}\right)$

Sol. Answer (2, 4)

$AP = \sqrt{S_1}$

$r = \sqrt{g^2 + f^2 - c}$



$$\cot\frac{\alpha}{2} = \frac{AP}{r} = \frac{\sqrt{S_1}}{\sqrt{g^2+f^2-c}}$$

$$\tan\frac{\alpha}{2} = \frac{\sqrt{g^2+f^2-c}}{\sqrt{S_1}}$$

3. The intersection points of $4x - 3y - 10 = 0$ and $x^2 + y^2 - 2x + 4y = 20$ must be

(1) (4, 2) (2) (–2, –6) (3) $\left(0, \frac{5}{2}\right)$ (4) $\left(\frac{5}{2}, -\frac{5}{2}\right)$

Sol. Answer (1, 2)

$$y = \frac{4x-10}{3} \quad \ldots(i)$$

Put in equation of circle

$$x^2 + \left(\frac{4x-10}{3}\right)^2 - 2x + 4\left(\frac{4x-10}{3}\right) = 20$$

$9x^2 + 16x^2 - 80x + 100 - 18x + 48x - 120 = 180$

$25x^2 - 50x - 200 = 0$

$x^2 - 2x - 8 = 0$

$(x-4)(x+2) = 0$

$x = -2$ and 4

$y = -6$ and 2

$(-2, -6), (4, 2)$

4. A square is inscribed in the circle $x^2 + y^2 - 2x + 4y - 93 = 0$ with its sides parallel to the axes of coordinates. The coordinates of the vertices are

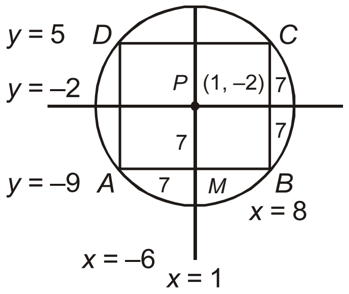
(1) (–6, –9) (2) (–6, 5) (3) (8, –9) (4) (8, 5)

Sol. Answer (1, 2, 3, 4)

Centre P (1, –2)

$r = \sqrt{1+4+93} = 7\cdot\sqrt{2}$

Side of square = $r\sqrt{2} = 14$



A (–6, –9)

B (8, –9)

C (8, 5)

D (–6, 5)

5. An equation of a line passing through the point (–2, 11) and touching the circle $x^2 + y^2 = 25$ is

(1) $4x + 3y = 25$ (2) $3x + 4y = 38$ (3) $24x - 7y + 125 = 0$ (4) $7x + 24y - 230 = 0$

Sol. Answer (1, 3)

Equation of line $y - 11 = m(x + 2)$

$mx - y + 2m + 11 = 0$

Distance of line from centre = radius

$\frac{2m+11}{\sqrt{m^2+1}} = 5 \quad \therefore \quad m = \frac{24}{7}, \frac{-4}{3}$

Tangents are $4x + 3y = 25$ and $24x - 7y + 125 = 0$

6. The equation of the tangent to the circle $x^2 + y^2 + 4x - 4y + 4 = 0$ which makes equal intercepts on the coordinate axes is given by

(1) $x - y = 2\sqrt{2}$ (2) $x + y = 2\sqrt{2}$ (3) $x - y + 2\sqrt{2} = 0$ (4) $x + y + 2\sqrt{2} = 0$

Sol. Answer (2, 4)

Equation of a line $x + y = a$ …(i)

Distance of line O from centre = radius

Centre (–2, 2), radius = $\sqrt{4+4-4} = 2$

$\left|\frac{-2+2+a}{\sqrt{2}}\right| = (2)$

$|a| = 2\sqrt{2}$

Equation of tangent $x + y = \pm 2\sqrt{2}$

7. If the line $y = x + c$ intersects the circle $x^2 + y^2 = r^2$ in two distinct points, then

(1) $-r\sqrt{2} < c \le 0$ (2) $0 \le c < r\sqrt{2}$ (3) $-c\sqrt{2} < r$ (4) $r < c\sqrt{2}$

Sol. Answer (1, 2)

Distance from centre < radius.

$\left|\frac{c}{\sqrt{2}}\right| < r$

$|c| < r\sqrt{2} \quad \Rightarrow \quad 0 \le c < r\sqrt{2}$

or $\quad -r\sqrt{2} < c \le 0$

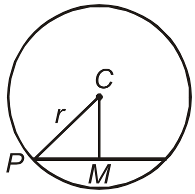
8. If a chord of the circle $x^2 + y^2 - 4x - 2y - k = 0$ is trisected at the points $\left(\frac{1}{3}, \frac{1}{3}\right)$ and $\left(\frac{8}{3}, \frac{8}{3}\right)$ then

(1) Length of the chord = $7\sqrt{2}$ (2) $k = 20$

(3) Radius of the circle = 5 (4) $k = 25$

Sol. Answer (1, 2, 3)

$A\left(\frac{1}{3}, \frac{1}{3}\right), B\left(\frac{8}{3}, \frac{8}{3}\right)$



$AB = \sqrt{\left(\frac{7}{3}\right)^2 + \left(\frac{7}{3}\right)^2} = \frac{7}{3} \cdot \sqrt{2}$

Length of chord = $3 \cdot AB = 7 \cdot \sqrt{2}$

Equation of chord $\frac{y - \frac{1}{3}}{\frac{8}{3} - \frac{1}{3}} = \frac{x - \frac{1}{3}}{\frac{8}{3} - \frac{1}{3}}$

$x - y = 0$

Distance of chord from centre (2, 1)

$CM = \frac{2-1}{\sqrt{2}} = \frac{1}{\sqrt{2}}$

Radius of circle $r = \sqrt{5+k}$

$r^2 = PM^2 + CM^2$

$5 + k = \left(\frac{7}{\sqrt{2}}\right)^2 + \frac{1}{2} = 25$

$k = 20$

Radius of circle, $r = \sqrt{5+20} = 5$

9. Equation of the circle with centre (4, 3) and touching the circle $x^2 + y^2 = 1$ is

(1) $x^2 + y^2 - 8x - 6y + 9 = 0$
(2) $x^2 + y^2 + 8x + 6y - 11 = 0$
(3) $x^2 + y^2 - 8x - 6y - 11 = 0$
(4) $x^2 + y^2 + 8x + 6y - 9 = 0$

Sol. Answer (1, 3)

$C_1C_2 = r_1 \pm r_2$

$\sqrt{16+9} = 1 \pm r$

$r = 4, 6$

Equation of circle $(x-4)^2 + (y-3)^2 = 16$ or $(x-4)^2 + (y-3)^2 = 36$

$\Rightarrow \quad x^2 + y^2 - 8x - 6y + 9 = 0$ or $x^2 + y^2 - 8x - 6y - 11 = 0$

10. An equation of a circle through the origin, making an intercept of $\sqrt{10}$ on the line $y = 2x + \frac{5}{\sqrt{2}}$, which subtends an angle of 45° at the origin is

(1) $x^2 + y^2 - 4x - 2y = 0$
(2) $x^2 + y^2 - 2x - 4y = 0$
(3) $x^2 + y^2 + 4x + 2y = 0$
(4) $x^2 + y^2 + 2x + 8y = 0$

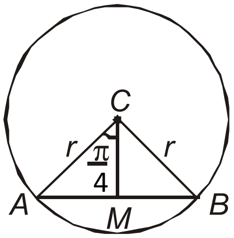
Sol. Answer (2, 4)

Chord subtend 90° angle at the centre of the circle.

$\therefore \quad AB^2 = r^2 + r^2 - 2r \cdot r \cos 90°$

$10 = 2r^2$

$r = \sqrt{5}$



Let centre of circle be (a, b)

$\therefore \quad r = \sqrt{a^2 + b^2} = \sqrt{5}$

$a^2 + b^2 = 5 \qquad \ldots(i)$

$CM = r \cdot \cos 45° = \frac{r}{\sqrt{2}}$

$$\frac{2a - b + \frac{5}{\sqrt{2}}}{\sqrt{5}} = \frac{\sqrt{5}}{\sqrt{2}}$$

$2a - b + \frac{5}{\sqrt{2}} = \frac{5}{\sqrt{2}} \quad \Rightarrow \quad 2a = b \qquad \ldots(ii)$

Solve Equation (i) and (ii), $a = 1$ and $b = 2$

$\therefore$ Equation of circle $x^2 + y^2 - 2x - 4y = 0$

11. The equation of common tangent to the circles $x^2 + y^2 - 2x - 6y + 9 = 0$ and $x^2 + y^2 + 6x - 2y + 1 = 0$ is

(1) $x = 0$ (2) $y - 4 = 0$ (3) $3x + 4y = 10$ (4) $4x - 3y = 0$

Sol. Answer (1, 2, 3, 4)

$C_1(1, 3)$, $r_1 = 1$

If distance of line from centre = radius.

$\therefore$ Line is a tangent of the circle.

$\therefore$ All lines are the tangents of the circles.

12. Equation of a common tangent to the circles $x^2 + y^2 - 6x = 0$ and $x^2 + y^2 + 2x = 0$ is

(1) $x = 1$ (2) $x = 0$ (3) $x + \sqrt{3}y + 3 = 0$ (4) $x - \sqrt{3}y + 3 = 0$

Sol. Answer (2, 3, 4)

$C_1(3, 0)$, $C_2(-1, 0)$

$r_1 = 3$, $r_2 = 1$

$C_1C_2 = \sqrt{16} = 4 = r_1 + r_2$

$\therefore$ Both circles touch externally

$\therefore$ $S_1 - S_2 = 0$ is a common tangent

$x = 0$ is a common tangent

Let $x+\sqrt{3}y+3=0$ is a common tangent

$$C_1M=\frac{3+3}{2}=3=r_1$$

$$C_2M=\frac{-1+3}{2}=1=r_2$$

This is a common tangent.

Similarly $x-\sqrt{3}y+3=0$ is a common tangent.

13. If the circles $x^2 + y^2 - 2ax - 2by - c^2 = 0$ and $x^2 + y^2 = K^2$ touch each other, then

(1) $K=\sqrt{a^2+b^2}-\sqrt{a^2+b^2+c^2}$

(2) $K=\sqrt{a^2+b^2+c^2}-\sqrt{a^2+b^2}$

(3) $K=\sqrt{a^2+b^2+c^2}-\sqrt{b^2+c^2}$

(4) $K=\sqrt{a^2+b^2+c^2}-\sqrt{a^2+c^2}$

Sol. Answer (1, 2)

$C_1C_2=r_1+r_2,\ C_1(a,\ b)$

$r_1=\sqrt{a^2+b^2+c^2}$

C_2 (0, 0)

$r_2 = k$

$\sqrt{a^2+b^2}=\sqrt{a^2+b^2+c^2}+k$

$k=\sqrt{a^2+b^2}-\sqrt{a^2+b^2+c^2}$

or $C_1C_2=r_1-r_2$

$\sqrt{a^2+b^2}=\sqrt{a^2+b^2+c^2}-k$

$k=\sqrt{a^2+b^2+c^2}-\sqrt{a^2+b^2}$

14. $C_1 : x^2 + y^2 = 25$, $C_2 : x^2 + y^2 - 2x - 4y - 7 = 0$ be two circles intersecting at A and B then

(1) Equation of common chord is $x + 2y - 9 = 0$

(2) Equation of common chord is $x + 2y + 7 = 0$

(3) Point of intersection of tangents at A and B to C_1 is $\left(\frac{25}{9},\frac{50}{9}\right)$

(4) C_1, C_2 have four common tangents

Sol. Answer (1, 3)

Equation of common chord

$C_1-C_2=0 \Rightarrow 2x+4y-18=0$

$x+2y-9=0$ …(i)

Let intersection point of tangents be $P(h, k)$.

$\therefore$ Equation of chord of contact $T = 0$

$$xh + yk = 25 \qquad \ldots\text{(ii)}$$

Line (i) and (ii) are coincident lines

$$\frac{h}{1} = \frac{k}{2} = \frac{+25}{9},\ h = \frac{25}{9},\ k = \frac{50}{9}$$

$$P\left(\frac{25}{9}, \frac{50}{9}\right)$$

15. Two circle $x^2 + y^2 + ax + ay - 7 = 0$ and $x^2 + y^2 - 10x + 2ay + 1 = 0$ will cut orthogonally if the value of a is

(1) 3 (2) 2 (3) –2 (4) –3

Sol. Answer (1, 2)

$$2g_1g_2 + 2f_1f_2 = c_1 + c_2$$

$$2\left(\frac{a}{2}\right)(-5) + 2\left(\frac{a}{2}\right)(a) = -7 + 1$$

$$a^2 - 5a + 6 = 0$$

$a = 2, 3$

16. The positive integral value of λ, for which line $4x + 3y - 16\lambda = 0$ lies between the circles $x^2 + y^2 - 4x - 4y + 4 = 0$ and $x^2 + y^2 - 20x - 2y + 100 = 0$, and does not intersect either of the circles, may be

(1) 27 (2) 30 (3) 33 (4) 36

Sol. Answer (1, 2, 3, 4)

$x^2 + y^2 - 4x - 4y + 4 = 0$ ... (1)

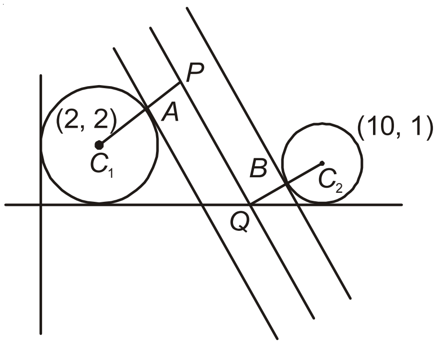
$x^2 + y^2 - 20x - 2y + 100 = 0$ ... (2)

$\because$ λ is positive

$\therefore$ line intersects x and y axes on positive side i.e. intercepts lies in 1st quadrant

Clearly from figure it will be between the tangents at A and B to the two circles.

Tangent parallel to $4x + 3y - \lambda = 0$ for circle (1) is



$$\left|\frac{8 + 6 - \lambda}{5}\right| = 2$$

$\Rightarrow$ 14 – = ± 10

$\Rightarrow$ $\lambda = 24, 4$

Tangent at A is $4x + 3y - 24 = 0$ … (3)

Tangent at B,

$$\left|\frac{40 + 3 - \lambda}{5}\right| = 1$$

$\Rightarrow$ 43 – = ± 5

$\Rightarrow$ $\lambda = 48, 38$

$\therefore$ Tangent at B is $4x + 3y - 38 = 0$

$\therefore$ $24 < \lambda < 38$

17. If the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ cuts each of the circles $x^2 + y^2 - 4 = 0$, $x^2 + y^2 - 6x - 8y + 10 = 0$ and $x^2 + y^2 + 2x - 4y - 2 = 0$ at the extremities of a diameters then,

(1) $c = -4$ (2) $g + f = c - 1$ (3) $g^2 + f^2 - c = 17$ (4) $gf = 6$

Sol. Answer (1, 2, 3, 4)

If circle S cuts S_1 at the extremities of diameter

$\therefore$ Centre of S_1 lie on common chord of the circles

Equation of common chord of S and S_1

$S - S_1 = 0$

$2gx + 2fy + c + 4 = 0$ centre of S_1 (0, 0) lie on this chord

$\therefore$ $c = -4$ ...(i)

Equation of common chord of S and S_2

$S - S_2 = 0$

$(2g + 6)x + (2f + 8)y + c - 10 = 0$

Centre of S_2 (3, 4) lie on this chord

$\therefore$ $3g + 4f + 18 = 0$ ...(ii)

Equation of common chord of S and S_3

$(2g - 2)x + (2f + 4)y + c + 2 = 0$

Centre (–1, 2) lie on this chord

$g - 2f - 4 = 0$ ...(iii)

Solve Equation (ii) and (iii)

$g = -2$

$f = -3$

$\therefore$ All answers are correct.

18. The lines $3x - 4y + 4 = 0$ and $6x - 8y - 7 = 0$ are tangents to the same circle, then

(1) Radius of the circle $= \dfrac{3}{4}$ (2) Radius of the circle $= \dfrac{3}{2}$

(3) Centre of the circle lies on $12x - 16y + 1 = 0$ (4) Centre of the circle lies on $12x - 16y + 31 = 0$

Sol. Answer (1, 3)

Lines are parallel.

$\therefore$ Diameter = Distance between lines

$$2r = \left|\frac{4 - \left(-\frac{7}{2}\right)}{\sqrt{9 + 16}}\right| = \frac{15}{2.5}$$

$$r = \frac{3}{4}$$

Equation of a line mid parallel to given lines are $\dfrac{L_1 + L_2}{2} = 0$

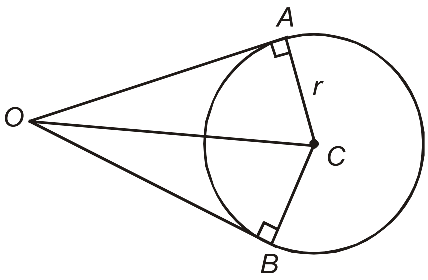
$12x - 16y + 1 = 0$

$\therefore$ Centre lie on line $12x - 16y + 1 = 0$.

19. If the area of a quadrilateral formed by the tangents from the origin to $x^2 + y^2 + 6x - 10y + \lambda = 0$ and the pair of radii at the points of contact of these tangents to the circle, is 8 cm^2, then the value of λ must be

(1) 2 (2) 4 (3) 16 (4) 32

Sol. Answer (1, 4)



$OA = \sqrt{S_1} = \sqrt{\lambda}$

$r = \sqrt{9 + 25 - \lambda}$

Area of $OACB$ = 2 Area of Δ OAC

$$= 2 \cdot \frac{1}{2} \cdot OA \cdot r = OA \cdot r$$

$8 = \sqrt{\lambda} \cdot \sqrt{34 - \lambda}$

$\therefore$ $\lambda = 2, 32$

20. If $(a \cos\theta_1, a \sin\theta_1)$, $(a \cos\theta_2, a \sin\theta_2)$, $(a \cos\theta_3, a \sin\theta_3)$ represents the vertices of an equilateral triangle inscribed in $x^2 + y^2 = a^2$, then

(1) $\Sigma \cos\theta_i = 0$ (2) $\Sigma \sin\theta_i = 0$ (3) $\Sigma \tan\theta_i = 0$ (4) $\Sigma \cot\theta_i = 0$

Sol. Answer (1, 2)

In equilateral Δ centroid and circumcentre are same.

$$\therefore \quad \left(\frac{\Sigma a \cos\theta_i}{3}, \frac{\Sigma a \sin\theta_i}{3}\right) = (0,\ 0)$$

$\Sigma \cos\theta_i = 0,\ \Sigma \sin\theta_i = 0$

21. Equation of tangents drawn from (0, 0) to $x^2 + y^2 - 6x - 6y + 9 = 0$ are

(1) $x = y$ (2) $y = 0$ (3) $x = 0$ (4) $x + y = 0$

Sol. Answer (2, 3)

Centre (3, 3) radius = 3

$\therefore$ Circle touches both the axis.

$\therefore$ Equal of tangents are $x = 0$ and $y = 0$

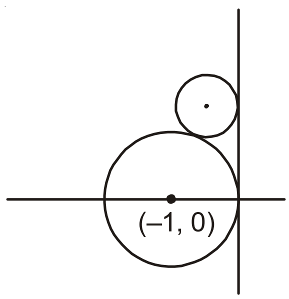
22. The locus of the centre of the circle which moves such that it touches the circle $(x + 1)^2 + y^2 = 1$ externally and also the y-axis is

(1) $y^2 = 4x, x \geq 0$ (2) $y^2 = -4x, x \leq 0$

(3) $y = 0, x > 0$ (4) $y = 0, \forall\, x \in R$

Sol. Answer (2, 3)

Let the centre of moving circle be (h, k)



$\because$ Circle touch externally

$\Rightarrow$ $(h + 1)^2 + k^2 = (1 + |h|)^2$

$\Rightarrow$ $h^2 + 2h + 1 + k^2 = 1 + h^2 + 2|h|$

$\Rightarrow$ $k^2 = 2(|h| - h)$

$$= \begin{cases} -4h \ ; & h \leq 0 \\ 0 \ ; & h > 0 \end{cases}$$

$\therefore$ Locus of (h, k) is $y^2 = -4x$, $x \leq 0$ and $y = 0$ if $x > 0$ or we can write as locus of (h, k) is the set S

where $S = \{(x, y); y^2 = -4x, x \leq 0\} \cup \{(x, 0); x > 0\}$

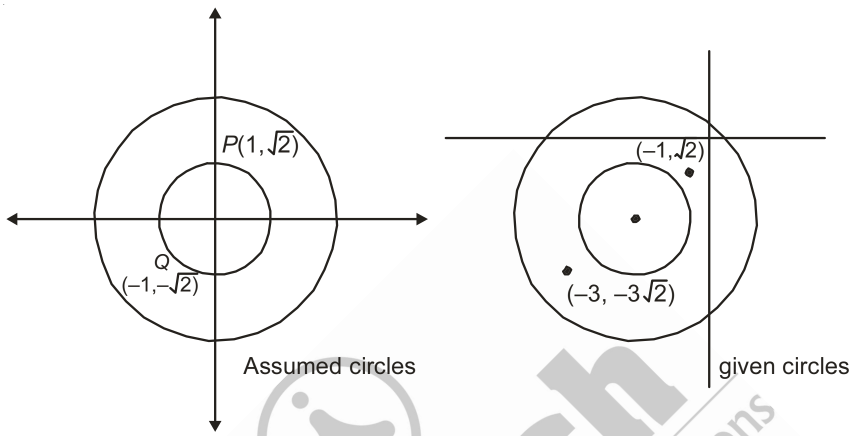
23. If a point $(a, \sqrt{2}a)$ lies in region bounded between the circles $x^2 + y^2 + 4x + 4y + 7 = 0$ and $x^2 + y^2 + 4x + 4y - 1 = 0$, then the number of integral values of a exceeds

(1) 0 (2) 1

(3) 2 (4) 3

Sol. Answer (1, 2)

Given circles are $(x + 2)^2 + (y + 2)^2 = 1$ and $(x + 2)^2 + (y + 2)^2 = 9$



Our objective is to find number of points $(a, \sqrt{2}a)$ (numbers only not the co-ordinates of point), therefore lets find points $(a, \sqrt{2}a)$ for $x^2 + y^2 = 1$ and $x^2 + y^2 = 9$ (concentric and with radii 1 and 3)

$1 < a^2 + 2a^2 < 9$

$\Rightarrow \dfrac{1}{3} < a^2 < 3$

$\Rightarrow a \in \left(-\sqrt{3}, -\dfrac{1}{\sqrt{3}}\right) \cup \left(\dfrac{1}{\sqrt{3}}, \sqrt{3}\right)$

Integral values of a are -1 and $+1$

$\therefore$ Two integral values of a exist.

24. Tangents and normal are drawn from a point (3, 1) to circle C whose equation is $x^2 + y^2 - 2x - 2y + 1 = 0$. Let the points of contact of tangents be $T_i(x_i, y_i)$, where $i = 1, 2$ and feet of normals be N_1 and N_2 (N_1 is near P). Tangents are drawn at N_1 and N_2 and normals are drawn at T_1 and T_2

(1) $x_1 + x_2 + y_1 + y_2 = 5$

(2) $x_1x_2y_1y_2 = -\dfrac{9}{16}$

(3) Normal at T_1 and tangents at T_2 and N_2 are concurrent

(4) Circle is incircle of the triangle formed by tangents from P and tangent at N_2

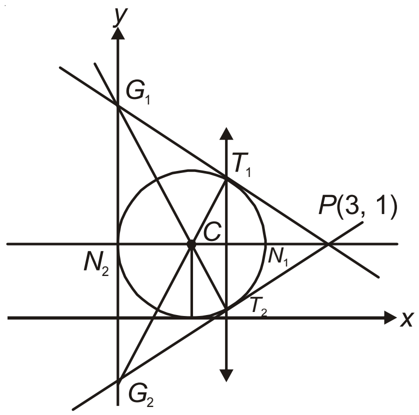
Sol. Answer (1, 3, 4)

Chord of contact of tangents drawn from (3, 1) to the circle

$x^2 + y^2 - 2x - 2y + 1 = 0$ is $3x + y - (x + 3) - (y + 1) + 1 = 0$

$\Rightarrow \quad 2x = 3$

$\Rightarrow \quad \boxed{x = \frac{3}{2}}$ equation of T_1T_2



Points of contact T_1, T_2 are

$$\frac{9}{4} + y^2 - 3 - 2y + 1 = 0$$

$$y^2 - 2y + \frac{1}{4} = 0$$

$$y = \frac{2 \pm \sqrt{4-1}}{2} = 1 \pm \frac{\sqrt{3}}{2}$$

$$\therefore \quad T_1 : \left(\frac{3}{2}, \frac{\sqrt{3}}{2} + 1\right) = (x_1, y_1)$$

$$T_2 : \left(\frac{3}{2}, 1 - \frac{\sqrt{3}}{2}\right) = (x_2, y_2)$$

Option (1) : $x_1 + x_2 + x_3 + x_4 = \frac{3}{2} + \frac{3}{2} + 1 + \frac{\sqrt{3}}{2} + 1 - \frac{\sqrt{3}}{2} = 5$ (given options is correct)

Option (2) : $x_1x_2y_1y_2 = \frac{9}{4}\left(1 - \frac{3}{4}\right) = \frac{9}{16}$ (gives options is wrong)

Option (3) : Equation of tangent at N_2 : $x = 0$ … (i)

Equation of tangent at T_2 : $y - 1 = \left(\dfrac{1 - \frac{\sqrt{3}}{2} - 1}{\frac{3}{2} - 3}\right)(x - 3)$

$$\Rightarrow \quad y - 1 = \frac{1}{\sqrt{3}}(x - 3) \qquad \text{… (ii)}$$

Equation of normal at T_1 : $y - 1 = \dfrac{1 + \frac{\sqrt{3}}{2} - 1}{\frac{3}{2} - 1}(x - 1)$

$$\Rightarrow \quad y - 1 = \sqrt{3}\,(x - 1) \qquad \text{… (iii)}$$

Clearly (i), (ii) and (iii) are concurrent at $(0, 1 - \sqrt{3})$ (Option (3) is correct)

Option (4) : True clear from figure

[Option (3) and (4) are true only if ΔPG_1G_2 is equilateral $\Delta \Rightarrow \angle$ between tangents from P is $\frac{\pi}{3}$]

Alter : Shift the centre of circle at (0, 0) and solve. Later change to given system. (For option (1) & (2) & (3) and (4) are verified by statement given above)

25. Let C_1 and C_2 be two concentric circles such that the radius of C_2 is double that of radius of C_1. Tangents PQ and PR are drawn from a point on C_2 to C_1.

(1) Centroid of ΔPQR lies on C_1

(2) Orthocentre of ΔPQR lies on C_1

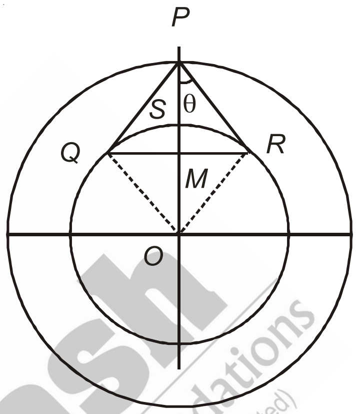
(3) If radius of C_1 is $\sqrt{3}$ then area of ΔPQR is $\frac{9\sqrt{3}}{4}$ sq. units

(4) If radius of C_1 is $\sqrt{3}$ then area of ΔPQR is $\frac{27}{4}$ sq. units

Sol. Answer (1, 2, 3)

$OR = OQ = r$ (say)

$OP = 2r$



From ΔOPR, $PR = \sqrt{4r^2 - r^2} = \sqrt{3}\,r$

Let $\angle OPR = Q \Rightarrow \tan\theta = \frac{OR}{PR}$

$$= \frac{r}{\sqrt{3}r} = \frac{1}{\sqrt{3}}$$

$\Rightarrow \quad \theta = \frac{\pi}{6}$

$\therefore \quad \angle QPR = \frac{\pi}{3}$

Also $PQ = PR \quad \Rightarrow \Delta PQR$ is equilateral

$MR = \frac{1}{2}QR = \frac{1}{2}PR$

$= \frac{\sqrt{3}}{2}r$

$\therefore \quad OM = \sqrt{r^2 - \frac{3}{4}r^2} = \frac{r}{2}$

$\Rightarrow \quad SM = \frac{r}{2},\ PS = r$

S divides PM in the ratio 2 : 1

$\Rightarrow$ Centroid of ΔPQR

Also, since it is equilateral Δ

$\therefore$ Orthocenter is also same.

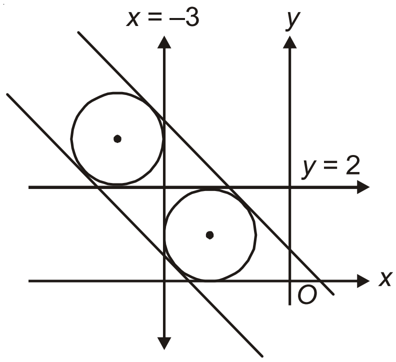
Area of $\Delta PQR = \frac{\sqrt{3}}{4}a^2 = \frac{\sqrt{3}}{4}(\sqrt{3}r)^2 = \frac{3\sqrt{3}}{4}r^2$

If $r = \sqrt{3} \Rightarrow$ Area $= \frac{9\sqrt{3}}{4}$ sq. units.

26. The equation(s) of common tangent(s) to the two circles $x^2 + y^2 + 4x - 2y + 4 = 0$ and $x^2 + y^2 + 8x - 6y + 24 = 0$ is /are

(1) $x + 3 = 0$ (2) $y - 2 = 0$ (3) $x + y = \sqrt{2} - 1$ (4) $x + y + \sqrt{2} + 1 = 0$

Sol. Answer (1, 2, 3, 4)



$(x + 2)^2 + (y - 1)^2 = 1$ ... (1)

$(x + 4)^2 + (y - 3)^2 = 1$ ... (2)

Let the tangents to (1) and (2) be

$y - 1 = m(x + 2) \pm \sqrt{1 + m^2}$ ...(3)

$y - 3 = m(x + 4) \pm \sqrt{1 + m^2}$ ...(4) respectively

If it be common tangents then these two equations are identical.

Equating constants on both sides

$1 + 2m \pm \sqrt{1 + m^2} = 3 + 4m \pm \sqrt{1 + m^2}$ taking +ve on both sides,

$2m + 2 = 0 \Rightarrow m = -1$ (two parallel tangents) taking +ve on left and –ve on right

$2\sqrt{1 + m^2} = 2m + 2$

$\Rightarrow$ $1 + m^2 = 1 + m^2 + 2m$

$\Rightarrow$ $m = 0$

$\Rightarrow$ One tangent is parallel to x-axis and since second degree is cancelled on both sides, therefore one tangent is vertical.

(Other the cases *i.e.*, both –ve or –ve and +ve will repeat these cases only)

$\therefore$ Tangents are

$m = -1$ in (3) $y - 1 = -x - 2 \pm \sqrt{2}$

$\Rightarrow$ $x + y = \sqrt{2} - 1$ & $x + y = -\sqrt{2} - 1$

$m = -1$ in (4), $y - 3 = -x - 4 \pm \sqrt{2} \Rightarrow x + y = \sqrt{2} - 1, x + y = -\sqrt{2} - 1$

$\Rightarrow$ Both are common tangents

$$\left.\begin{array}{l} m = 0 \text{ in (3) } y - 1 = \pm 1 \quad \Rightarrow \quad y = 2,\ y = 0 \\ m = 0 \text{ in (4) } y - 3 = \pm 1 \quad \Rightarrow \quad y = 4,\ y = 2 \end{array}\right] \Rightarrow \boxed{y = 2} \text{ is tangent}$$

$\frac{1}{m} = 0$ in (3)

$\Rightarrow$ $\frac{y - 1}{m} = x + 2 \pm \sqrt{\frac{1}{m^2} + 1} \Rightarrow$ $x + 2 \pm 1 = 0$

$\Rightarrow$ $x + 3 = 0, x + 1 = 0$

$\frac{1}{m} = 0$ in (4)

$\Rightarrow$ $\frac{y - 3}{m} = x + 4 \pm \sqrt{\frac{1}{m^2} + 1}$

$\Rightarrow$ $x + 4 \pm 1 = 0$

$\Rightarrow$ $x + 5 = 0, x + 3 = 0$

$\therefore$ $x + 3 = 0$ is common tangent

$\therefore$ 4 common tangents are,

$x + 3 = 0$

$y - 2 = 0$

$x + y = \sqrt{2} - 1$

$x + y + \sqrt{2} + 1 = 0$

Alter :

From figure it is clear that one tangent is parallel to x-axis and one is parallel to y-axis these are $\boxed{y = 2}$ & $\boxed{x = -3}$

$\because$ Radii are equal other two common tangents are parallel. Slope = slope of line joining centers = $\dfrac{3-1}{-4+2} = -1$

$\therefore$ Tangents is $x + y = \lambda$

Length of perpendicular from centre = Radius

$$\Rightarrow \left|\frac{-2+1-\lambda}{\sqrt{2}}\right| = 1$$

$$\Rightarrow 1 + \lambda = \pm\sqrt{2}$$

$$\Rightarrow \lambda = \pm\sqrt{2} - 1$$

$\therefore$ Tangents are $\boxed{x + y = \pm\sqrt{2} - 1}$

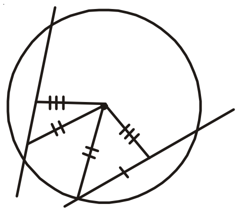
27. Chords are drawn to the circle $x^2 + y^2 - 2x - 2y - 8 = 0$ from the point $(-1, -1)$. If circle cuts off equal intercept on these chords and the line $3x + 4y + 3 = 0$, then the equation of chord(s) may be

(1) $x + 1 = 0$ (2) $x + y + 2 = 0$ (3) $y + 1 = 0$ (4) $2x + y + 3 = 0$

Sol. Answer (1, 3)

Let the line be

$y + 1 = m(x + 1)$



$\because$ Circle cuts equal intercept on this line and the line $3x + 4y + 3 = 0$

$\Rightarrow$ Length of perpendicular from centre (1, 1) to these lines and equal (from figure)

$$\Rightarrow \left|\frac{2m-2}{\sqrt{1+m^2}}\right| = \left|\frac{3+4+3}{\sqrt{9+16}}\right|$$

$$\Rightarrow 2(m-1) = 2\sqrt{1+m^2}$$

$\Rightarrow$ $m^2 + 1 - 2m = 1 + m^2$

$\Rightarrow$ $m = 0$, also one line is vertical ($\because$ m^2 is cancelled)

Chords are $x + 1 = 0$ and $y + 1 = 0$

28. The equation(s) of circle(s) touching $12x - 5y = 7$ at (1, 1) and having radius 13 is/are

(1) $x^2 + y^2 - 22x + 12y - 12 = 0$ (2) $x^2 + y^2 + 22x - 12y - 12 = 0$

(3) $x^2 + y^2 - 26x - 8y + 16 = 0$ (4) $x^2 + y^2 - 26x + 8y + 16 = 0$

Sol. Answer (2, 4)

Equation of required circle can be written as

$S_P + L_T = 0$ where S_P Point circle at the point of contact

$L_T \rightarrow$ tangent line at that point

$(x-1)^2 + (y-1)^2 + \lambda\,(12x - 5y - 7) = 0$

$x^2 + y^2 + x\,(-2 + 12\lambda) + y\,(-2 - 5\lambda) + (2 - 7\lambda) = 0$

If this circle is the required one then the radius = 13

$$\Rightarrow \sqrt{(-1+6\lambda)^2 + \left(-1-\frac{5}{2}\lambda\right)^2 - (2-7\lambda)} = 13$$

$$\Rightarrow 1 + 36\lambda^2 - 12\lambda + 1 + \frac{25}{4}\lambda^2 + 5\lambda - 2 + 7\lambda = 169$$

$$\Rightarrow \frac{169}{4}\lambda^2 = 169$$

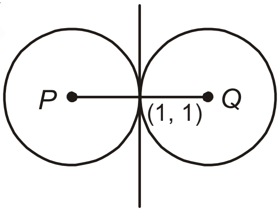
$\Rightarrow \lambda = \pm 2$

$\therefore$ Required circles are

$\lambda = 2 \qquad \Rightarrow x^2 + y^2 + 22x - 12y - 12 = 0$

and $\lambda = -2 \qquad \Rightarrow x^2 + y^2 - 26x + 8y + 16 = 0$

Alter :



$PQ \rightarrow$ line joining the centres of two possible circles

It is perpendicular to given tangent

$\therefore$ Slope of $PQ = -\dfrac{5}{12}$

$\therefore$ Equation of PQ

$$\frac{x-1}{\frac{-12}{13}} = \frac{y-1}{\frac{5}{13}} = \pm r \text{ (say)}$$ to get points P and Q on either side of (1, 1) put $r = 13$

$\Rightarrow x - 1 = \mp 12,\ y - 1 = \pm 5$

$\Rightarrow x = -11,\ x = 13$

and $y = 6,\ y = -4$

$\therefore$ Center are (–11, 6) and (13, –4)

$\therefore$ Circles are $(x + 11)^2 + (y - 6)^2 = 169$ and $(x - 13)^2 + (y + 4)^2 = 169$

29. Let a circle cuts orthogonally each of the three circles $x^2 + y^2 + 3x + 4y + 11 = 0$, $x^2 + y^2 - 3x + 7y - 1 = 0$ and $x^2 + y^2 + 2x = 0$

(1) The centre of the circle is (–3, –2)

(2) Radius of the circle is 3

(3) Equation of chord of contact of tangents drawn from (2, 4) is $5x + 6y - 18 = 0$

(4) Length of tangent from (2, 4) to the circle is $\sqrt{13}$

Sol. Answer (1, 2)

(1) R.A. of (1) and (3) is

$x + 4y + 11 = 0$

R.A of (2) and (3) is

$5x - 7y + 1 = 0$

Solving these two equations we get radical centre, which is the cube of required circle.

$$\frac{x}{4+77} = \frac{y}{55-1} = \frac{1}{-7-20}$$

$x = -3,\ y = -2$

$\therefore$ Centre is $(-3, -2)$

(2) Radius of the circle = length of tangent from $(-3, -2)$ to any of the three circles

$\sqrt{9+4-4} = 3$

(3) $\therefore$ Equation the circle is

$(x + 3)^2 + (y + 2)^2 = 9$

$x^2 + y^2 + 6x + 4y + 4 = 0$

Chord of contact of tangents drawn from (2, 4) is

$2x + 4y + 3(x + 2) + 2(y + 4) + 4 = 0$

$5x + 6y + 18 = 0$

(4) Length of tangent $= \sqrt{4+16+12+16+4} = 2\sqrt{13}$

30. The equation of a circle touching x-axis at $(-4, 0)$ and cutting off an intercept of 6 units on y-axis can be

(1) $x^2 + y^2 + 8x + 12y + 16 = 0$

(2) $x^2 + y^2 + 8x - 12y + 16 = 0$

(3) $x^2 + y^2 + 8x + 10y + 16 = 0$

(4) $x^2 + y^2 + 8x - 10y + 16 = 0$

Sol. Answer (3, 4)

Let the circle be $x^2 + y^2 + 2gx + 2fy + c = 0$

$2\sqrt{g^2 - c} = 0 \quad \Rightarrow g^2 = c$ … (1)

$2\sqrt{f^2 - c} = 6 \quad \Rightarrow f^2 - c = 9$ … (2)

Passes through $(-4, 0)$

$\Rightarrow$ $16 + 0 - 8g + 0 + c = 0$

Putting $c = g^2$ from (1)

$\Rightarrow$ $16 - 8g + g^2 = 0$

$\Rightarrow$ $(g - 4)^2 = 0$

$g = 4$

$\Rightarrow$ $c = 16$

$\therefore$ $f^2 = 9 + 16 = 25$

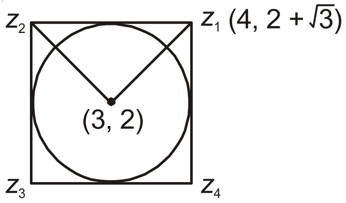
$\Rightarrow$ $f = \pm 5$

$\therefore$ Equations of circles are $x^2 + y^2 + 8x \pm 10y + 16 = 0$

31. Let one of the vertices of the square circumseribing the circle $x^2 + y^2 - 6x - 4y + 11 = 0$ be $(4,\ 2+\sqrt{3})$. The other vertices of the square may be

(1) $(3-\sqrt{3},\ 3)$ (2) $(2,\ 2-\sqrt{3})$ (3) $(3+\sqrt{3},\ +1)$ (4) $(0, 0)$

Sol. Answer (1, 2, 3)



$z_2(3+2i) = [4+(2+\sqrt{3})i-(3+2i)]e^{i\frac{\pi}{2}}$

$z_2 = (1+\sqrt{3}i)i + 3 + 2i$

$= -\sqrt{3} + i + 3 + 2i$

$= (3-\sqrt{3}) + 3i$

$z_2 = (3-\sqrt{3},\ 3)$

$\dfrac{z_3 + z_1}{2} = z_0 \Rightarrow z_3 = (6+4i) - (4+(2+\sqrt{3})i)$

$= 2 + (2-\sqrt{3})i$

$\Rightarrow\ z_3 = (2,\ 2-\sqrt{3})$

$\dfrac{z_4 + z_2}{2} = z_0 \Rightarrow z_4 = (6+4i) - (3-\sqrt{3}+3i)$

$= 3 + \sqrt{3} + i$

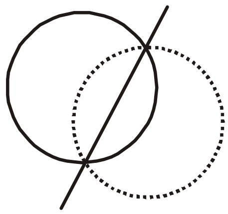
$\therefore\ z_4 = (3+\sqrt{3},\ 1)$

32. If $x^2 + y^2 - 2y - 15 + \lambda(2x + y - 9) = 0$ represents family of circles for different values of λ, then the equation of the circle(s) along these circles having minimum radius is/are

(1) $3x^2 + 3y^2 - 2x - 7y - 36 = 0$ (2) $x^2 + y^2 - 2y - 15 = 0$

(3) $5x^2 + 5y^2 - 32x - 26y + 69 = 0$ (4) $x^2 + y^2 - 10x - 7y + 30 = 0$

Sol. Answer (3)



The circle will be of minimum radius if chord $2x + y - 9 = 0$ is diameter

$x^2 + y^2 + 2\lambda x + y(\lambda - 2) - (15 + 9\lambda) = 0$

$\Rightarrow$ Centre $\left(-\lambda,\ \dfrac{2-\lambda}{2}\right)$ lies on $2x + y - 9 = 0$

$\Rightarrow\ -2\lambda + \dfrac{2-\lambda}{2} - 9 = 0$

$-5\lambda - 16 = 0$

$\Rightarrow\ \lambda = -\dfrac{16}{5}$

$\therefore$ Required circle is

$x^2 + y^2 - \dfrac{32}{5}x - \dfrac{26}{5}y + \dfrac{69}{5} = 0$

$5x^2 + 5y^2 - 32x + 69 = 0$

33. The straight line $\ell x + my = 1$ intersects $px^2 + 2qxy + ry^2 = s$ at AB. Chord AB subtends a right angle at the origin. If $\ell x + my = 1$ is a tangent to a circle $x^2 + y^2 = a^2$, then $a =$

(1) $\sqrt{\dfrac{s}{p+r}}$ (2) $\sqrt{\dfrac{p+r}{s}}$ (3) $\sqrt{\ell^2+m^2}$ (4) $\dfrac{1}{\sqrt{\ell^2+m^2}}$

Sol. Answer (1, 4)

Combined equation of OA and OB is

$x^2 + 2qxy + ry^2 = s(lx + my)^2$

$\because$ OA is perpendicular to OB

$\therefore$ Coefficient of x^2 + co-efficient of $y^2 = 0$

$(p - sl^2) + (r - sm^2) = 0$

$\Rightarrow$ $p + r = s(l^2 + m^2)$ ... (1)

$lx + my = 1$ touches $x^2 + y^2 = a^2$

$\therefore$ $\left|\dfrac{1}{\sqrt{l^2+m^2}}\right| = a$

$\therefore$ $a = \dfrac{1}{\sqrt{l^2+m^2}}$

Also from (1) $l^2 + m^2 = \dfrac{p+r}{s}$

$\therefore$ $a = \dfrac{1}{\sqrt{l^2+m^2}} = \sqrt{\dfrac{s}{p+r}}$

34. Equation(s) of circle(s) concentric with $x^2 + y^2 + 2x + 4y = 0$ and touching the circle $x^2 + y^2 - 2x - 1 = 0$ is/are

(1) $x^2 + y^2 + 2x + 4y + 3 = 0$

(2) $x^2 + y^2 + 2x + 4y - 13 = 0$

(3) $x^2 + y^2 + 2x + 4y - 3 = 0$

(4) $x^2 + y^2 + 2x + 4y + 13 = 0$

Sol. Answer (1, 2)

Centre of the required circle is $(-1, -2)$, let its radius be 'r'

It touches $x^2 + y^2 - 2x - 1 = 0$, (may be internally or externally)

$\Rightarrow$ Distance between centers = $|r \pm \sqrt{2}|$

$r + \sqrt{2} = \sqrt{(1+1)^2 + (+2)^2} = 2\sqrt{2}$

$\Rightarrow$ $r = \sqrt{2}$

$\therefore$ Required circles are, $(x + 1)^2 + (y + 2)^2 = 2$

$\Rightarrow$ $x^2 + y^2 + 2x + 4y + 3 = 0$ and $(x + 1)^2 + (y + 2)^2 = 18$

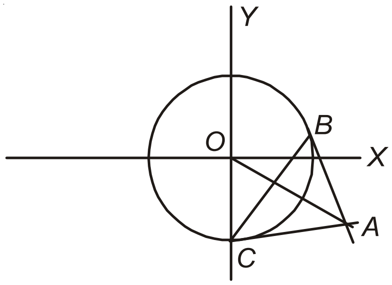
$\Rightarrow$ $x^2 + y^2 + 2x + 4y - 13 = 0$

$|r - \sqrt{2}| = 2\sqrt{2}$

$r - \sqrt{2} = 2\sqrt{2}$ or $r - \sqrt{2} \neq -2\sqrt{2}$

$\Rightarrow$ $r = 3\sqrt{2}$

35. Let the midpoint of the chord of contact of tangents drawn from A to the circle $x^2 + y^2 = 4$ be $M(1, -1)$ and the points of contact be B and C



(1) The area of ΔABC is 2 sq. units

(2) The area of ΔABC is $\frac{1}{2}$ sq. units

(3) Co-ordinate of point A is (2, –2)

(4) ΔABC is right angled triangle

Sol. Answer (1, 3, 4)

Equation of chord BC is

$x - y = 2$ …. (1)

Let A be (h, k), the equation BC i.e. chord of contact of tangents drawn from (h, k) is

$hx + ky = 4$ … (2)

From (1) & (2)

$$\frac{h}{1} = \frac{k}{-1} = \frac{4}{2}$$

$\Rightarrow$ $(h, k) = (2, -2)$

$$AM = \left|\frac{2+2-2}{\sqrt{2}}\right| = \sqrt{2}$$

$$BM = \sqrt{OB^2 - OM^2} = \sqrt{4 - \left|\frac{2}{\sqrt{2}}\right|^2} = \sqrt{2}$$

$\therefore$ Area of $\Delta ABC = \frac{1}{2} AM \times BC = \frac{1}{2} \times \sqrt{2} \times 2\sqrt{2} = 2$ sq. units

Point (2, –2) lies on director circle $x^2 + y^2 = 8$

$\therefore$ $\angle A = \frac{\pi}{2}$

36. A straight line through the vertex P of a triangle PQR intersects the side QR at the point S and the circumcircle of the triangle PQR at the point T. If S is not the centre of the circumcircle, then **[IIT-JEE 2008]**

(1) $\frac{1}{PS} + \frac{1}{ST} < \frac{2}{\sqrt{QS \times SR}}$

(2) $\frac{1}{PS} + \frac{1}{ST} > \frac{2}{\sqrt{QS \times SR}}$

(3) $\frac{1}{PS} + \frac{1}{ST} < \frac{4}{QR}$

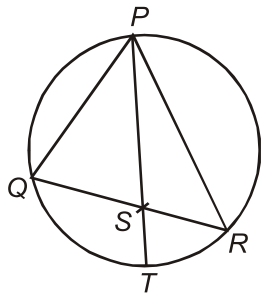
(4) $\frac{1}{PS} + \frac{1}{ST} > \frac{4}{QR}$

Sol. Answer (2, 4)

Let a straight line through the vertex P of a given ΔPQR intersects the side QR at the point S and the circumcircle of ΔPQR at the point T.

Points P, Q, R, T are concyclic, hence

$PS \,.\, ST = QS \,.\, SR$



Now $\dfrac{PS+ST}{2} \ge \sqrt{PS \cdot ST}$

and $\dfrac{1}{PS}+\dfrac{1}{ST} \ge \dfrac{2}{\sqrt{PS \cdot ST}} = \dfrac{2}{\sqrt{QS.SR}}$

Also $\dfrac{SQ+SR}{2} \ge \sqrt{SQ \cdot SR}$

$\Rightarrow \quad \dfrac{QR}{2} \ge \sqrt{SQ \cdot SR}$

$\Rightarrow \quad \dfrac{1}{\sqrt{SQ \cdot SR}} \ge \dfrac{2}{QR}$

$\Rightarrow \quad \dfrac{2}{\sqrt{SQ \cdot SR}} \ge \dfrac{4}{QR}$

Hence $\dfrac{1}{PS}+\dfrac{1}{ST} \ge \dfrac{2}{\sqrt{QS \cdot SR}} \ge \dfrac{4}{QR}$

37. Circle(s) touching x-axis at a distance 3 from the origin and having an intercept of length $2\sqrt{7}$ on y-axis is (are) **[JEE(Advanced) 2013]**

(1) $x^2 + y^2 - 6x + 8y + 9 = 0$
(2) $x^2 + y^2 - 6x + 7y + 9 = 0$
(3) $x^2 + y^2 - 6x - 8y + 9 = 0$
(4) $x^2 + y^2 - 6x - 7y + 9 = 0$

Sol. Answer (1, 3)

Clearly centre is $(3, \alpha)$ and radius is $|\alpha|$

So, $x^2 + y^2 - 6x - 2\alpha y + c = 0$

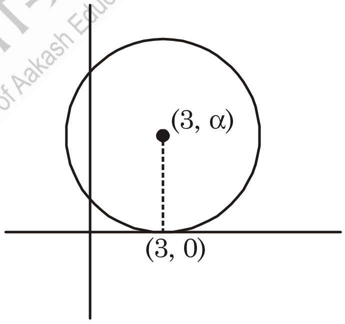
Radius $9 + \alpha^2 - c = \alpha^2$

$c = 9$

Intercept on y axis, $2\sqrt{\alpha^2 - c} = 2\sqrt{7}$

$\alpha^2 - 9 = 7$

$\alpha = \pm 4$



So equation becomes $x^2 + y^2 - 6x \pm 8y + 9 = 0$

38. A circle S passes through the point (0, 1) and is orthogonal to the circles $(x - 1)^2 + y^2 = 16$ and $x^2 + y^2 = 1$. Then **[JEE(Advanced)-2014]**

(1) Radius of S is 8
(2) Radius of S is 7
(3) Centre of S is (–7, 1)
(4) Centre of S is (–8, 1)

Sol. Answer (2, 3)

Let the circle be $x^2 + y^2 + 2gx + 2fy + C = 0$.

It is orthogonal with $(x - 1)^2 + y^2 = 16$

$\therefore \quad 2(-g + 0) = -15 + C \quad \Rightarrow \quad -2g = -15 + C$

It is also orthogonal with $x^2 + y^2 = 1$

$\therefore \quad 0 = -1 + C \qquad \Rightarrow \qquad C = 1$

$\therefore \quad g = 7$

This circle passes through (0, 1)

$\therefore \quad 1 + 2f + 1 = 0 \qquad \Rightarrow \qquad f = -1$

$\therefore$ The circle is $x^2 + y^2 + 14x - 2y + 1 = 0$

$(x + 7)^2 + (y - 1)^2 = 49$

$\therefore$ centre : (–7, 1) and radius : 7

39. Let T be the line passing through the points $P(-2, 7)$ and $Q(2, -5)$. Let F_1 be the set of all pairs of circles (S_1, S_2) such that T is tangent to S_1 at P and tangent to S_2 at Q, and also such that S_1 and S_2 touch each other at a point, say, M. Let E_1 be the set representing the locus of M as the pair (S_1, S_2) varies in F_1. Let the set of all straight line segments joining a pair of distinct points of E_1 and passing through the point $R(1, 1)$ be F_2. Let E_2 be the set of the mid-points of the line segments in the set F_2. Then, which of the following statement(s) is (are) TRUE? **[JEE(Advanced) 2018]**

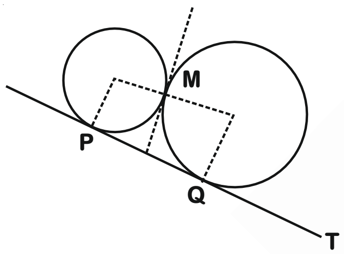
(1) The point (–2, 7) lies in E_1

(2) The point $\left(\frac{4}{5}, \frac{7}{5}\right)$ does NOT lie in E_2

(3) The point $\left(\frac{1}{2}, 1\right)$ lies in E_2

(4) The point $\left(0, \frac{3}{2}\right)$ does NOT lie in E_1

Sol. Answer (2, 4)



$\angle PMQ = 90°$

$\Rightarrow \quad \frac{\beta+5}{\alpha-2} \times \frac{\beta-7}{\alpha+2} = -1$

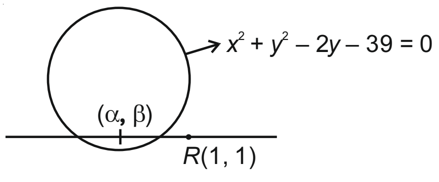
Locus of M

$x^2 + y^2 - 2y - 39 = 0 \qquad$ ...(1)

Equation of chord of circle whose midpoint is (α, β) is

$S_1 = T$

$\Rightarrow \quad x\alpha + y\beta - (y + \beta) - 39 = \alpha^2 + \beta^2 - 2\beta - 39$



It passes through (1, 1)

$\Rightarrow \quad \alpha^2 + \beta^2 - 2\beta - \alpha + 1 = 0$

Locus : $x^2 + y^2 - x - 2y + 1 = 0 \qquad$ ...(2)

Option (1) is incorrect although it satisfies eq. (1) otherwise the line T would touch the second circle on two points. Also (4/5, 7/5) satisfies eq. (2) but again in this case one end of the chord would be (–2, 7) which is not included in E_1. Therefore $\left(\frac{4}{5}, \frac{7}{5}\right)$ does not lie in E_2. (1/2, 1) does not satisfy equation (2), therefore it does not lie in E_2. (0, 3/2) does not satisfy (1), so does not lie in E_1.

SECTION - C

Linked Comprehension Type Questions

Comprehension-I

A circle C_1 of radius 2 units rolls on the outerside of the circle $C_2 : x^2 + y^2 + 4x = 0$, touching it externally.

1. The locus of the centre of C_1 is

(1) $x^2 + y^2 + 4y - 12 = 0$ (2) $x^2 + y^2 + 4x - 12 = 0$

(3) $x^2 + y^2 + 4x + 4y - 4 = 0$ (4) $x^2 + y^2 - 4x = 0$

Sol. Answer (2)

Let $C_1(h,\ k),\ r_1 = 2$

$C_2(-2,\ 0),\ r_2 = \sqrt{4} = 2$

$C_1C_2 = r_1 + r_2$

$\sqrt{(h+2)^2 + k^2} = 2 + 2$

$(h+2)^2 + k^2 = 16$

Locus of (h, k), $x^2 + y^2 + 4x - 12 = 0$

2. Area of a quadrilateral found by a pair of tangents from a point of $x^2 + y^2 + 4x - 12 = 0$ to C_2 with a pair of radii at the points of contact of the tangents is (in sq. units)

(1) $2\sqrt{3}$ (2) $4\sqrt{3}$

(3) $\sqrt{3}$ (4) $3\sqrt{3}$

Sol. Answer (2)

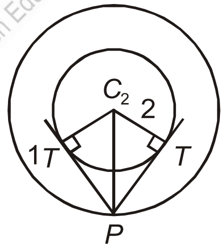
Let $P(x, y)$ be any point on $x^2 + y^2 + 4x - 12 = 0$

Length of tangent from $P(x, y)$ to $x^2 + y^2 + 4x = 0$

$PT = \sqrt{x^2 + y^2 + 4x} = \sqrt{12} = 2\sqrt{3}$

Area of $\Delta PTC_2 = \frac{1}{2} \cdot 2\sqrt{3} \cdot 2 = 2\sqrt{3}$

$\therefore$ Area of quadrilateral $= 2\,(2\sqrt{3}) = 4\sqrt{3}$ sq. units



3. Square of the length of the intercept made by $x^2 + y^2 + 4x - 12 = 0$ on any tangent to C_2 is

(1) 12 (2) 24 (3) 16 (4) 48

Sol. Answer (4)

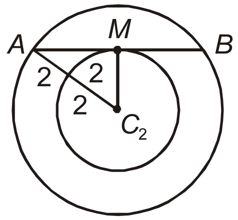
$AC_2 = 4$

$MC_2 = 2$

$\therefore\ AM = \sqrt{16 - 4} = 2\sqrt{3}$

$AB = 4\sqrt{3}$

$AB^2 = 48$



Comprehension-II

$C : x^2 + y^2 - 2x - 2ay - 8 = 0$, a is a variable

1. C represents a family of circles passing through two fixed points whose co-ordinates are

(1) (–2, 0), (4, 0) (2) (2, 0), (4, 0) (3) (–4, 0), (4, 0) (4) (2, 0), (–4, 0)

Sol. Answer (1)

$(x^2 + y^2 - 2x - 8) - 2ay = 0$

Represents family of circles which passes through the fixed point.

Point of intersection of $x^2 + y^2 - 2x - 8 = 0$ and $y = 0$.

Solve the equation $x^2 - 2x - 8 = 0 \Rightarrow x = 4, -2$ fixed points are (–2, 0) and (4, 0).

2. Equation of a circle C_1 of this family tangents to which at these fixed points intersects on the line $2y + x + 5 = 0$ is

(1) $x^2 + y^2 - 2x - 8y - 8 = 0$ (2) $x^2 + y^2 - 2x + 6y - 8 = 0$

(3) $x^2 + y^2 - 2x + 8y - 8 = 0$ (4) $x^2 + y^2 - 2x - 6y - 8 = 0$

Sol. Answer (4)

Fixed points are $A(-2, 0)$ and $B(4, 0)$.

Let line AB passes through a fixed point $P(h, k)$.

Equation of AB is $y = 0$

$P(h, k)$ lie on $y = 0$

$\therefore$ $k = 0$

Tangents at A and B intersects on polar.

$\therefore$ Equation of polar $T = 0$

$$x \cdot h + y \cdot k - (x + h) - a(y + k) - 8 = 0$$

Put $k = 0$

$\therefore$ $(h - 1)x - ay - (h + 8) = 0$ ...(i)

Equation of polar

$$x + 2y + 5 = 0 \quad \text{...(ii)}$$

These lines are coincident lines

$$\frac{h-1}{1} = \frac{-a}{2} = \frac{-(h+8)}{5}$$

$$5h - 5 = -h - 8$$

$$6h = -3 \qquad \frac{-a}{2} = -\frac{3}{2}$$

$$h = -\frac{1}{2} \qquad a = 3$$

$\therefore$ Equation of C_1

$$x^2 + y^2 - 2x - 6y - 8 = 0$$

3. If the chord joining the fixed points subtends an angle θ at the centre of the circle C_1 then θ equals

(1) 30° (2) 45° (3) 60° (4) 90°

Sol. Answer (4)

$A(-2, 0)$ and $B(4, 0)$

Centre $C_1(1, 3)$

Slope of $AC_1 = \dfrac{3-0}{1+2} = 1$

Slope of $BC_1 = \dfrac{3-0}{1-4} = \dfrac{3}{-3} = -1$

$\therefore$ Fixed points AB subtends 90° at the centre.

Comprehension-III

Let QR be the chord of contact of tangents drawn from $P(2, 0)$ to the circle $x^2 + y^2 = 1$. Let S be the orthocentre of the ΔPQR

1. S lies

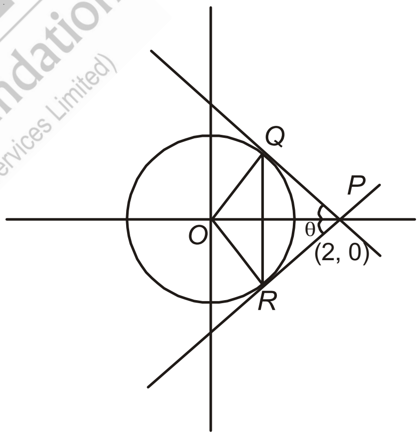
(1) Inside the circle (2) On the circle

(3) Outside the circle (4) Cannot be determined with given data

Sol. Answer (2)

Chord of contact of tangents from (2, 0) to $x^2 + y^2 = 1$ is $2x + 0 = 1$

$\Rightarrow \boxed{x = \frac{1}{2}} \rightarrow$ Equation of QR



$Q:\left(\dfrac{1}{2}, \dfrac{\sqrt{3}}{2}\right)$, $R:\left(\dfrac{1}{2}, -\dfrac{\sqrt{3}}{2}\right)$

Slope of $PR = \dfrac{\frac{\sqrt{3}}{2}}{\frac{3}{2}} = \dfrac{1}{\sqrt{3}}$

$\Rightarrow \theta = \dfrac{\pi}{6}$ (from figure)

$\therefore \angle QPR = \dfrac{\pi}{3}$

$\Rightarrow \Delta PQR$ is equilateral

$\therefore$ Orthocenter = Centroid

$= \left(\dfrac{\frac{1}{2}+\frac{1}{2}+2}{3}, \dfrac{\frac{\sqrt{3}}{2}-\frac{\sqrt{3}}{2}+0}{3}\right)$

$= (1, 0)$

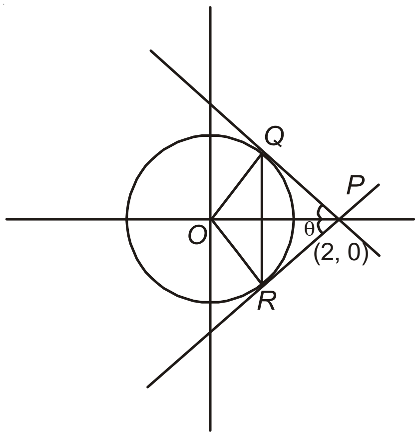
$\Rightarrow$ Lies on circle

2. The orthocentre of the ΔPQS is

(1) Origin (2) R (3) Inside the ΔPQS (4) Inside the circle

Sol. Answer (2)

Chord of contact of tangents from (2, 0) to $x^2 + y^2 = 1$ is $2x + 0 = 1$



$\Rightarrow \boxed{x = \dfrac{1}{2}} \rightarrow$ Equation of QR

$Q:\left(\dfrac{1}{2}, \dfrac{\sqrt{3}}{2}\right),\ R:\left(\dfrac{1}{2}, -\dfrac{\sqrt{3}}{2}\right)$

Slope of $PR = \dfrac{\frac{\sqrt{3}}{2}}{\frac{3}{2}} = \dfrac{1}{\sqrt{3}}$

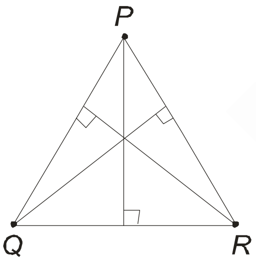
$\Rightarrow \theta = \dfrac{\pi}{6}$ (from figure)

$\therefore \angle QPR = \dfrac{\pi}{3}$

$\Rightarrow \Delta PQR$ is equilateral

In a ΔPQR if S is orthocenter then orthocenter of Δ formed by any two vertices of ΔPQR and S is the third vertex

$\Rightarrow$ Orthocenter of ΔPQS is R

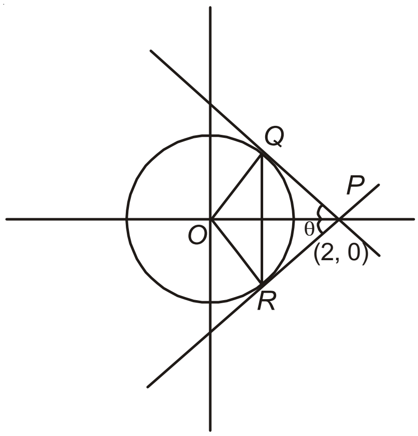


3. The angle subtended by the chord QR at the centre of the circle is

(1) $\dfrac{\pi}{6}$ (2) $\dfrac{\pi}{3}$ (3) $\dfrac{\pi}{2}$ (4) $\dfrac{2\pi}{3}$

Sol. Answer (4)

Chord of contact of tangents from (2, 0) to $x^2 + y^2 = 1$ is $2x + 0 = 1$



$\Rightarrow \boxed{x = \dfrac{1}{2}} \rightarrow$ Equation of QR

$Q:\left(\dfrac{1}{2}, \dfrac{\sqrt{3}}{2}\right),\ R:\left(\dfrac{1}{2}, -\dfrac{\sqrt{3}}{2}\right)$

Slope of $PR = \dfrac{\frac{\sqrt{3}}{2}}{\frac{3}{2}} = \dfrac{1}{\sqrt{3}}$

$\Rightarrow \quad \theta = \dfrac{\pi}{6}$ (from figure)

$\therefore \quad \angle QPR = \dfrac{\pi}{3}$

$\Rightarrow \quad \Delta PQR$ is equilateral

Angle subtended by chord QR is $\angle QOR = \pi - \angle QPR$

$$= \pi - \frac{\pi}{3} = \frac{2\pi}{3}$$

Comprehension-IV

Let the circle $x^2 + y^2 + 2ax + 2by + c = 0$ is orthogonal to each of the circles $x^2 + y^2 + 4x + 2y + 1 = 0$, $2x^2 + 2y^2 + 8x + 6y - 3 = 0$, $x^2 + y^2 + 6x - 2y - 3 = 0$

1. The value of $\dfrac{|abc|}{7}$ is

(1) 85 (2) 83 (3) 81 (4) None of these

Sol. Answer (1)

Radical centre of the 3 circle is the centres of the circle

$\therefore$ Radical axis of $x^2 + y^2 + 4x + 2y + 1 = 0$ and $x^2 + y^2 + 6x - 2y - 3 = 0$ is $2x - 4y - 4 = 0$

$\Rightarrow \quad x - 2y - 2 = 0 \quad \ldots (1)$

Radical axis of $x^2 + y^2 + 4x + 2y + 1 = 0$ and $x^2 + y^2 + 4x + 3y - \dfrac{3}{2} = 0$ is

$\Rightarrow \quad y - \dfrac{5}{2} = 0 \quad \ldots (2)$

From (1) and (2),

Radical centre is $\left(7, \dfrac{5}{2}\right)$

Radius $= \sqrt{49 + \dfrac{25}{4} + 28 + 5 + 1} = \sqrt{\dfrac{357}{4}}$

Circle is $(x-7)^2 + \left(y - \dfrac{5}{2}\right)^2 = \dfrac{357}{4}$

$x^2 + y^2 - 14x - 5y - 34 = 0$

$a = -7, \; b = -\dfrac{5}{2}, \; c = -34$

$$\frac{|abc|}{7} = \frac{7 \times \frac{5}{2} \times 34}{7}$$

$= 85$

2. The centre of the circle

(1) Is $\left(7, \frac{5}{2}\right)$ (2) Lies on $x - 2y + 2 = 0$

(3) Is (7, 5) (4) Lies on $3x - 2y - 16 = 0$

Sol. Answer (1, 4)

Radical centre of the 3 circle is the centres of the circle

$\therefore$ Radical axis of $x^2 + y^2 + 4x + 2y + 1 = 0$ and $x^2 + y^2 + 6x - 2y - 3 = 0$ is $2x - 4y - 4 = 0$

$\Rightarrow$ $x - 2y - 2 = 0$ ... (1)

Radical axis of $x^2 + y^2 + 4x + 2y + 1 = 0$ and $x^2 + y^2 + 4x + 3y - \frac{3}{2} = 0$ is

$\Rightarrow$ $y - \frac{5}{2} = 0$ ... (2)

From (1) and (2),

Radical centre is $\left(7, \frac{5}{2}\right)$

Radius $= \sqrt{49 + \frac{25}{4} + 28 + 5 + 1}$

$= \sqrt{\frac{357}{4}}$

Circle is $(x - 7)^2 + \left(y - \frac{5}{2}\right)^2 = \frac{357}{4}$

$x^2 + y^2 - 14x - 5y - 34 = 0$

Center is $\left(7, \frac{5}{2}\right)$ it lies on $3x - 2y - 16 = 0$

$21 - 5 - 16 = 0$

3. Line $3x - 4y + 11 = 0$ is

(1) A tangent to the circle (2) A chord of the circle

(3) A diameter of the circle (4) Is no contact with the circle

Sol. Answer (3)

Radical centre of the 3 circle is the centres of the circle

$\therefore$ Radical axis of $x^2 + y^2 + 4x + 2y + 1 = 0$ and $x^2 + y^2 + 6x - 2y - 3 = 0$ is $2x - 4y - 4 = 0$

$\Rightarrow$ $x - 2y - 2 = 0$ ... (1)

Radical axis of $x^2 + y^2 + 4x + 2y + 1 = 0$ and $x^2 + y^2 + 4x + 3y - \frac{3}{2} = 0$ is

$\Rightarrow$ $y - \frac{5}{2} = 0$ ... (2)

From (1) and (2), radical centre is $\left(7, \frac{5}{2}\right)$

Radius $= \sqrt{49+\frac{25}{4}+28+5+1} = \sqrt{\frac{357}{4}}$

Circle is $(x-7)^2 + \left(y-\frac{5}{2}\right)^2 = \frac{357}{4}$

$x^2 + y^2 - 14x - 5y - 34 = 0$

Line $3x - 4y - 11 = 0$

Length of perpendicular from centre to the line $= \left|\frac{21-10-11}{5}\right| = 0$

$\therefore$ Line is a diameter

Comprehension-V

L_1 is a line intersecting x and y-axis at $P(a, 0)$ and $Q(0, b)$. L_2 is a line perpendicular to L_1 intersecting x and y-axis at R and S respectively.

1. Locus of the point of intersection of the lines PS and QR is a circle passing through

(1) (0, 0) (2) (a, a) (3) (b, b) (4) (b, a)

Sol. Answer (1)

$P(a, 0)$, $Q(0, b)$

Slope of $PQ = -\frac{b}{a}$ $\therefore$ Slope of $L_2 = \frac{a}{b}$

Equation of L_2, $y = \frac{a}{b}x + \lambda$ $\therefore$ $R\left(\frac{-b\lambda}{a}, 0\right)$ and $S(0, \lambda)$

Equation of PS, $\frac{x}{a}+\frac{y}{\lambda}=1$

$\frac{y}{\lambda} = 1 - \frac{x}{a}$ ...(i)

Equation of QR, $\frac{x \cdot a}{-b\lambda}+\frac{y}{b}=1$

$\frac{xa}{b\lambda} = \frac{y}{b} - 1$ ...(ii)

Eliminate λ

$\therefore$ $\frac{yb}{xa} = \frac{(a-x)b}{(y-b)a}$

$y(y-b) = x(a-x)$

$x^2 + y^2 - ax - by = 0$

Circle passes through (0, 0) and (a, b).

2. Common chord of the circles on QS and PR as diameters passes through the point (a, b) if

(1) $a = 2b$ (2) $2a = b$ (3) $a = b$ (4) $2a = -b$

Sol. Answer (3)

$Q(0, b)$, $S(0, \lambda)$

Equation of circle $x \cdot x + (y - b)(y - \lambda) = 0$

$$x^2 + y^2 - (b + \lambda)y + b\lambda = 0 \quad \ldots\text{(i)}$$

$$P(a, 0),\ R\left(-\frac{b\lambda}{a}, 0\right)$$

$$(x - a)\left(x + \frac{b\lambda}{a}\right) + y^2 = 0$$

$$x^2 + y^2 + \left(\frac{b\lambda}{a} - a\right)x - b\lambda = 0 \quad \ldots\text{(ii)}$$

Equation of common chord $S_1 - S_2 = 0$

$$-(b + \lambda)y - \left(\frac{b\lambda}{a} - a\right)x + 2b\lambda = 0$$

$$\lambda\left(2b - y - \frac{b}{a}x\right) + (ax - by) = 0$$

Common chord passes through (a, b).

$$\lambda(2b - b - b) + (a^2 - b^2) = 0$$

$$a^2 = b^2$$

$$a = b$$

3. If the area of ΔORS in 4 times the area of ΔOPQ then equation of PS is, where O is the origin

(1) $x + 2y = 2b$ (2) $2x + y = 2a$ (3) $x - 2y = -2b$ (4) $2x - y + 2a = 0$

Sol. Answer (2)

$\Delta ORS = 4\ \Delta OPQ$

$$\frac{1}{2}\left(\frac{b\lambda}{a}\right)\lambda = 4 \cdot \frac{1}{2}a \cdot b$$

$$\lambda^2 = 4a^2 \Rightarrow \lambda = 2a$$

$$P(a, 0),\ S(0, \lambda)$$

$$P(a, 0),\ S(0, 2a)$$

Equation of PS, $\dfrac{x}{a} + \dfrac{y}{2a} = 1$

$$2x + y = 2a$$

Comprehension-VI

A circle C of radius 1 is inscribed in an equilateral triangle PQR. The points of contact of C with the sides PQ, QR, RP are D, E, F respectively. The line PQ is given by the equation $\sqrt{3x} + y - 6 = 0$ and the point D is $\left(\frac{3\sqrt{3}}{2}, \frac{3}{2}\right)$. Further, it is given that the origin and the centre of C are on the same side of the line PQ.

[IIT-JEE 2008]

1. The equation of circle C is

(1) $(x-2\sqrt{3})^2 + (y-1)^2 = 1$ (2) $(x-2\sqrt{3})^2 + \left(y+\frac{1}{2}\right)^2 = 1$

(3) $(x-\sqrt{3})^2 + (y+1)^2 = 1$ (4) $(x-\sqrt{3})^2 + (y-1)^2 = 1$

Sol. Answer (4)

Let centre of C be (h, k)

Then,

$$\left|\frac{\sqrt{3}h + k - 6}{\sqrt{3+1}}\right| = 1$$

$\Rightarrow \quad \sqrt{3}h + k - 6 = 2, -2$

$\Rightarrow \quad \sqrt{3}h + k = 4$ ...(i)

(Rejecting '2' because origin and centre of C are on the same side of PQ)

The point $(\sqrt{3}, 1)$ satisfies equation (i)

$\therefore$ Equation of circle C is $(x-\sqrt{3})^2 + (y-1)^2 = 1$.

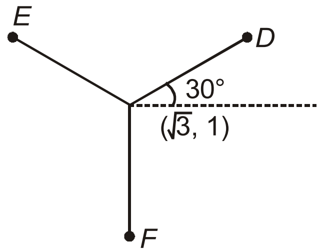
2. Points E and F are given by

(1) $\left(\frac{\sqrt{3}}{2}, \frac{3}{2}\right), (\sqrt{3}, 0)$ (2) $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right), (\sqrt{3}, 0)$ (3) $\left(\frac{\sqrt{3}}{2}, \frac{3}{2}\right), \left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$ (4) $\left(\frac{3}{2}, \frac{\sqrt{3}}{2}\right), \left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$

Sol. Answer (1)

Slope of line joining centre of circle to point D is $\dfrac{\frac{3}{2} - 1}{\frac{3\sqrt{3}}{2} - \sqrt{3}} = \dfrac{1}{\sqrt{3}}$ (makes an angle 30° with x-axis)

$\therefore$ Points E and F will make angle 150° and – 90° with x-axis.



$\therefore$ E and F are given by

$$\frac{x-\sqrt{3}}{\cos 150^\circ} = \frac{y-1}{\sin 150^\circ} = 1, \text{ and } \frac{x-\sqrt{3}}{\cos(-90^\circ)} = \frac{y-1}{\sin(-90^\circ)} = 1$$

$\therefore \quad E = \left(\frac{\sqrt{3}}{2}, \frac{3}{2}\right)$

$F = (\sqrt{3}, 0)$

3. Equations of the sides QR, RP are

(1) $y = \frac{2}{\sqrt{3}}x + 1, y = -\frac{2}{\sqrt{3}}x - 1$ (2) $y = \frac{1}{\sqrt{3}}x, y = 0$

(3) $y = \frac{\sqrt{3}}{2}x + 1, y = -\frac{\sqrt{3}}{2}x - 1$ (4) $y = \sqrt{3}x, y = 0$

Sol. Answer (4)

Clearly, point E and F satisfy the equations given in option (4).

Comprehension-VII

A tangent PT is drawn to the circle $x^2 + y^2 = 4$ at the point $P(\sqrt{3},1)$. A straight line L, perpendicular to PT is a tangent to the circle $(x - 3)^2 + y^2 = 1$. **[IIT-JEE 2012]**

1. A common tangent of the two circles is

(1) $x = 4$ (2) $y = 2$ (3) $x + \sqrt{3}y = 4$ (4) $x + 2\sqrt{2}y = 6$

Sol. Answer (4)

Let m be slope of the common tangent

$y = mx \pm 2\sqrt{1+m^2}$ ... (i)

$y = m(x-3) \pm \sqrt{1+m^2}$ ... (ii)

(i) and (ii) are identical

$$\frac{1}{1} = \frac{m}{m} = \frac{\pm 2\sqrt{1+m^2}}{-3m \pm \sqrt{1+m^2}}$$

$$-3m \pm \sqrt{1+m^2} = \pm 2\sqrt{1+m^2}$$

$\Rightarrow \quad m = \pm\frac{1}{2\sqrt{2}}$ or one tangent is vertical.

$\therefore$ Equation of common tangent is $x = 2$

For $m = -\frac{1}{2\sqrt{2}}$

$$y = \frac{-1}{2\sqrt{2}}x \pm 2\sqrt{1+\frac{1}{8}}.$$

$$2\sqrt{2}y + x = 6$$

Option (4) is correct

2. A possible equation of L is

(1) $x - \sqrt{3}y = 1$ (2) $x + \sqrt{3}y = 1$ (3) $x - \sqrt{3}y = -1$ (4) $x + \sqrt{3}y = 5$

Sol. Answer (1)

Equation of tangent at $P(\sqrt{3}, 1)$ is $\sqrt{3}x + y - 4 = 0$

Equation of L is $y = \dfrac{1}{\sqrt{3}}(x-3) \pm 1\sqrt{1+\dfrac{1}{3}}$

$\sqrt{3}y = +(x-3) \pm 2$

$\sqrt{3}y - x = -5$ or $\sqrt{3}y - x = -1$

Option (1) is correct

Comprehension-VIII

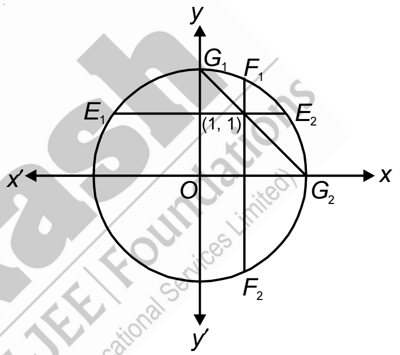
Let S be the circle in the xy-plane defined by the equation $x^2 + y^2 = 4$. **[JEE(Advanced)-2018]**

1. Let E_1E_2 and F_1F_2 be the chords of S passing through the point $P_0(1, 1)$ and parallel to the x-axis and the y-axis, respectively. Let G_1G_2 be the chord of S passing through P_0 and having slope –1. Let the tangents to S at E_1 and E_2 meet at E_3, the tangents to S at F_1 and F_2 meet at F_3, and the tangents to S at G_1 and G_2 meet at G_3. Then, the points E_3, F_3, and G_3 lie on the curve

(1) $x + y = 4$ (2) $(x-4)^2 + (y-4)^2 = 16$

(3) $(x-4)(y-4) = 4$ (4) $xy = 4$

Sol. Answer (1)



Equation of chord G_1G_2

$(y-1) = (-1)(x-1)$

$y - 1 = -x + 1$

$y + x = 2$

$G_1(0, 2)$ $G_2(2, 0)$

$E_1\left(-\sqrt{3}, 1\right)$ $E_2\left(\sqrt{3}, 1\right)$

$F_1\left(1, \sqrt{3}\right)$ $F_2\left(1, -\sqrt{3}\right)$

Intersection point of tangents at point F_1 and F_2 lies on x-axis and point F_3 is (4, 0). Intersection point of tangents at point E_1 and E_2 lies on y-axis and point E_3 is (0, 4).

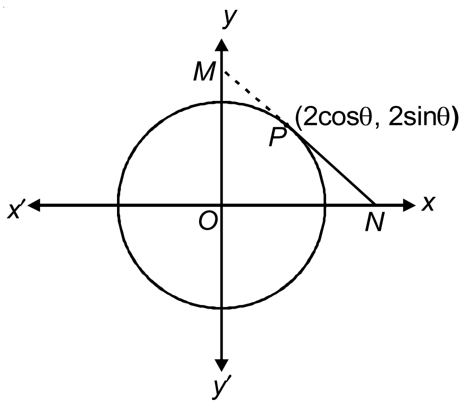
Intersection point of tangents at point G_1 and G_2 is $G_3(2, 2)$. Equation of curve which passes through (4, 0), (0, 4) and (2, 2) is $x + y = 4$.

2. Let P be a point on the circle S with both coordinates being positive. Let the tangent to S at P intersect the coordinate axes at the points M and N. Then, the mid-point of the line segment MN must lie on the curve

(1) $(x+y)^2 = 3xy$ (2) $x^{2/3} + y^{2/3} = 2^{4/3}$

(3) $x^2 + y^2 = 2xy$ (4) $x^2 + y^2 = x^2y^2$

Sol. Answer (4)



Let point $P(2\cos\theta, 2\sin\theta)$

Equation of tangent is $x\cos\theta + y\sin\theta = 2$

$N = \left(\dfrac{2}{\cos\theta}, 0\right)$

$M = \left(0, \dfrac{2}{\sin\theta}\right)$

Locus of mid-point is $\dfrac{1}{x^2} + \dfrac{1}{y^2} = 1$

i.e., $x^2 + y^2 = x^2y^2$

SECTION - D

Assertion-Reason Type Questions

1. STATEMENT-1 : The equation of chord of circle $x^2 + y^2 - 6x + 10y - 9 = 0$, which is bisected at (–2, 4) must be $x + y = 2$.

   and

   STATEMENT-2 : The equation of chord with mid-point (x_1, y_1) to the circle $x^2 + y^2 = r^2$ is $xx_1 + yy_1 = x_1^2 + y^2$.

Sol. Answer (4)

Equation of chord bisected at (–2, 4).

$T = S_1$

$$x \cdot (-2) + y \cdot 4 - 3(x-2) + 5(y+4) - 9 = 4 + 16 + 12 + 40 - 9$$

$$-5x + 9y - 46 = 0$$

$\therefore$ Statement-1 is false.

2. STATEMENT-1 : If two circles $x^2 + y^2 + 2gx + 2fy = 0$ and $x^2 + y^2 + 2g'x + 2f'y = 0$ touch each other then $f'g = fg'$.

   and

   STATEMENT-2 : Two circles touch each other if distance between their centres is equal to difference of their radii.

Sol. Answer (1)

$C_1(-g, -f)$, $C_2(-g', -f')$

$$r_1 = \sqrt{g^2 + f^2},\ r_2 = \sqrt{g'^2 + f'^2}$$

Circles touch each other

$$C_1C_2 = r_1 + r_2$$

$$\sqrt{(g-g')^2 + (f-f')^2} = \sqrt{g^2 + f^2} + \sqrt{g'^2 + f'^2}$$

Squaring both sides, we get

$$-2gg' - 2ff' = 2\sqrt{g^2 + f^2} \cdot \sqrt{g'^2 + f'^2}$$

Again squaring both sides, we get

$$2gg'ff' = g^2f'^2 + f^2g'^2$$

$$(gf' - fg')^2 = 0$$

$$gf' = fg'$$

$\therefore$ Statement-2 is correct explanation for statement-1.

3. STATEMENT-1 : If a line $L = 0$ is a tangent to the circle $S = 0$ then it will also be a tangent to the circle $S + \lambda L = 0$.

   and

   STATEMENT-2 : If a line touches a circle then perpendicular distance from the centre of the circle to the line must be equal to the radius.

Sol. Answer (2)

Equation of a circle passes through the intersection of a line

$L = 0$ and circle $S = 0$ be $S + \lambda L = 0$.

The tangent of this circle is also a line $L = 0$.

4. STATEMENT-1 : The farthest point on the circle $x^2 + y^2 - 2x - 4y + 4 = 0$ from (0, 0) is (1, 3).

and

STATEMENT-2 : The farthest and nearest points on a circle from a given point are the end points of the diameter through the point.

Sol. Answer (4)

Centre of circle (1, 2) and $r = \sqrt{1+4-4} = 1$

Farthest point on the circle from origin lie on a line joining O(0, 0) centre C(1, 2), i.e., $y = 2x$

But (1, 3) does not lie on $y = 2x$.

Statement-1 is false

5. STATEMENT-1 : The angle between the tangents drawn from the point (6, 8) to the circle $x^2 + y^2 = 50$ is 90°.

and

STATEMENT-2 : The locus of point of intersection of perpendicular tangents to the circle $x^2 + y^2 = r^2$ is $x^2 + y^2 = 2r^2$.

Sol. Answer (1)

Equation of director circle $x^2 + y^2 = 2r^2$

$x^2 + y^2 = 100$ (6, 8) lie on this circle.

$\therefore$ Statement-1 is true and statement-2 is a correct explanation for statement-1.

6. STATEMENT-1 : Let $x^2 + y^2 = a^2$ and $x^2 + y^2 - 6x - 8y - 11 = 0$ be two circles. If $a = 5$ then two common tangents are possible.

and

STATEMENT-2 : If two circles are intersecting then they have two common tangents.

Sol. Answer (1)

$C_1(0,\ 0),\ r_1 = a$

$C_2(3,\ 4),\ r_2 = 6$

Condition of intersecting circles

$r_1 \sim r_2 < c_1c_2 < r_1 + r_2$

$|a-b| < 5 < a+6 \quad \Rightarrow \quad -1 < a < 11$

$\therefore$ if $a = 5$ two common tangents are possible.

7. STATEMENT-1 : The shortest distance of the point (1, 1) form the circle $x^2 + y^2 - 4x - 6y + 4 = 0$ is $3 - \sqrt{5}$.

and

STATEMENT-2 : Shortest distance of a point = distance of the point from the centre – radius of the circle.

Sol. Answer (3)

Since point lies inside the circle.

8. STATEMENT-1 : Number of circle passing through (0, 3), (–1, 2) and (2, 5) is 1.

and

STATEMENT-2 : Through three non-collinear points, an unique circle can be drawn.

Sol. Answer (4)

Given points are collinear

9. STATEMENT-1 : If n circles ($n \geq 3$), no two circles are con-centric and no three centre are collinear and number of radical centre is equal to number of radical axes, then $n = 5$.

and

STATEMENT-2 : If no three centres are collinear and no two circles are concentric, then number of radical centre is ${}^nC_3$ and number of radical axes is ${}^nC_2$.

Sol. Answer (1)

Statement-2 is true

Now ${}^nC_3 = {}^nC_2$

$n = 3 + 2 = 5$

10. STATEMENT-1 : Area of an equilateral triangle inscribed in the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is $\frac{3\sqrt{3}}{4}(g^2 + f^2 - c)$.

and

STATEMENT-2 : Area of an equilateral triangle is $\frac{\sqrt{3}}{4}a^2$.

Sol. Answer (2)

Side of triangle = $2R \cos 30^\circ = \sqrt{3}\left(\sqrt{g^2 + f^2 - c}\right)$

11. STATEMENT-1 : If O is the origin and OP and OQ are tangents to the circle $x^2 + y^2 + 2x + 4y + 1 = 0$, the circumcentre of the triangle is $\left(\frac{-1}{2}, -1\right)$.

and

STATEMENT-2 : $OP.OQ = PQ^2$.

Sol. Answer (3)

Centre of circumcircle is mid point of O and centre of circle $x^2 + y^2 + 2x + 4y + 1 = 0$ i.e. $\left(-\frac{1}{2}, -1\right)$

12. Let C_1 and C_2 be two given circles and C be a moving circle which touches both the circles.

STATEMENT-1 : The locus of centre of circle C must be an ellipse.

and

STATEMENT-2 : The locus of a moving point whose sum of distances from two given points is always constant, is called an ellipse.

Sol. Answer (4)

Let centres of circles C, C_1 and C_2 are C, C_1 and C_2 and radius r, r_1 and r_2 respectively

$C_1C = r + r_1$ ... (i)

$C_2C = r + r_2$ ... (ii)

r is variable

(i) – (ii)

$C_1C - C_2C = r_1 - r_2$

Locus of C is a hyperbola.

$\therefore$ Statement-1 is false, statement-2 is true.

13. Let $S_1 : x^2 + y^2 = 25$ and $S_2 : x^2 + y^2 - 2x - 2y - 14 = 0$ be two circles.

STATEMENT-1 : S_1 and S_2 have exactly two common tangents.

and

STATEMENT-2 : If two circles touches each other internally they have one common tangent.

Sol. Answer (2)

$C_1(0, 0)$ $r_1 = 5$

$C_2(1, 1)$ $r_2 = \sqrt{1+1+14} = 4$

$C_1C_2 = \sqrt{1+1} = \sqrt{2}$

$r_1 + r_2 = 9$

$r_1 - r_2 = 1$

$r_1 - r_2 < C_1 C_2 < r_1 + r_2$

$\therefore$ Two common tangents are possible

Statement-1 is true, statement-2 is true but statement-2 is not a correct explanation of statement-1.

14. Tangents are drawn from the point (17, 7) to the circle $x^2 + y^2 = 169$ **[IIT-JEE 2007]**

STATEMENT-1 : The tangents are mutually perpendicular.

and

STATEMENT-2 : The locus of the points from which mutually perpendicular tangents can be drawn to the given circle is $x^2 + y^2 = 338$

Sol. Answer (1)

The locus of points from which mutually perpendicular tangents can be drawn to the circle $x^2 + y^2 = a^2$ is given by $x^2 + y^2 = 2a^2$

15. Consider **[IIT-JEE 2008]**

$L_1 : 2x + 3y + p - 3 = 0$

$L_2 : 2x + 3y + p + 3 = 0$

where p is a real number, and $C : x^2 + y^2 + 6x - 10y + 30 = 0$.

STATEMENT-1 : If line L_1 is a chord of circle C, then line L_2 is not always a diameter of circle C.

and

STATEMENT-2 : If line L_1 is a diameter of circle C, then line L_2 is not a chord of circle C.

Sol. Answer (3)

$$(x+3)^2 + (y-5)^2 = 9 + 25 - 30 = 4$$

$$(x+3)^2 + (y-5)^2 = 2^2$$

centre = (3, –5)

If L_1 is diameter, then, $2(3) + 3(-5) + p - 3 = 0$

$p = 12$

$\therefore$ L_1 is $2x + 3y + 9 = 0$

L_2 is $2x + 3y + 15 = 0$

Distance of centre of circle from L_2 equals

$$\left|\frac{2(3)+3\,(-5)+15}{\sqrt{2^2+3^2}}\right| = \frac{6}{\sqrt{13}} < 2 \text{ (radius of circle)}$$

$\therefore$ L_2 is a chord of circle C.

Statement (2), false.

16. STATEMENT-1 : A circle of smallest radius passing through A and B must be of radius $\frac{1}{2}$ AB.

and

STATEMENT-2 : A straight line is a shortest distance between two points.

Sol. Answer (2)

A circle of smallest radius passing through A and B.

Diameter = AB

$2r = AB$

$r = \frac{1}{2} \cdot AB$

$\therefore$ Statement-2 is not a correct explanation for statement-1.

SECTION - E

Matrix-Match Type Questions

1. If $S \equiv x^2 + y^2 + x - y - 2 = 0$, then

Column-I		**Column-II**
(A) (–2, 1) lies	(p)	On S
(B) (2, –1) lies	(q)	Outside S
(C) (0, 1) lies	(r)	On the tangent at (1, 0) to S
(D) (2, 3) lies	(s)	Inside the circle S

Sol. Answer : A(p), B(q), C(s), D(q, r)

(A) (–2, 1), $S_1 = 0$ lie on the circle

(B) (2, –1), $S_1 = 6$ lie outside the circle

(C) (0, 1), $S_1 = -2$ lie inside the circle

(D) Equation of tangent at (1, 0)

$$3x - y - 3 = 0$$

(2, 3) lie on this tangent.

2. $x^2 + y^2 - 14x - 10y + 24 = 0$, makes an

Column-I	Column-II
(A) Intercept on x-axis of length	(p) 0
(B) Intercept on y-axis of length	(q) 2
(C) Intercept on $y = x$ of length	(r) $8\sqrt{3}$
(D) Intercept on $7x + y - 4 = 0$ of length	(s) 10

Sol. Answer : A(s), B(q), C(r), D(p)

$C\,(7, 5),\ r = \sqrt{49+25-24} = 5\sqrt{2}$

(A) Intercept on x-axis = $2\sqrt{g^2 - C} = 2\sqrt{49-24} = 10$

(B) Intercept on y-axis = $2\sqrt{f^2 - C} = 2\sqrt{25-24} = 2$

(C) Put $y = x$ in the eq. of circle $x^2 - 12x + 12 = 0$

Let x_1, x_2 are the roots.

$x_1 + x_2 = 12$

$x_1 x_2 = 12$

Let $y = x$ intersect the circle at $A(x_1 y_1)$ and $B(x_2 y_2)$

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(x_2 - x_1)^2 + (x_2 - x_1)^2}$$

$$= \sqrt{(x_2 - x_1)^2} \cdot \sqrt{2} = \sqrt{2} \cdot \sqrt{(x_1 + x_2)^2 - 4x_1 x_2}$$

$$= \sqrt{2} \cdot \sqrt{144 - 48} = \sqrt{2}\sqrt{96}$$

$$= \sqrt{2} \cdot 4\sqrt{6} = 8\sqrt{3} \quad \left[\begin{array}{l} \because \ y = x \\ \therefore \ y_1 - y_1 = x_2 - x_1 \end{array}\right]$$

(D) $CM = \dfrac{49+5-4}{\sqrt{49+1}} = \sqrt{50} = 5\sqrt{2} = r$

$\therefore$ Line $7x + y - 4 = 0$ is a tangent.

$\therefore$ Intercept on this line is equal to zero.

3. If $S \equiv (x - 2)^2 + (y + 1)^2 = 1$ is a circle, then the equation of a

Column-I	Column-II
(A) Tangent to S	(p) $3x + 4y - 2 = 0$
(B) Diameter of S	(q) $x = 0$
(C) Line perpendicular to a tangent to S	(r) $y = 0$
(D) Chord of S	(s) $x - y - 2 = 0$

Sol. Answer : A(r), B(p), C(p, q, r, s), D(p, s)

$c(2, -1)$, $r = 1$

(A) Distance of $y = 0$ from $c(2, -1)$ is equal to radius. $\therefore$ $y = 0$ is a tangent.

(B) Centre $(2, -1)$ lie on $3x + 4y - 2 = 0$ $\therefore$ $3x + 4y - 2 = 0$ is a diameter.

(C) $y = 0$ is a tangent $\therefore$ Perpendicular line to this tangent is $x = 0$.

(D) Distance of line $x - y - 2 = 0$ from centre $= \dfrac{2+1-2}{\sqrt{2}} = \dfrac{1}{\sqrt{2}} < 1$ (radius)

$\therefore$ $x - y - 2 = 0$ is a chord.

4. Match column-I to column-II according to given conditions.

Column-I	Column-II
(A) Length of common tangents of $x^2 + y^2 = 1$ and $x^2 + y^2 - 10x + 21 = 0$ is	(p) $2\sqrt{6}$
(B) Length of common tangents of $x^2 + y^2 = 1$ and $x^2 + y^2 - 4x + 3 = 0$ is	(q) 4
(C) Length of common tangents of $x^2 + y^2 = 1$ and $x^2 + y^2 - 2x - 3 = 0$ is always less than	(r) 2
(D) Length of common tangents of $x^2 + y^2 = 1$ and $x^2 + y^2 - 2x = 0$ is always less than	(s) 0
	(t) 1

Sol. Answer A(p, q), B(r, s), C(p, q, r, t), D(p, q, r, t)

(A) $C_1 = (0, 0)$, $C_2 = (5, 0)$, $r_1 = 1$, $r_2 = 2$

$C_1C_2 = 5$, $r_1 + r_2 = 3$, $|r_1 - r_2| = 1$

$C_1C_2 > r_1 + r_2$, Hence there exists four common tangents.

Length of direct common tangents $= \sqrt{(C_1C_2)^2 - (r_1 - r_2)^2} = \sqrt{24} = 2\sqrt{6}$

Length of transverse common tangent $= \sqrt{(C_1C_2)^2 - (r_1 + r_2)^2} = 4$

(B) $C_1 = (0, 0)$, $C_2 = (2, 0)$, $r_1 = 1$, $r_2 = 1$, $C_1C_2 = r_1 + r_2$

Hence circles externally touches

Length of direction common tangent $= \sqrt{(C_1C_2)^2 - (r_1 - r_2)^2} = \sqrt{2}$

Length of transverse common tangent is 0

(C) $C_1 = (0, 0)$, $C_2 = (1, 0)$, $r_1 = 1$, $r_2 = 2$

$\Rightarrow C_1C_2 = |r_2 - r_1|$

Hence circles internally touches. Hence the length of direct tangent is zero.

Transverse common tangent in this case does not exist.

(D) $C_1 = (0, 0)$, $C_2 = (0, 0)$

Circles are concentric hence no common tangent exist. Hence length is zero.

5. Match column-I to column-II according to the given conditions.

Column-I	Column-II
(A) The area of triangle formed by the tangents and the chord of contact drawn from (2, 3) to the circle $x^2 + y^2 = 1$, is	(p) $\frac{12\sqrt{12}}{13}$
(B) The minimum distance of (0, 0) from the circle $x^2 + y^2 - 10x + 24 = 0$ is	(q) 4
(C) The radius of smallest circle passing through (4, 0) and (0, 4) is	(r) 6
(D) The maximum number of normals of a circle passing through the centre is always greater than	(s) $2\sqrt{2}$
	(t) 1

Sol. Answer A(p), B(q), C(s), D(p, q, r, s, t)

(A) Equation of chord of contact $\Rightarrow$ $2x + 3y = 1$

Length of chord of contact = $\sqrt{\frac{12}{13}}$

Length of perpendicular from (2, 3) to chord of contact = $\frac{12}{\sqrt{13}}$

Area of triangle = $\frac{12\sqrt{12}}{13}$

(B) $C \equiv (5, 0)$

Let $P \equiv (0, 0)$

$CP = 5$

Radius $r = 1$

Minimum distance = 5 – 1 = 4

Maximum distance = 1 + 5 = 6

Hence the distance varies from 4 to 6.

(C) Radius = $\frac{4\sqrt{2}}{2} = 2\sqrt{2}$, for smallest circle (4, 0) and (0, 4) will be the end of diameter.A(p), B(q, r), C(s), D(p, q, r, s, t)

(D) From centre infinite normals can be drawn.

6. Match the following

Column-I	Column-II
(A) For what value of k the length of intercept cuts by the circle $x^2 + y^2 + kx - 7y - 4 = 0$ on the x-axis will be $\sqrt{20}$	(p) 1
(B) Number of common tangents can be drawn to the circles $x^2 + y^2 - 4x - 6y - 3 = 0$ and $x^2 + y^2 + 2x + 2y + 1 = 0$ more than or equal to	(q) 2

(C)	If the common chord of the circles $x^2 + (y-k)^2 = 16$ and $x^2 + y^2 = 16$ subtend a right angle at the origin then value of $[k]$ is equal to	(r)	3
(D)	If the straight line $y = mx$ touches or lies outside the circle $x^2 + y^2 - 20y + 90 = 0$, then $\lvert m \rvert$ can be equal to	(s)	4
		(t)	5

Sol. Answer A(q), B(p, q, r), C(t), D(p, q, r)

(A) Length of intercepton x-axis is

$$2\sqrt{g^2 - c} = \sqrt{20} \quad \text{(given)}$$

$$2\sqrt{\frac{k^2}{4} + 4} = \sqrt{20}$$

$$\frac{k^2}{4} + 4 = \frac{20}{4}$$

$$\frac{k^2}{4} = 1$$

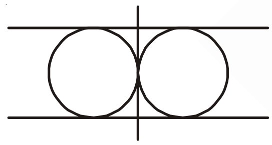
$k = \pm 2$

(B) Let $S_1 = x^2 + y^2 - 4x - 6y - 3 = 0$

and $S_2 = x^2 + y^2 + 2x + 2y + 1 = 0$

Centre and radius of S_1 are $C_1(2, 3)$ and $r_1 = 4$ and radius of S_2 are $C_2(-1, 1)$ and $r_2 = 1$

Hence $C_1C_2 = \sqrt{a + 16} = 5 = r_1 + r_2$ so total number of common tangents can be 3



(C) The common chord is

$x^2 + (y - k)^2 - 16 - \{x^2 + y^2 - 16\} = 0$

i.e., $y = \dfrac{k}{2}$

Now the pair of straight lines joining the origin to the points of intersection of $y = \dfrac{k}{2}$ and

$x^2 + y^2 = 16$ is $x^2 + y^2 = 16\left(\dfrac{2y}{k}\right)^2$ (make homogeneous)

i.e., $k^2x^2 + (k^2 - 64)y^2 = 0$

These lines are at right angles if

$k^2 + k^2 - 64 = 0$

i.e., $k = \pm 4\sqrt{2}$

$[k] = [4\sqrt{2}] = 5$

$[k] = [-4\sqrt{2}] = -6$

(D) Distance from the centre (0, 10) on the line $y = mx$

$$= \frac{10}{\sqrt{1+m^2}} \geq \text{radius} = \sqrt{100-90} = \sqrt{10}$$

$\Rightarrow$ $100 \geq 10 + 10\,m^2$

$90 \geq 10\,m^2$

$\Rightarrow$ $m^2 \geq 9$

$\Rightarrow$ $|m| \leq 3$

$\therefore$ $|m| = 0, 1, 2, 3$

7. Match the following columns

Column-I	**Column-II**
(A) A circle is inscribed in an equilateral triangle of side a, then the area of any square inscribed in the circle is	(p) $\frac{a^2}{6}$
(B) The equation of circle $x^2 + y^2 = 4a^2$ with origin as centre, passing through the vertices of an equilateral triangle whose median is of length	(q) a
(C) If a chord of the circle $x^2+y^2=\frac{9a^2}{2}$ makes equal intercept of length k on the co-ordinate axes then the value of k can be	(r) $2a$
(D) If the line $hx + ky = 1$ touches $x^2+y^2=\frac{1}{4a^2}$, then the locus of the point (h, k) is a circle of radius	(s) $3a$

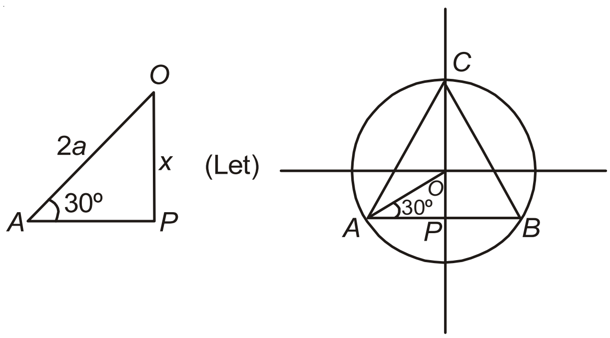
Sol. Answer A(p), B(s), C(p, q, r, s), D(r)

(A) $r = \frac{\Delta}{s} = \frac{a}{2\sqrt{3}}$

$\therefore$ Area of square inscribed $= \frac{2a^2}{12} = \frac{a^2}{6}$

(B) In ΔOAP

$$\frac{x}{2a} = \sin 30^\circ$$



$$x = 2a \times \frac{1}{2}$$

$x = a$

Length of medium $CO + OP = 2a + a$

$= 3a$

(C) Let the equation of the chord $x \pm y = \pm k$ and the length of the perpendicular from the centre (0, 0) of the circle

$x^2 + y^2 = \frac{9a^2}{2}$ must be less than the radius $\frac{3a}{\sqrt{2}}$ of the circle

$$\Rightarrow \frac{|\pm k|}{\sqrt{1+1}} \le \frac{\sqrt{a}}{\sqrt{2}}a$$

$$\frac{|k|}{\sqrt{2}} < \frac{3}{\sqrt{2}}a$$

$k < 3a$

(D) Since $hx + ky = 1$ touches $x^2 + y^2 = \frac{1}{4a^2}$

So, $\left|\frac{-1}{\sqrt{h^2+k^2}}\right| = \frac{1}{2a}$

$h^2 + k^2 = 4a^2$

So locus of (h, k) is $x^2 + y^2 = 4a^2$ which is a circle of radius $2a$

SECTION - F

Integer Answer Type Questions

1. If the locus of middle points of chords of the circle $x^2 + y^2 = 4$, which subtend a right angle at the point $(a, 0)$ is $x^2 + y^2 - ax + \frac{a^2 - p}{2} = 0$, then the value of p is _____.

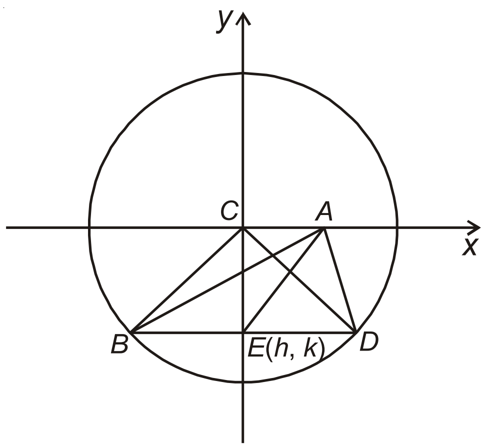
Sol. Answer (4)

Let middle point $(h, k) \equiv E$

As we know that the middle point of hypotenuse in a right angle triangle is equidistant from all vertices.

$\Rightarrow$ $AE = BE = DE = (h - a)^2 + (k - 0)^2$

Now in triangle ECD



$EC^2 + ED^2 = CD^2$

$\Rightarrow$ $(h^2 + k^2) + (h - a)^2 + k^2 = 4$

$\Rightarrow$ $h^2 + k^2 + h^2 + a^2 - 2ha + k^2 = 4$

$2(h^2 + k^2) - 2ah + a^2 - 4 = 0$

$$h^2 + k^2 - \frac{2ah}{2} + \frac{a^2 - 4}{2} = 0$$

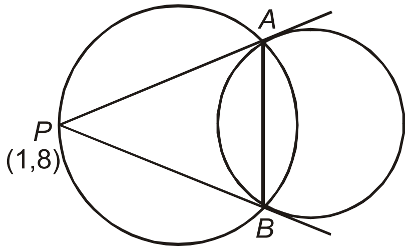
Locus of (h, k) is $x^2 + y^2 - ax + \frac{a^2 - 4}{2} = 0$

$\Rightarrow$ $p = 4$

2. Tangents are drawn from the point P (1, 8) to the circle $x^2 + y^2 - 6x - 4y - 11 = 0$ touch the circle at the points A and B. If the radius of circumcircle of triangle PAB is k then the value of $[k]$, where [ ] represents the greatest integer function.

Sol. Answer (3)

Equation of AB is given by



$$x.1 + y.8 - 6\left(\frac{x+1}{2}\right) - 4\left(\frac{y+8}{2}\right) - 11 = 0$$

$x + 8y - 3(x + 1) - 2(y + 8) - 11 = 0$

$x + 8y - 3x - 3 - 2y - 16 - 11 = 0$

$6y - 2x - 30 = 0$

$3y - x - 15 = 0$

$x - 3y + 15 = 0$

Equation of circumcircle of triangle PAB is

$x^2 + y^2 - 6x - 4y - 11 + \lambda(x - 3y + 15) = 0, \lambda \in R$ ...(ii)

Putting $x = 1$, $y = 8$

$1 + 64 - 6 - 32 - 11 + \lambda(1 - 24 + 11) = 0$

$\Rightarrow$ $16 - 8\lambda = 0$

$\Rightarrow$ $\lambda = 2$

Putting $\lambda = 2$ in (ii)

$x^2 + y^2 - 6x - 4y + 2(x - 3y + 15) - 11 = 0$

$x^2 + y^2 - 4x - 10y + 19 = 0$

Radius $k = \sqrt{4 + 25 - 19} = \sqrt{10}$ $\Rightarrow k = \sqrt{10}$

$[k] = [\sqrt{10}] = 3$

3. If the limiting point of coaxial system of circles are (2, 3) and (6, 7) and the equation of radical axis of the system is $x + y = \lambda$, then the value of λ is _____.

Sol. Answer (9)

If the limiting points of coaxial system of circles are A(2, 3) and B(6, 7) then the equation of radical axis will be the perpendicular bisector of AB.

$$M = \left(\frac{2+6}{2}, \frac{3+7}{2}\right) = (4, 5)$$

Slope $AB = \frac{7-3}{6-2} = 1$ $\Rightarrow$ Slope of radical axis = –1

Hence the equation of radical axis.

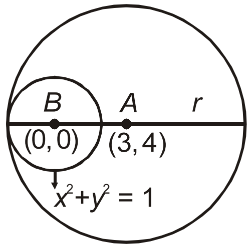
$y - 5 = -1(x - 4)$, $y + x = 9$

$\Rightarrow$ $\lambda = 9$

4. If the centre of a circle is (3, 4) and its size is just sufficient to contain to circle $x^2 + y^2 = 1$. Then the radius of the required circle is _____.

Sol. Answer (6)

In this case the figure may be shown as following.



In this case the radius of required circle is

$AB + 1 = 5 + 1 = 6$

5. Two tangents are drawn from a point P to the given circle each of length l units. These tangents touch the circle at the points A and B. Two tangents are drawn again from at point Q on the chord of contact AB, each of length m units. If m and l are connected by the relation $ml - 2 = 0$ then the minimum distance between P and Q will be

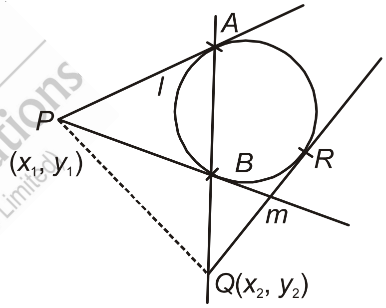
Sol. Answer (2)

Let the equation of the circle is $x^2 + y^2 = a^2$ and P is any point whose coordinates are (x_1, y_1)

Then $l = \sqrt{x_1^2 + y_1^2 - a^2}$

$\Rightarrow \quad l^2 + a^2 = x_1^2 + y_1^2$

Now, equation of chord of contact AB is $xx_1 + yy_1 = a^2$



Let $Q(x_2, y_2)$ is any point on AB then $x_1x_2 + y_1y_2 = a^2$ and $m = \sqrt{x_2^2 + y_2^2 - a^2}$

$\Rightarrow$ $m^2 + a^2 = x_2{}^2 + y_2{}^2$ if d is the distance between P and Q then

$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$

$d = \sqrt{x_1^2 + y_1^2 + x_2^2 + y_2^2 - 2(x_1x_2 + y_1y_2)}$

$= \sqrt{l^2 + a^2 + m^2 + a^2 - 2a^2}$ $\quad [\because\ x_1x_2 + y_1y_2 = a^2]$

$d = \sqrt{l^2 + m^2}$

Now we know that $\dfrac{l^2 + m^2}{2} \geq (l^2m^2)^{1/2}$

$\Rightarrow \quad \ell^2 + m^2 \geq 2\ell m$

So, $d \geq \sqrt{2\ell m}$

But given that $ml - 2 = 0 \quad \Rightarrow \quad ml = 2$

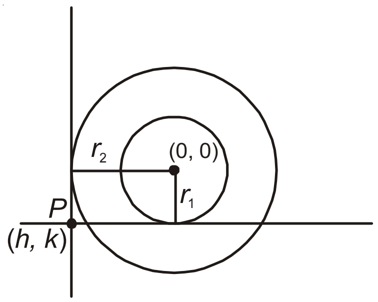
So minimum distance between P and Q will be 2 units.

6. There are two perpendicular lines, one touches to the circle $x^2 + y^2 = r_1{}^2$ and other touches to the circle $x^2 + y^2 = r_2{}^2$ if the locus of the point of intersection of these tangents is $x^2 + y^2 = 9$, then the value of $r_1{}^2 + r_2{}^2$ is.

Sol. Answer (9)

From figure $h^2 + k^2 = r_1{}^2 + r_2{}^2$

So locus of (h, k) is $x^2 + y^2 = r_1{}^2 + r_2{}^2$



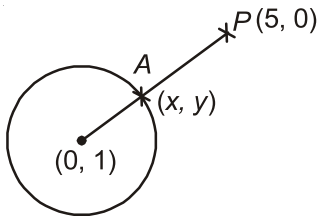
But given locus is $x^2 + y^2 = 9$

So, $r_1{}^2 + r_2{}^2 = 9$

7. If the minimum value of $r = \sqrt{(1+\sqrt{4-x^2})^2 + (x-5)^2}$ is $\sqrt{26} - 2k$, then the value of k is.

Sol. Answer (1)

Let $r = \sqrt{(1+\sqrt{4-x^2})^2 + (x-5)^2}$



Suppose $y = 1 + \sqrt{4-x^2}$

then $(y-1) = 4 - x^2$

$\Rightarrow$ $x^2 + (y-1)^2 = 4$ which is a circle

So, $r = \sqrt{y^2 + (x-5)^2}$

So it is clear r is the distance between the points (x, y) and $(5, 0)$ as shown in the figure.

Now we have to find the minimum value of PA

From figure minimum value of PA

$= \sqrt{26} - 2$

But given value = $\sqrt{26} - 2k$

So, $k = 1$

8. Suppose the equation of circle is $x^2 + y^2 - 8x - 6y + 24 = 0$ and let (p, q) is any point on the circle, Then the no. of possible integral values of $|p + q|$ is.

Sol. Answer (3)

The equation can be written as $(x-4)^2 + (y-3)^2 = 1$

and the coordinate $(p, q) \equiv (4 + \cos\theta, 3 + \sin\theta)$

So $|p + q| = |7 + \sin\theta + \cos\theta|$ it is clear $7 - \sqrt{2} \le |p+q| \le 7 + \sqrt{2}$

and hence the integral values of $|p + q| = 6, 7, 8$

and number of integral values = 3

9. Let the equation of the circle is $x^2 + y^2 = 4$. Find the total no. of points on $y = |x|$ from which perpendicular tangents can be drawn are.

Sol. Answer (2)

We know that equation of director circle is $x^2 + y^2 = 8$ and the number of points of intersection of $y = |x|$ and $x^2 + y^2 = 8$ are 2.

So total number of points from which perpendicular tangents are drawn = 2.

10. Let the equation of the circle is $x^2 + y^2 - 2x - 4y + 1 = 0$. A line through $P(\alpha, -1)$ is drawn which intersect the given circle at the point A and B. if PA. PB has the minimum value then the value of α is.

Sol. Answer (1)

We know that $PA.PB = (PT)^2$

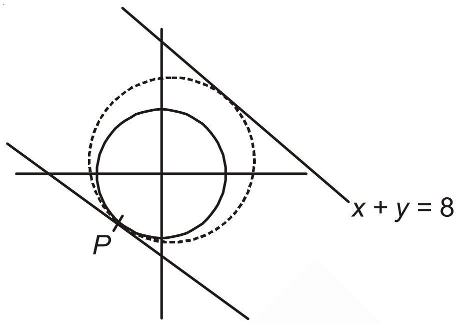
$\Rightarrow$ $PA.PB = \alpha^2 + 1 - 2\alpha + 4 + 1 = (\alpha - 1)^2 + 5$

$PA.PB$ will be minimum at $\alpha = 1$

11. Let the equation of the circle is $x^2 + y^2 = 25$ and the equation of the line $x + y = 8$. If the radius of the circle of maximum area and also touches $x + y = 8$ and $x^2 + y^2 = 25$ is $\frac{(4\sqrt{2}+5)}{\lambda}$. Then the value of λ is.

Sol. Answer (2)

The possible figure is shown in the figure it is clear the tangent at P will be $x + y + 5\sqrt{2} = 0$



So radius of the required circle will be = $\frac{4\sqrt{2}+5}{2}$

So the value of $\lambda = 2$.

12. If the circles $x^2 + y^2 + 2b_1y + 1 = 0$ and $x^2 + y^2 + 2b_2y + 1 = 0$ cuts orthogonally then the value of b_1b_2 is.

Sol. Answer (1)

We know that if two circle cuts orthogonally

then $2g_1g_2 + 2f_1f_2 = c_1 + c_2$

So, $2b_1b_2 = 1 + 1$

$b_1b_2 = 1$

13. Let the equation of circle is $x^2 + y^2 = 9$ and the equation of the line is $x + y = a \ \forall \ a \in N$ and makes the intercepts AB by the circle $x^2 + y^2 = 9$. How many such intercepted portions are possible?

Sol. Answer (4)

Since AB is the intercept so its length will be $2\sqrt{9-\frac{a^2}{2}} = \sqrt{2}\ \sqrt{18-a^2}ac$

Since $a \in N$

so $a = 1, 2, 3, 4$ the number of such intercepted portions = 4

14. The centres of two circles C_1 and C_2 each of unit radius are at a distance of 6 units from each other. Let P be the mid point of the line segment joining the centres of C_1 and C_2 and C be a circle touching circles C_1 and C_2 externally. If a common tangent to C_1 and C passing through P is also a common tangent to C_2 and C, then the radius of the circle C is **[IIT-JEE 2009]**

Sol. Answer (8)

The figure below is according to the given situation

Let $\angle SPO_2 = \theta$

$\sin\theta = \frac{1}{3}$

Now $\dfrac{PO_2}{QO_2} = \sin\theta$

$\Rightarrow \dfrac{3}{QO_2} = \dfrac{1}{3}$

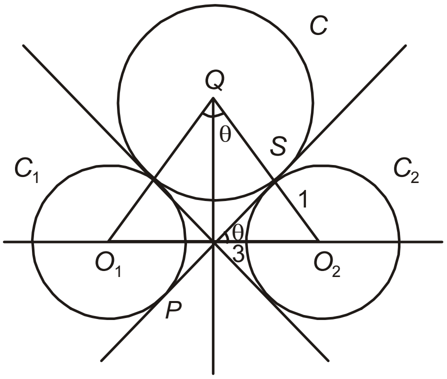
$\Rightarrow QO_2 = 9$

$QS = QO_2 - SO_2$

$= 9 - 1$

$= 8$

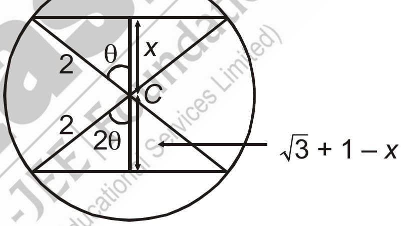
Hence the radius of circle C is 8.



15. Two parallel chords of a circle of radius 2 are at a distance $\sqrt{3}+1$ apart. If the chords subtend at the center, angles of $\dfrac{\pi}{k}$ and $\dfrac{2\pi}{k}$, where $k > 0$, then the value of $[k]$ is **[IIT-JEE 2010]**

[**Note** : $[k]$ denotes the largest integer less than or equal to k]

Sol. Answer (3)



Let $\theta = \dfrac{\pi}{2k}$

$\cos\theta = \dfrac{x}{2}$

$\Rightarrow \cos 2\theta = \dfrac{\sqrt{3}+1-x}{2}$

$\Rightarrow 2\cos^2\theta - 1 = \dfrac{\sqrt{3}+1-x}{2}$

$\Rightarrow 2\left(\dfrac{x^2}{4}\right) - 1 = \dfrac{\sqrt{3}+1-x}{2}$

$\Rightarrow x^2 + x - 3 - \sqrt{3} = 0$

$x = \dfrac{-1 \pm \sqrt{1+12+4\sqrt{3}}}{2}$

$= \dfrac{-1 \pm \sqrt{13+4\sqrt{3}}}{2}$

$= \dfrac{-1+2\sqrt{3}+1}{2} = \sqrt{3}$

$\therefore \cos\theta = \dfrac{\sqrt{3}}{2} \Rightarrow \theta = \dfrac{\pi}{6}$

$\therefore$ Required angle $= \dfrac{\pi}{k} = 2\theta = \dfrac{\pi}{3}$

$\Rightarrow k = 3$

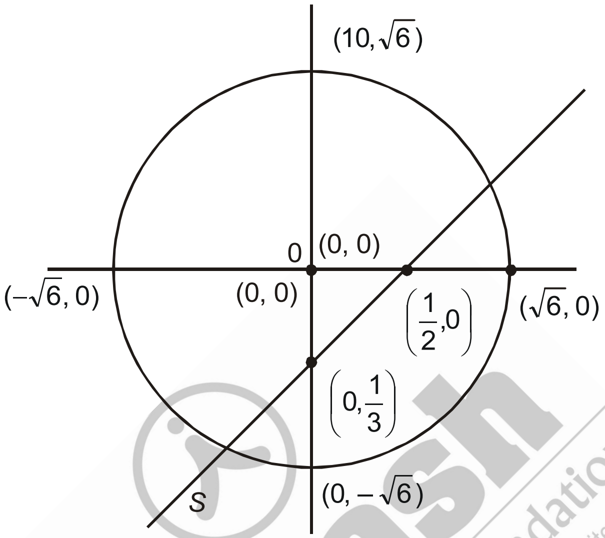
16. The straight line $2x - 3y = 1$ divides the circular region $x^2 + y^2 \le 6$ into two parts. If

$$S = \left\{\left(2, \frac{3}{4}\right), \left(\frac{5}{2}, \frac{3}{4}\right), \left(\frac{1}{4}, -\frac{1}{4}\right), \left(\frac{1}{8}, \frac{1}{4}\right)\right\},$$

then the number of point(s) in S lying inside the smaller part is **[IIT-JEE 2011]**

Sol. Answer (2)

We observe that the points $\left(2, \frac{3}{4}\right), \left(\frac{5}{2}, \frac{3}{4}\right), \left(\frac{1}{4}, -\frac{1}{4}\right)$



lie on the opposite side of origin with respect to the given line. But 5/2, 3/4 lied outside the circle. Hence there are exactly two points of the set lie in the smaller part

SECTION - G

Multiple True-False Type Questions

1. STATEMENT-1 : If two circles intersect, then the common chord of the circles is also the radical axis completely.

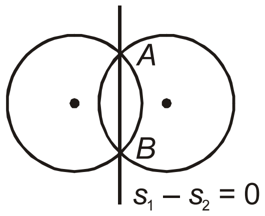
STATEMENT-2 : The radical centre of three circles drawn on three sides of a triangle by considering them as diameter is circumcentre of the triangle.

STATEMENT-3 : If $3x + 4y + 5 = 0$ and $3x + 4y + 10 = 0$ are two tangents of a circle then its radius is equal to 1.

(1) T F T (2) T T T (3) F F F (4) F F T

Sol. Answer (3)

Statement-1:



If two circles intersect then the equation of common chord end radical axis will be same but actually they are not same. Because chord AB cannot be the part of radical axis as no tangent can be drawn from AB. Hence statement-1 is false.

Statement-2:

In this radical centre will be orthocenter

Hence statement-2 is false

Statement-3:

The distance between lines = diameter

$\Rightarrow$ diameter = $\left|\dfrac{10-5}{\sqrt{9+16}}\right| = 1$

Hence radius = $\dfrac{1}{2}$

Statement-3 is also false.

2. STATEMENT-1 : The area of equilateral triangle inscribed in the circle $x^2 + y^2 + 6x + 8y + 24 = 0$ is $\dfrac{3\sqrt{3}}{4}$.

STATEMENT-2 : The abscissae of two points A and B are the roots of the equation $x^2 + 3ax + b^2 = 0$ and their ordinates are the roots of $x^2 + 3bx + a^2 = 0$. Then the equation of the circle with AB as diameter is $x^2 + y^2 + 3ax + 3by + a^2 + b^2 = 0$.

STATEMENT-3 : If the circle $x^2 + y^2 + 2gx + 2fx + c = 0$ always passes through exactly three quadrants then $c > 0$.

(1) T F T (2) T T T (3) F F F (4) F F T

Sol. Answer (2)

Statement-1:

The area of equilateral triangle inscribed in the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is

$\dfrac{3\sqrt{3}}{4}\left(g^2+f^2-c\right)$

In the given problem $g = 3$, $f = 4$, $c = 24$

Area = $\dfrac{3\sqrt{3}}{4}(9+16-24) = \dfrac{3\sqrt{3}}{4}$

Hence statement-1 is true.

Statement-2 :

Let $A \equiv (x_1, y_1)$, $B \equiv (x_2, y_2)$

Equation of the required circle

$(x - x_1)(x - x_{-2}) + (y - y_1)(y - y_2) = 0$

$(x^2 - (x_1 + x_2)x + x_1x_2) + (y^2 - (y_1 + y_2)y + y_1y_2) = 0$

$\Rightarrow (x^2 + 3ax + b^2) + (y^2 + 3by + a^2) = 0$

$x^2 + y^2 + 3ax + 3by + a^2 + b^2 = 0$

Hence statement-2 is also true

Statement-3:

In this case (0, 0) lies outside the circle

Hence 0 + 0 + 0 + 0 + 1 > 0

$c > 0$

Statement-3 is true.

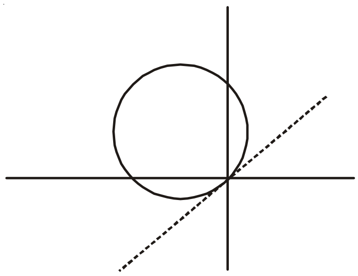
3. STATEMENT-1 : Reflexion of circle $x^2 + y^2 + 2x - 2y + 1 = 0$ in line mirror $y = x$ is $x^2 + y^2 - 2x + 2y + 1 = 0$.

STATEMENT-2 : If circle $x^2 + y^2 + 2x - 2y + 1 = 0$ is rolled along the line $y = x$ by $\sqrt{2}$ units then new circle is $x^2 + y^2 - 4y + 3 = 0$.

STATEMENT-3 : If circle $x^2 + y^2 + 2x - 2y + 1 = 0$ is rolled along the line $y = x$ by $\sqrt{2}$ units then new circle is $x^2 + y^2 + 4x + 3 = 0$.

(1) T T T (2) T T F (3) T F T (4) F F F

Sol. Answer (1)



Centre of circle is (–1, 1) and radius 1 units new positions of centre of circle is rolled by $\sqrt{2}$ units

$$\frac{x+1}{\cos 45} = \frac{y-1}{\sin 45} = \pm\sqrt{2}$$

Centre are (0, 2), (–2, 0)

Hence circles are $x^2 + y^2 - 4y + 3 = 0$

$x^2 + y^2 + 4x + 3 = 0$

4. STATEMENT-1 : If coordinates of the vertices of a triangle are rational then triangle must be equilateral.

STATEMENT-2 : If coordinate of the vertices of a triangle are rational then centroid, circumcentre, orthocentre must be rational.

STATEMENT-3 : Three rational points lying on a circle may have irrational centre.

(1) F T T (2) F T F (3) F F F (4) T T T

Sol. Answer (2)

If (x_1, y_1), (x_2, y_2), (x_3, y_3) are three vertices then

$$\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = \frac{\sqrt{3}}{4}a^2$$, where a is side of equilateral triangle

Which show contradiction for rational numbers

For (2), (3) statements, equation of line must have rational coefficients if it is bisector of line segments having rational ends.

5. STATEMENT-1 : Tangents from origin to the circle $A(x - 2)^2 + y^2 = 1$, then circle B touching the circle A and tangents have radius $\frac{1}{3}$ units.

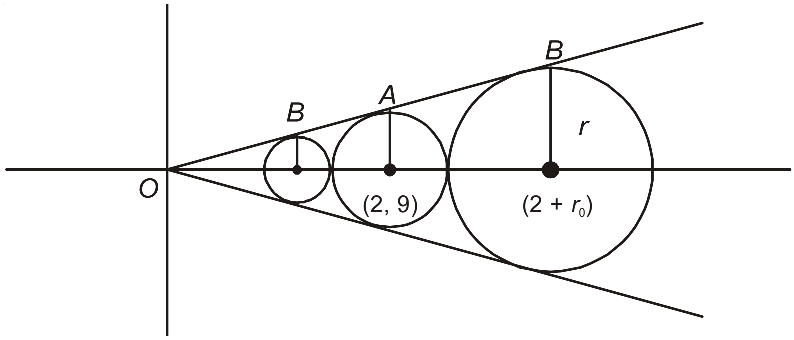
STATEMENT-2 : Circle B touching the circle A and tangents have radius 2 units.

STATEMENT-3 : Length of common tangents between circle A and B is $\sqrt{3}$.

(1) T T F (2) T F F (3) F F F (4) T T T

Sol. Answer (4)

Two circles possible



$\frac{r_2}{2+r} = \frac{1}{2}$ $r = 2$

$\frac{r}{1-r} = \frac{1}{2}$ $r = \frac{1}{3}$

$OA = \sqrt{3}$

$OB = 2\sqrt{3}$ hence $AB = \sqrt{3}$

6. STATEMENT-1 : Equation of circle passing through (1, 1), (2, 2) of least radius in first quadrant is $x^2 + y^2 - 3x - 3y + 4 = 0$.

STATEMENT-2 : Equation of circle passing through (1, 1), (2, 2) of maximum radius in first quadrant is $x^2 + y^2 - 4x - 2y + 4 = 0$.

STATEMENT-3 : Equation of circle passing through (1, 1), (2, 2) of maximum radius in first quadrant is $x^2 + y^2 - 2x - 4y + 4 = 0$.

(1) T T F (2) T F F (3) F F F (4) T T T

Sol. Answer (4)

For least radius (1, 1), (2, 2) shall be ends of a diameter, hence equation of circle is $(x-1)(x-2) +$

$(y-1)(y-2) = 0$ for maximum radius it must touch one of the coordinate axis for

$x^2 + y^2 + 2gx + 2fy + c = 0$

$1 + 1 + 2g + 2f + c = 0$

$4 + 4 + 4g + 4f + c = 0$

$g^2 = c$

Hence $c = 4$, $g = -2$, $f = -1$

Hence equation follows similarly for $f^2 = c$

7. STATEMENT-1 : Equation of circle which touches the circle $x^2 + y^2 - 6x + 6y + 17 = 0$ externally and to which the lines $x^2 - 3xy - 3x + 9y = 0$ are normal is $x^2 + y^2 - 6x - 2y + 1 = 0$.

STATEMENT-2 : Equation of circle which touches the circle $x^2 + y^2 - 6x + 6y + 17 = 0$ internally and to which the lines $x^2 - 3xy - 3x + 9y = 0$ are normal is $x^2 + y^2 - 6x - 2y - 15 = 0$.

STATEMENT-3 : Equation of circle which is orthogonal to circle $x^2 + y^2 - 6x + 6y + 17 = 0$ and have normals along $x^2 - 3xy - 3x + 9y = 0$ is $x^2 + y^2 - 6x - 2y - 5 = 0$.

(1) T T F (2) T F F (3) F F F (4) T T T

Sol. Answer (4)

Normals are

$(x - 3y)(x - 3) = 0$

Centre is (3, 1)

Radius to externally touch the circle is $r = 3$

Hence circle is $(x-3)^2 + (y-1)^2 = 9$ radius to internally touch = 5

Hence circle is $(x-3)^2 + (y-1)^2 = 25$

For orthogonal intersection

$2g_1g_2 + 2f_1f_2 = c_1 + c_2$

SECTION - H

Aakash Challengers Questions

1. If P_1, P_2, P_3 are the perimeters of the three circles.

$S_1 : x^2 + y^2 + 8x - 6y = 0$

$S_2 : 4x^2 + 4y^2 - 4x - 12y - 186 = 0$ and

$S_3 : x^2 + y^2 - 6x + 6y - 9 = 0$ respectively, then the relation amongst P_1, P_2 and P_3 is ....

Sol. $S_1 : x^2 + y^2 + 8x - 6y = 0$

$\Rightarrow \quad (x^2 + 8x + 16) + (y^2 - 6y + 9) = 16 + 9$

$\Rightarrow \quad (x + 4)^2 + (y - 3)^2 = 5^2$

$\therefore \quad P_1 = 2\pi r = 2\pi \times 5 = 10\pi$

$S_2 : 4x^2 + 4y^2 - 4x - 12y - 186 = 0$

$$\Rightarrow \quad x^2 + y^2 - 2x - 2y - 6y = \frac{93}{2}$$

$$\Rightarrow \quad (x^2 - 2x + 1) + (y^2 - 6y + 9) = \frac{93}{2} + 10$$

$$\Rightarrow \quad (x - 1)^2 + (y - 3)^2 = \frac{113}{2}$$

$$\therefore \quad P_2 = 2\pi r = 2\pi \times \frac{\sqrt{113}}{\sqrt{2}} = \pi\sqrt{226}$$

$S_3 : x^2 + y^2 - 6x + 6y - 9 = 0$

$\Rightarrow \quad (x^2 - 6x + 8) + (y^2 + 6y + 9) = 9 + 9 + 9$

$\Rightarrow \quad (x - 3)^2 + (y + 3)^2 = 27 = (3\sqrt{3})^2$

$P_3 = 2\pi r = 2\pi \times 3\sqrt{3} = 6\pi\sqrt{3}$

$\therefore \quad P_1 < P_3 < P_2$

2. A point (x, y) in the Euclidean plane is called as lattice point when x and y both are integers. If a circle of radius r is drawn with any of these lattice point as centre, then the minimum value of r for which the circle has at least one point common with a line of slope $\frac{2}{5}$ is $\frac{\sqrt{k}}{k'}$, where k, k' are integers and $k' \neq 0$, then $\frac{k'}{k}$ is equal to

(1) 1 (2) 2

(3) 3 (4) 4

Sol. Answer (2)

Let the straight line be

$2x - 5y - k = 0$

Let the lattice point be (m, n) where $m, n \in Z$. Now the line and the circle must intersect at at least one point.

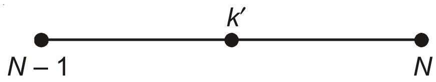
$$\Rightarrow \quad \left|\frac{2m - 5n - k}{\sqrt{(2)^2 + (-5)^2}}\right| \le r$$

$\Rightarrow \quad |2m - 5n - k| \le \sqrt{29}r$

Now as the value of *m, n* vary, $2m - 5n$ can take all integral values. In fact

$N = 2(3N) - 5(N)$

$\Rightarrow \quad |N - k| \le \sqrt{29}r$



Now *N* is an integer, *K* is a real number

$\Rightarrow \quad |N - K| \le \dfrac{1}{2}$

$\Rightarrow \quad \sqrt{29}r \ge \dfrac{1}{2}$

$\Rightarrow \quad r \ge \dfrac{\sqrt{29}}{58}$

$\Rightarrow \quad r_{\min} = \dfrac{\sqrt{29}}{58}$

$\Rightarrow \quad \dfrac{k'}{k} = 2$

3. If a circle of radius *c* touches the circle, $x^2 + y^2 = a^2$ and $x^2 - 2ax + a^2 + y^2 - 2bx + 2ab = 0$, where a, $b > 0$ externally and also their common tangents then

(1) $\dfrac{1}{\sqrt{a}} = \dfrac{1}{\sqrt{c}} + \dfrac{1}{\sqrt{b}}$

(2) $\dfrac{1}{\sqrt{b}} = \dfrac{1}{\sqrt{c}} + \dfrac{1}{\sqrt{a}}$

(3) $\dfrac{1}{\sqrt{c}} = \dfrac{1}{\sqrt{a}} + \dfrac{1}{\sqrt{b}}$

(4) $c = (ab)^{\frac{1}{2}}$

Sol. Answer (3)

The given circles are $x^2 + y^2 = a^2$ and $x^2 - 2ax + a^2 + y^2 - 2bx + 2ab = 0$

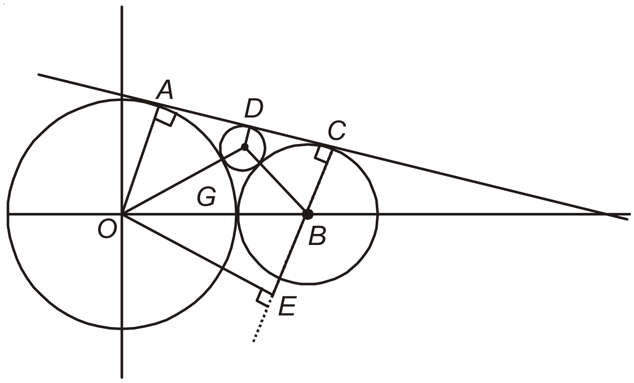
i.e., $x^2 + y^2 = a^2$ and $(x - (a + b))^2 + y^2 = b^2$

Clearly these two circles touch externally, as the distance between their center is equal to the sum of radii.

$AD + DC = AC$

$\Rightarrow \quad OE^2 = OB^2 - BE^2$

$= (a + b)^2 - (a - b)^2 = 4ab$



$OE = 2\sqrt{ab} = AC$

Similarly $CD = 2\sqrt{bc}$

and $AD = 2\sqrt{ac}$

$AC = AD + DC$

$\Rightarrow \quad 2\sqrt{ab} = 2\sqrt{ac} + 2\sqrt{bc}$

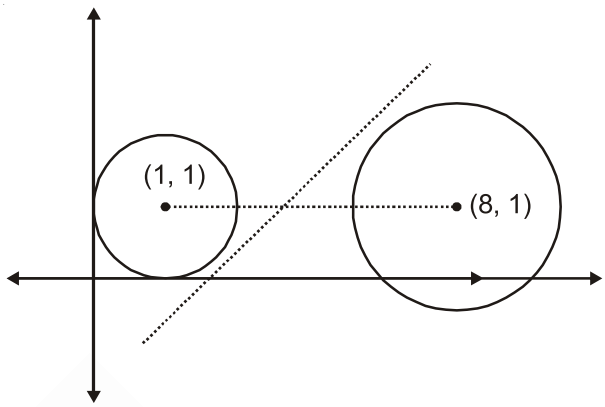
$\Rightarrow \quad \dfrac{1}{\sqrt{c}} = \dfrac{1}{\sqrt{a}} + \dfrac{1}{\sqrt{b}}$

4. Find the range of parameter a for which the variable line $y = 2x + a$ lies between the circles $x^2 + y^2 - 2x - 2y + 1 = 0$ and $x^2 + y^2 - 16x - 2y + 61 = 0$ without intersecting or touching either circle.

Sol. Now as the variable line $y = 2x + a$ lies between the circles, so the centers of the circle must lie on two different side of the lines

$\Rightarrow \quad (a + 1)(a + 15) < 0$

$\Rightarrow \quad -15 < a < -1$



and $\left|\dfrac{a+1}{\sqrt{5}}\right| > 1 \quad \Rightarrow |a+1| > \sqrt{5}$

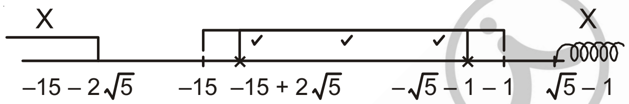
$\Rightarrow \quad a+1 > \sqrt{5}$ or $a+1 < -\sqrt{5}$

$\Rightarrow \quad a > \sqrt{5} - 1$ or $a < -\sqrt{5} - 1$

and $\dfrac{|a+15|}{\sqrt{5}} > 2 \Rightarrow |a+15| > 2\sqrt{5}$

$a + 15 > 2\sqrt{5} \mid a + 15 < -2\sqrt{5}$

$a > -15 + 2\sqrt{5} \mid a < -15 - 2\sqrt{5}$



$\Rightarrow \quad a \in (-15 + 2\sqrt{5}, \ -\sqrt{5} - 1)$

5. A triangle has two of its sides along the axes, its third side touches the circle $x^2 + y^2 - 2ax - 2ay + a^2 = 0$. If the locus of the circumcentre of the triangle passes through the point (38, –37), then $a^2 - 2a$ is equal to ________.

Sol. Let equation of line $\dfrac{x}{p} + \dfrac{y}{q} = 1$.

Condition of tangency. Distance from centre (a, a) is equal to radius

$$\frac{\dfrac{a}{p} + \dfrac{a}{q} - 1}{\sqrt{\dfrac{1}{p^2} + \dfrac{1}{q^2}}} = |a| \qquad \ldots(1)$$

Let centre of circumcircle be (h, k).

$p = 2h$ and $q = 2k$ locus passes through (38, –37)

$p = 76$ and $q = -74$

from (1)

$qa + pa - pq = |a|\sqrt{p^2 + q^2}$

$a(76 - 74) + 76 \times 74 = a\sqrt{(76)^2 + (74)^2}$

$2a + 76 \times 74 = a\sqrt{11252} = 2a\sqrt{2813}$

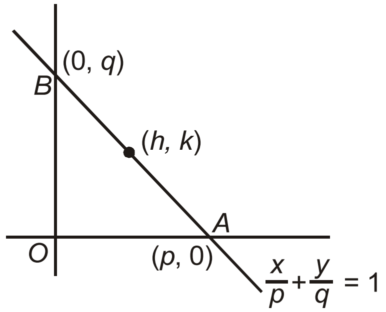
$a + 76 \times 37 = a\sqrt{2813}$

Squaring both sides

$a^2 + (76 \times 37)^2 + 76 \times 74a = a^2(2813)$

$2812a^2 - 5624a = (76 \times 37)^2$

$a^2 - 2a = \dfrac{(76 \times 37)^2}{2812} = 2812$



6. Two circles are inscribed and circumscribed about a square $ABCD$, length of each side of the square is 32. P and Q are two points respectively on these circles, then $\Sigma(PA)^2 + \Sigma(QA)^2$ equals _________.

Sol. $AB = BC = CD = AD = 32$

Let P is a mid-point of AB.

$\therefore \quad AP = 16$

$\Sigma PA^2 = PA^2 + PB^2 + PC^2 + PD^2$

$= (16)^2 + (16)^2 + (16)^2 + (32)^2 + (16)^2 + (32)^2$

$= 3072$

Q lie on outer circle

$\therefore \quad$ Let Q lie at A.

$\therefore \quad QA^2 + QB^2 + QC^2 + QD^2 = 0 + (32)^2 + 2(32)^2 + (32)^2 = 4096$

$\Sigma PA^2 + \Sigma QA^2 = 3072 + 4096 = 7168$

7. The least distance between two points P and Q on the circle $x^2 + y^2 - 8x + 10y + 37 = 0$ and $x^2 + y^2 + 16x + 55 = 0$ is ....

Sol. $S_1 : x^2 + y^2 - 8x - 10y + 37 = 0$

$\Rightarrow \quad (x^2 - 8x) + (y^2 - 10y) + 37 = 0$

$\Rightarrow \quad (x^2 - 8x + 16) + (y^2 - 10y + 25) = 16 + 25 - 37 = 4$

$\Rightarrow \quad (x - 4)^2 + (y - 5)^2 = 2^2$

$\therefore \quad C_1 : (4, 5), r_1 = 2$

Also, $S_2 : x^2 + y^2 + 16x + 55 = 0$

$\Rightarrow \quad (x^2 + 16x) + (y^2) + 55 = 0$

$\Rightarrow \quad x^2 + 16x + 64 + y^2 = 64 - 55 = 9$

$\Rightarrow \quad (x + 8)^2 + y^2 = 3^2$

$\therefore \quad C_2 : (-8, 0), r_2 = 3$

$\therefore \quad C_1C_2 = \sqrt{(4+8)^2 + 5^2} = \sqrt{144+25} = 13$

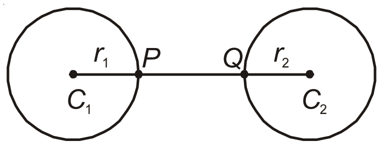
$\Rightarrow \quad C_1C_2 > r_1 + r_2$

$\therefore \quad PQ = C_1C_2 - (r_1 + r_2)$

$= 13 - (2 + 3)$

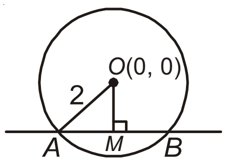
$= 13 - 5$

$= 8$



8. The sum of the squares of the lengths of the chords intercepted by the lines $x + y = n$, $n \in N$, on the circle $x^2 + y^2 = 4$ is ....

Sol. $OM = \left|\frac{n}{\sqrt{1^2+1^2}}\right| = \frac{n}{\sqrt{2}}$



$\Rightarrow \quad OM^2 = \frac{n^2}{2}$

$\Rightarrow \quad AM^2 = OA^2 - OM^2$

$= 4 - \frac{n^2}{2}$

$AM^2 = \frac{8-n^2}{2} \Rightarrow AM = \sqrt{\frac{8-n^2}{2}}$

$AB = 2AM = 2\sqrt{\frac{8-n^2}{2}}$

$AB^2 = 2(8-n^2)$

Hence, the sum of the squares of the lengths

$= 2(8 - 1^2) + 2(8 - 2^2)$

$= 2(8 - 1) + 2(8 - 4)$

$= 2 \times 7 + 2 \times 4$

$= 14 + 8$

$= 22$

9. The chord of contact of tangents from 3 points A, B, C to the circle $x^2 + y^2 = 4$ are concurrent, then the points A, B and C are

Sol. Let $A : (x_1, y_1)$, $B : (x_2, y_2)$ and $C : (x_3, y_3)$

Chord of contacts are

$xx_1 + yy_1 = 4$

$xx_2 + yy_2 = 4$

$xx_3 + yy_3 = 4$

These chord of contacts are concurrent if

$$\begin{vmatrix} x_1 & y_1 & -4 \\ x_2 & y_2 & -4 \\ x_3 & y_3 & -4 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = \frac{0}{-4} = 0$$

A, B and *C* are collinear

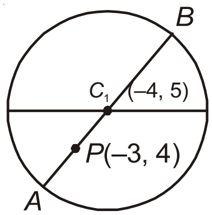
10. Let x, y be real variable satisfying $x^2 + y^2 + 8x - 10y - 59 = 0$. Let $a = \max\{(x + 3)^2 + (y - 4)^2\}$ and $b = \min\{(x + 3)^2 + (y - 4)^2\}$, then $a + b + 2010$ is ....

Sol. S : $x^2 + y^2 + 8x - 10y - 59 = 0$

$\Rightarrow$ $(x^2 + 8x) + (y^2 - 10y) = 59$

$\Rightarrow$ $(x^2 + 8x + 16) + (y^2 - 10y + 25) = 59 + 25 + 16 = 100 = 10^2$

$\Rightarrow$ $(x + 4)^2 + (y - 5)^2 = 10^2$



C_1 : (–4, 5), $r = 10$

$C_1P = \sqrt{(1)^2 + 1^2} = \sqrt{2}$

Here, $a = 2 + 10 = 12$

$b = -2 + 10 = 8$

Thus, $a + b + 2010 = 12 + 8 + 2010 = 20 + 2010 = 2030$

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